

Finite-time breakdown of Euler flows from smooth, compactly supported initial data in every fixed dimension $d \geq 3$

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Abstract

For every fixed integer $d \geq 3$, we construct smooth, compactly supported, divergence-free initial data for the incompressible Euler equation on $\mathbb{R}^d$ whose maximal smooth Sobolev solution has finite lifetime. The Frobenius norm of its velocity gradient has infinite endpoint limsup, and the time integral of the L^∞ norm of $(\Omega_{ij})_{i < j}$ diverges. The pressure has the standard Newton normalization. The proof extends the localized oscillatory construction and scalar amplification mechanism of OpenAI. The higher-dimensional argument retains the full transverse space and the full pressure coupling, replaces curl by antisymmetric potentials, uses reflection only for the central endpoint selection, and proves a quantitative ambient relative estimate. A variational history solver, polynomial bounds for the lifted particle flow, and a fixed physical mean cutoff yield a single frequency sequence for which the initial increments are summable at every Sobolev order. Constants may depend on the fixed dimension.

1 Statement and conventions

For each fixed integer $d \geq 3$, let $C_{c,\sigma}^\infty(\mathbb{R}^d)$ denote smooth compactly supported vector fields with zero divergence. All dimension-dependent constants, derivative orders, starting indices and scale thresholds may depend on this fixed d .

Theorem 1.1 (H_d). *There exists $u_0 \in C_{c,\sigma}^\infty(\mathbb{R}^d)$ whose unique maximal smooth Sobolev solution of*

$$u_t + (u \cdot \nabla)u + \nabla P = 0, \quad \operatorname{div} u = 0, \quad u(0) = u_0$$

has $0 < T^(u_0) < \infty$. Here the pressure is exactly*

$$P = (-\Delta)^{-1} \sum_{i,j=1}^d \partial_i \partial_j (u_i u_j),$$

and, on every $[0, S]$ with $S < T^$, the solution lies in $C([0, S]; H^m) \cap C^1([0, S]; H^{m-1})$ for every integer $m \geq 1$. With $\Omega_{ij} = \partial_i u_j - \partial_j u_i$, it satisfies*

$$\limsup_{t \uparrow T^*} \operatorname{ess\,sup}_x \left(\sum_{i,j} |\partial_i u_j(t, x)|^2 \right)^{1/2} = \infty, \quad \int_0^{T^*} \operatorname{ess\,sup}_x \left(\sum_{i < j} |\Omega_{ij}(t, x)|^2 \right)^{1/2} dt = \infty.$$

The three-dimensional conclusion is the theorem of OpenAI [1], with the Newton pressure and smooth Sobolev class specified above. The present argument applies in that dimension with a transverse space of dimension two. For $d > 3$, every velocity and every localization is

defined on $\mathbb{R}^d$. In particular, the construction retains compact support and finite energy in all spatial variables.

The scalar amplification equation, the variational history method, and the finite oscillatory expansion follow the structure of [1]. Their scalar and Hilbert-space estimates are reproduced below with their hypotheses. The dimension-dependent steps are the antisymmetric-potential localization, the full $(d - 1)$ -dimensional transverse evolution, its coupling to the pressure, and the Sobolev orders in the nonlinear correction and continuation argument. These steps are proved in the ambient dimension. The parameter estimates then provide one frequency sequence for which all initial Sobolev increments are summable.

Throughout, C_d denotes a positive constant depending on the fixed dimension and on fixed auxiliary cutoffs. Constants may change between estimates. Dependence on a derivative order is indicated explicitly. The notation $A \lesssim_d B$ means $A \leq C_d B$, and $A \asymp_d B$ means both inequalities. The principal stage parameters are listed below; each is defined precisely at its first use.

t_0, τ	physical activation time and scaled forward time
$\rho, k, \kappa = k^{-1}$	physical localization length and frequency parameters
δ, f_δ	angular width and periodic primary profile
P	fixed angular period
h_*, h_{child}	inherited shear size and prescribed new shear size
Θ, e	scaled time bound and geometric comparison error
α, A_{tar}	primary amplitude and unnormalized target size
$\mathcal{P}$	bound for the fixed source coefficients and inverse estimates
ℓ_*, r_*	fixed physical mean-cutoff parameters

2 The ambient equation and the endpoint criteria

Fix an integer $d \geq 3$. Constants may depend on d . We use the Euclidean norm on vectors, the operator norm on linear maps when estimating compositions, and the Frobenius norm in the statement of the singularity conclusions. Put $s = d + 2$. In particular $s > d/2 + 1$, so $H^s(\mathbb{R}^d)$ embeds continuously into the space of fields with bounded continuous velocity gradients. We use the ordered-word Sobolev energy

$$E_m(u) = \|u\|_{H^m}^2 = \sum_{r=0}^m \sum_{i_1, \dots, i_r=1}^d \|\partial_{i_1} \cdots \partial_{i_r} u\|_2^2.$$

It is equivalent, with constants depending only on d, m , to the usual multi-index norm. The logarithmic estimate below uses the higher fixed order $s_{\log} = d + 5$. Both orders depend on d .

Lemma 2.1 (Local theory and persistence). *Every solenoidal datum in all integer Sobolev spaces has a unique smooth Sobolev Euler solution on an interval of length bounded below in terms of its H^s norm. The solution belongs to $C_t H^m \cap C_t^1 H^{m-1}$ on every shorter closed interval, for every $m \geq 1$. For $m \geq s$ it satisfies*

$$\frac{d}{dt} \|u\|_{H^m} \leq C_{d,m} \|\nabla u\|_\infty \|u\|_{H^m}. \quad (1)$$

The same solution is obtained at every regularity level.

Proof. Let J_N be convolution with an even, nonnegative, smooth compactly supported kernel of integral one and support radius tending to zero. Set $S_N = J_N \Pi$, where Π is the orthogonal solenoidal projection, and solve on $L^2(\mathbb{R}^d)$

$$\partial_t u_N = -S_N((S_N u_N \cdot \nabla) S_N u_N), \quad u_N(0) = u_0.$$

The smoothed quadratic right-hand side is locally Lipschitz on L^2 . The operators S_N are self-adjoint contractions, commute with every coordinate derivative, and have solenoidal range. Hence the solution remains solenoidal. The integral equation and smoothing give all spatial Sobolev derivatives for this fixed-scale solution. If $v_N = S_N u_N$, then for every derivative word D^w the energy pairing is

$$\langle D^w u_N, D^w \partial_t u_N \rangle = -\langle D^w v_N, D^w ((v_N \cdot \nabla) v_N) \rangle.$$

This identity follows from self-adjointness of the mollifier. The top differentiated transport term cancels by $\operatorname{div} v_N = 0$. For a smooth solution of the limiting Euler equation the same cancellation holds, and the pressure term pairs to zero against each solenoidal velocity derivative. At order one the remaining integral is bounded directly by $C_d \|\nabla u\|_\infty \|\nabla u\|_2^2$. For $m \geq 2$ the remaining terms are products with derivative orders j and $m-j+1$. The Gagliardo–Nirenberg inequalities with endpoints $\nabla u \in L^\infty$ and $D^m u \in L^2$ give, for $1 \leq j \leq m$,

$$\|D^j u\|_{L^{2(m-1)/(j-1)}} \leq C_{d,m} \|\nabla u\|_\infty^{1-(j-1)/(m-1)} \|D^m u\|_2^{(j-1)/(m-1)}.$$

The exponent is interpreted as infinity at $j = 1$. The two reciprocal Lebesgue exponents sum to $1/2$, and the two interpolation parameters sum to one. This proves the homogeneous top-order estimate. Summing the lower orders and retaining conservation of L^2 energy proves (1). The argument uses coordinate differentiation and integration by parts in the ambient dimension.

Combining the preceding pairings, Sobolev embedding, and contraction of S_N in every derivative word gives, for $m \geq s$,

$$E_m(u_N)' \leq C_{d,m} \sqrt{E_s(u_N)} E_m(u_N), \quad E_s(u_N)' \leq C_d (1 + E_s(u_N))^2.$$

An explicit common lifespan is

$$T_0 = \frac{1}{2(1 + C_d)(1 + E_s(u_0))}.$$

On this closed interval $E_s(u_N) \leq 2E_s(u_0) + 1$, uniformly in N . Gronwall then bounds every higher $E_m(u_N)$ on the same interval. These bounds include the L^2 norm required for Banach-space continuation of the regularized quadratic equation.

To pass to the limit, translation followed by the fundamental theorem of calculus gives $\|J_N f - f\|_2 \leq \epsilon_N \|\nabla f\|_2$, with $\epsilon_N \rightarrow 0$. Apply this estimate to both inner smoothed factors and the outer smoothing of the quadratic term. The uniform bounds through order $d+4$ imply that

$$r_N := \partial_t u_N + \Pi((u_N \cdot \nabla) u_N) \quad \text{satisfies} \quad \sup_{t \leq T_0} \|r_N(t)\|_2 \longrightarrow 0.$$

For $u_N - u_M$, the solenoidal transport cancellation leaves an L^2 difference inequality with uniformly bounded gradient coefficient and forcing $r_N - r_M$. All initial values equal u_0 , so Gronwall's inequality implies that u_N is Cauchy in $C_t L^2$. Interpolation with the uniform higher bounds gives convergence in $C_t H^m$ for every fixed m . The limits of the derivative words are the weak derivatives of the same velocity field; Sobolev embedding at arbitrarily high orders supplies its smooth representative. Continuity of the projected quadratic term then gives the limiting integral equation and its time derivative in every H^{m-1} . It also supplies the one-sided derivatives at the endpoints. The same difference estimate proves uniqueness. Consequently the solutions produced using different cutoff orders or Sobolev indices agree on their common intervals. This proves persistence and defines a single maximal smooth Sobolev solution. $\square$

Lemma 2.2 (Newton pressure with fixed normalization). *For a smooth solenoidal u in all Sobolev spaces, set $F = \sum_{i,j} \partial_i \partial_j (u_i u_j)$ and $N_d(x) = |x|^{2-d}/((d-2)|\mathbb{S}^{d-1}|)$ for $x \neq 0$. Since $|\mathbb{S}^{d-1}| = d|B_1|$, this coefficient equals $[d(d-2)|B_1|]^{-1}$. The integral $P = N_d * F$ is absolutely convergent at every point and smooth. Its gradient is the L^2 gradient part of $-(u \cdot \nabla)u$. Thus the projected equation gives the Euler equation with the pressure normalization specified by this convolution.*

Proof. Every derivative of F is both bounded and integrable: expand by the product rule and use the L^2 bounds for the two factors of each product. The kernel is locally integrable and bounded outside the unit ball, so the Newton integral is absolutely convergent. Equivalently, with $G_t(x) = (4\pi t)^{-d/2} e^{-|x|^2/(4t)}$,

$$N_d * F = \int_0^\infty G_t * F dt.$$

Indeed $\int_0^\infty G_t(x) dt = N_d(x)$ for $x \neq 0$, by the substitution $r = |x|^2/(4t)$ and the Gamma integral. Absolute Fubini follows from $\|G_t * |D^\gamma F|\|_\infty \leq \min(\|D^\gamma F\|_\infty, C_d t^{-d/2} \|D^\gamma F\|_1)$. This envelope is time integrable because $d > 2$. It also permits differentiation under the integral to every spatial order.

For each coordinate derivative, the small-time L^2 bound is $\|\partial_i F\|_2$, while the large-time bound is $\|\partial_i G_t\|_2 \|F\|_1 \leq C_d t^{-d/4-1/2} \|F\|_1$. The latter is integrable at infinity for $d > 2$. Hence $\nabla P \in L^2$, and the same argument applies to all its derivatives. The heat equation and its limits at zero and infinity give $-\Delta P = F$. Writing $B = (u \cdot \nabla)u$, incompressibility gives $\operatorname{div} B = F$, so $\nabla P + B$ is solenoidal. Moreover ∇P lies in the closed L^2 gradient space. Indeed its distributional curl vanishes. After Fourier transformation, its L^2 transform is parallel to ξ for almost every $\xi \neq 0$; this is exactly the range of the orthogonal gradient multiplier $\xi \otimes \xi / |\xi|^2$. The zero frequency is a set of measure zero. Cutting off in frequency away from zero and then approximating the potentials by compact smooth functions identifies this range with the closed L^2 gradient space. Orthogonal Helmholtz projection therefore yields $\nabla P = \Pi B - B$.

This normalization is also preserved under the limiting procedure in Lemma 2.1. More generally, suppose $u_n \rightarrow u$ in $C([0, S]; H^m)$ for every finite m , and define F_n, P_n by the same formulas. For each multi-index γ , the product rule, Cauchy–Schwarz, and Sobolev embedding give

$$\sup_{t \leq S} (\|D^\gamma(F_n - F)\|_1 + \|D^\gamma(F_n - F)\|_\infty + \|\nabla D^\gamma(F_n - F)\|_2) \rightarrow 0.$$

Splitting the heat integral at time one in the two estimates above yields

$$\begin{aligned} \|D^\gamma(P_n - P)\|_\infty &\leq C_d (\|D^\gamma(F_n - F)\|_\infty + \|D^\gamma(F_n - F)\|_1), \\ \|\nabla D^\gamma(P_n - P)\|_2 &\leq C_d (\|\nabla D^\gamma(F_n - F)\|_2 + \|D^\gamma(F_n - F)\|_1). \end{aligned}$$

These estimates are uniform on $[0, S]$. Thus the Newton scalars converge with every spatial derivative uniformly, and their gradients converge in every Sobolev space. The same estimates applied to two nearby times give their time continuity. In particular the normalization is specified by the convolution before and after passage to the limit. $\square$

Lemma 2.3 (Stability on varying intervals). *Let u be a smooth Euler solution on $[0, S]$ and let v_n be smooth solutions on $[0, S_n]$, where $0 < S_n \leq S$. If $v_n(0) \rightarrow u(0)$ in H^s , then*

$$\sup_{0 \leq t \leq S_n} \|v_n(t) - u(t)\|_{H^s} \rightarrow 0.$$

No uniform higher norm of v_n is assumed.

Proof. Put $w = v_n - u$. Its equation is $w_t + (u + w) \cdot \nabla w + w \cdot \nabla u + \nabla q = 0$. Differentiation through order s , the transport cancellation, and the Sobolev product estimate give

$$y' \leq C_d \|u\|_{H^{s+1}} y + C_d y^2, \quad y = \|w\|_{H^s}.$$

All coefficients involving u are bounded on $[0, S]$ independently of n . While $y \leq 1$, Gronwall gives $y(t) \leq y(0)e^{CS}$. For large n this is less than $1/2$; continuity excludes a first time when $y = 1$. The estimate therefore holds on the full interval $[0, S_n]$. Sobolev embedding also gives uniform convergence of the velocity gradients there. $\square$

Lemma 2.4 (Continuation and the exact vorticity norm). *Let u be the maximal solution from Lemma 2.1, and suppose its maximal lifetime T^* is finite. Write $\Omega_{ij} = \partial_i u_j - \partial_j u_i$ and $|\Omega| = (\sum_{i < j} |\Omega_{ij}|^2)^{1/2}$. Then*

$$\limsup_{t \uparrow T^*} \|\nabla u(t)\|_{L^\infty, F} = \infty, \quad \int_0^{T^*} \|\Omega(t)\|_\infty dt = \infty.$$

Proof. If $\|\nabla u\|_\infty$ were bounded, (1) would bound the H^s norm up to T^* . Restarting the local solution at times approaching T^* gives a common positive extension length. The same energy estimate propagates every higher norm on a smaller common interval, so uniqueness gives an extension in the full smooth Sobolev class, contrary to maximality. A finite endpoint limsup would imply such a bound, since the solution is smooth on shorter intervals. The inequalities $\|A\|_{\text{op}} \leq \|A\|_F \leq \sqrt{d} \|A\|_{\text{op}}$ identify the required Frobenius conclusion.

For the second assertion, incompressibility gives $\sum_i \partial_i \Omega_{ij} = \Delta u_j$. Use the Gaussian averages

$$Q_t f(x) = (\pi t)^{-d/2} \int_{\mathbb{R}^d} e^{-|y|^2/t} f(x+y) dy, \quad \partial_i Q_t f = \frac{1}{4} Q_t \Delta f.$$

For $f = u_j$, $g_i = \Omega_{ij}$, and $0 < \epsilon < 1$, write

$$\partial_k f = Q_1 \partial_k f + (Q_\epsilon - Q_1) \partial_k f + (I - Q_\epsilon) \partial_k f.$$

The first term is at most $C_d \|f\|_2$, by moving its derivative to the Gaussian kernel and using its L^2 norm. In the middle term use $\Delta f = \sum_i \partial_i g_i$ and move two spatial derivatives to the Gaussian kernel. Its second derivative has L^1 norm at most C_d/t , so time integration from ϵ to one bounds this term by $C_d W(-\log \epsilon)$ whenever $|g_i(x)| \leq W$. The final term is bounded by $C_d \epsilon \|D^2 \partial_k f\|_\infty$: integrate the displayed heat identity from zero to ϵ , using contraction of Gaussian averaging on bounded functions. All integrations by parts are justified by the Gaussian decay and the bounded smooth derivatives of the fields.

Let $\mathcal{H}(u) = \sum_{r=0}^{d+5} \|D^r u\|_2$, with the operator tensor norm in each L^2 term. Sobolev embedding applied to each third derivative gives $\|D^2 \partial_k u_j\|_\infty \leq C_d \mathcal{H}(u)$. Moreover $\mathcal{H}(u) \leq C_d \sqrt{E_{s_{\log}}(u)}$. Choose $\epsilon = (e + C_d \mathcal{H}(u))^{-4}$ in the three bounds. The high remainder is bounded by a dimensional constant, while the middle term is logarithmic. Each entry of Ω , including a lower-triangular entry by antisymmetry, is bounded by the once-counted norm $|\Omega| = (\sum_{i < j} |\Omega_{ij}|^2)^{1/2}$. Summing the finite coordinate estimates therefore proves

$$\|\nabla u\|_{\infty, \text{op}} \leq C_d (1 + \|u(0)\|_2 + \|\Omega\|_\infty \log(e + \mathcal{H}(u))). \quad (2)$$

Here L^2 energy conservation has replaced $\|u(t)\|_2$ by $\|u(0)\|_2$.

Put $K(t) = \|\nabla u(t)\|_{\infty, \text{op}}$, $W(t) = \|\Omega(t)\|_\infty$, and $G(t) = \int_0^t K(r) dr$. The gradient energy inequality at order $s_{\log}$ implies

$$E_{s_{\log}}(u(t)) \leq E_{s_{\log}}(u(0)) e^{C_d G(t)}, \quad \log(e + \mathcal{H}(u(t))) \leq C_0 + C_d G(t),$$

where C_0 depends only on the initial high norm and d . Consequently (2) gives $G' \leq C(1 + W)(1 + G)$, and hence

$$1 + G(t) \leq \exp\left(C\left(t + \int_0^t W(r) dr\right)\right).$$

A finite vorticity integral on $[0, T^*)$ would bound G uniformly. The energy inequality at the lower order $s = d + 2$ would then bound E_s uniformly, permitting the restart already proved and contradicting maximality. This argument differentiates the integral G , and does not require differentiability of a supremum norm or of $\mathcal{H}$.

On every shorter closed time interval, the bounded gradient and vorticity tensors depend continuously on time in the uniform norm: their differences are controlled by the H^{d+2} difference of the velocities. Their norm paths are therefore continuous and nonnegative. The ordinary integrals on these shorter intervals equal the corresponding nonnegative Lebesgue integrals; monotonicity makes a finite integral on $[0, T^*)$ a common bound for all of them. Thus the contradiction proves divergence in the extended nonnegative sense used in the statement. For smooth fields the pointwise and essential suprema agree: a strict pointwise lower bound persists on an open ball of positive measure. Thus these are the essential-supremum norms in Theorem 1.1. $\square$

3 Full-dimensional solenoidal geometry

Let $E = \text{span}(e_1, e_2, e_3)$ and let $\mathcal{R} = I_E \oplus (-I_{E^\perp})$. The fields used below have odd parity and satisfy $u(\mathcal{R}x) = \mathcal{R}u(x)$. The fields are constructed directly on $\mathbb{R}^d$.

Lemma 3.1 (Antisymmetric potentials). *If $A_{ij} = -A_{ji}$ is a smooth compactly supported tensor and $v_j = \sum_i \partial_i A_{ij}$, then $\text{div } v = 0$. For any trace-free matrix L ,*

$$A_{ij}(x) = \frac{x_i(Lx)_j - x_j(Lx)_i}{d} \implies \sum_i \partial_i A_{ij} = (Lx)_j.$$

Multiplying this tensor by an even compact Gevrey cutoff equal to one near zero produces an odd compact solenoidal velocity equal to Lx near zero. If $LR = \mathcal{R}L$ and the cutoff is radial, the velocity is $\mathcal{R}$ -equivariant.

Proof. Commutation of mixed derivatives and exchange of i, j make $\sum_{i,j} \partial_j \partial_i A_{ij}$ equal to its negative. For the displayed tensor, differentiation of the first numerator term gives $(d + 1)(Lx)_j$, and differentiation of the second gives $(Lx)_j + x_j \text{tr } L$. Their difference is $d(Lx)_j$. The cutoff preserves antisymmetry and compact support. The tensor is even, so its divergence is odd. Under $\mathcal{R}$ the tensor transforms as a two-tensor; applying divergence gives the stated equivariance. Gevrey bounds follow from the finite product rule with a polynomial. $\square$

Antisymmetric potentials also provide oscillatory solenoidal corrections. Put $D_i = \kappa \partial_{y_i} + m_i \partial_\theta$. These constant-coefficient operators commute. Thus $\sum_i D_i \sum_j D_j A_{ji} = 0$ for every antisymmetric A . When $m \cdot v = 0$, $|m| = 1$ and $F' = f$,

$$A_{ij} = (m_i v_j - m_j v_i) F(\theta) \implies \sum_i D_i A_{ij} = v_j f(\theta) + \kappa \sum_i \partial_{y_i} A_{ij}.$$

The last term is the spatial corrector and is retained at the terminal truncation order. This identity treats every ambient component.

Lemma 3.2 (Symmetry of the ambient pressure and central blocks). *If u is $\mathcal{R}$ -equivariant, its Newton pressure obeys $P(\mathcal{R}x) = P(x)$. At $x \in E$, both $\nabla u(x)$ and $\nabla^2 P(x)$ preserve E and $E^\perp$.*

Proof. The chain rule and orthogonality show that $\sum_{i,j} \partial_i \partial_j (u_i u_j)$ is invariant under $\mathcal{R}$. The Newton kernel is radial and Lebesgue measure is orthogonally invariant, so a change of variables in its absolutely convergent convolution proves the pressure identity. Differentiate the velocity identity once and the pressure identity twice at a fixed point. The resulting maps commute with $\mathcal{R}$, and therefore preserve its two eigenspaces. The trace of the full velocity gradient vanishes; its restriction to E need not be trace free. $\square$

Lemma 3.3 (The endpoint operator preserves the active plane). *Along the central trajectory of the equivariant preceding solution, suppose the normal $m(t)$ belongs to E . By Lemma 3.2, its gradient $M(t)$ and pressure Hessian $H(t)$ commute with $\mathcal{R}$. For the stationary displacement problem on $[0, t_0]$ with $m(t) \cdot \eta(t) = 0$, $\eta(0) = 0$, prescribed $\eta(t_0) = Y$, and coercive action $\mathcal{A}(\eta) = \int_0^{t_0} (|\eta'|^2 - \langle H\eta, \eta \rangle) dt$, the endpoint map $\Lambda Y = \text{proj}_{m(t_0)^\perp} \eta'(t_0)$ commutes with $\mathcal{R}$. In particular a prescribed endpoint in $E \cap m(t_0)^\perp$ produces a displacement and velocity in E .*

Proof. Reflection preserves the constraint, the two endpoints and the action. Uniqueness for the coercive variational problem therefore gives $\eta_{\mathcal{R}Y} = \mathcal{R}\eta_Y$. Taking the terminal derivative and its orthogonal projection proves the claim about Λ . If $Y = \mathcal{R}Y$, uniqueness also gives $\eta_Y = \mathcal{R}\eta_Y$ at every time. The identity remains true for the physical velocity $\eta' - M\eta$ because M commutes with the same reflection. Hence its $E^\perp$ components vanish, including after endpoint normalization. $\square$

4 Polynomial bounds from the lifted particle flow

The estimates in this section control the particle flow through the small lifted transport field. The resulting bounds depend polynomially on the physical frequency.

Write $\mathcal{L} = \mathbb{R}^d \times \mathbb{R}$, with its product norm, and put $J_k x = (x, k \langle m_0, x \rangle)$, $|m_0| = 1$. For a lifted packet solution $z(t, y, \theta)$ the lifted transport field is

$$b(t, y, \theta) = (\kappa z(t, y, \theta), \langle m_0, z(t, y, \theta) \rangle), \quad k\kappa = 1.$$

Throughout this section, z is the finite packet together with the correction that eliminates its residual. The normal component in the second entry retains the cancellation that gives the small drift estimate.

Lemma 4.1 (Invariance of the physical graph). *Let b be a smooth bounded Lipschitz field on $\mathcal{L}$, continuous in time, of the displayed form. Its global flow $\Psi_{s,t}$ satisfies*

$$\Psi_{s,t}(J_k x) = J_k \psi_{s,t}(x), \quad \psi_{s,t}(x) = \pi \Psi_{s,t}(J_k x),$$

where $\pi(y, \theta) = y$. The map ψ is the flow of $v(t, x) = \pi b(t, J_k x)$ and $\psi_{t,s}$ is its inverse. For any $\ell > 0$, the flow of $v_\ell(t, x) = \ell v(t, x/\ell)$ is $\ell \psi_{s,t}(x/\ell)$.

Proof. The linear functional $q(y, \theta) = \theta - k \langle m_0, y \rangle$ satisfies $q(b) = 0$. Its derivative along every trajectory is zero. Therefore the level set $q = 0$, which is precisely the image of J_k , is invariant. Projection of the lifted ODE gives the ODE for v ; uniqueness identifies its flow and inverse. Substitution and the chain rule prove the scaling identity in every ambient dimension. $\square$

Lemma 4.2 (Small lifted flow bounds). *Suppose, for $0 \leq t \leq T$ and every $n \geq 0$,*

$$\|D^n b(t)\|_\infty \leq B R^n (n!)^2, \quad B \geq 0, \quad R > 0, \quad B R T \leq \frac{1}{8}.$$

Then the displacement $f_t = \Psi_{0,t} - \text{id}$ obeys

$$\|D^n f_t\|_\infty \leq B t (4R)^n (n!)^2 \quad (n \geq 0). \quad (3)$$

Put $\mathcal{R}(S) = (4R+1)((1+BT)S+2)$. If also $\|D^n b_t\|_\infty \leq B_1 R_1^n (n!)^2$ with $B_1 \geq 0$ and $R_1 \geq 0$, then the material velocity and acceleration satisfy

$$\|D^n(b(t) \circ \Psi_{0,t})\|_\infty \leq B\mathcal{R}(R)^n (n!)^2, \quad (4)$$

$$\|D^n((b_t + Db b)(t) \circ \Psi_{0,t})\|_\infty \leq (B_1 + 3B^2 R)\mathcal{R}(4R + R_1)^n (n!)^2. \quad (5)$$

These estimates hold in every finite-dimensional normed space.

Proof. Apply the finite generating-sum argument of [1, §3.8] to the integral flow equation. For a finite derivative cutoff N , let

$$F_N(t) = \sum_{n=1}^N \frac{r^n}{(n!)^2} \|D^n f_t\|_\infty, \quad r = (4R)^{-1}.$$

The differentiated equation $f_t = \int_0^t b(s) \circ (\text{id} + f_s) ds$ and the finite Faà di Bruno convolution give

$$F_N(t) \leq \int_0^t \frac{BR(r + F_N(s))}{1 - R(r + F_N(s))} ds$$

as long as the denominator is positive. The algebra behind this inequality is the scalar factorial majorant for a composition: after division by $(n!)^2$, each partition coefficient is bounded by the corresponding geometric-series coefficient. It applies to operator norms of multilinear derivatives, independently of the ambient dimension. Under the bootstrap assumption $F_N \leq Bt$, one has $R(r + F_N) \leq 1/4 + BRT \leq 3/8$, so the integrand is at most $3B/5$. Continuity and a first-contact argument give $F_N(t) \leq Bt$ for all $t \leq T$, uniformly in N . The n th summand gives (3) for $n > 0$; order zero follows by directly integrating $\|b\|_\infty \leq B$. The finite generating sums may be estimated pointwise before taking the supremum, so differentiation of the supremum is unnecessary.

Adding the identity gives, for $n > 0$, $\|D^n \Psi_{0,t}\|_\infty \leq (1 + BT)(4R + 1)^n (n!)^2$. The same finite composition formula implies that composing a field bounded by $CS^n (n!)^2$ with this map is bounded by $C\mathcal{R}(S)^n (n!)^2$. This gives (4). Differentiating a factorial bound once and using $(n+1)^2 \leq 4^n$ bounds Db by $BR(4R)^n (n!)^2$. The factorial Leibniz convolution is at most three times the product of the two amplitudes. At the common radius $4R + R_1$ this bounds $b_t + Db b$ by $(B_1 + 3B^2 R)(4R + R_1)^n (n!)^2$. The composition estimate proves (5). $\square$

Lemma 4.3 (Physical labels and their inverse). *The graph and physical scaling in Lemma 4.1 convert a bound $\|D^n f\|_\infty \leq CS^n (n!)^2$ into*

$$\|D^n[\ell\pi f(J_k(\cdot/\ell))]\|_\infty \leq C(\ell^{-1}S(1 + |k|))^n (n!)^2 \quad (0 < \ell \leq 1).$$

Consequently the physical displacement, material velocity and material acceleration have factorial bounds with amplitudes and radii polynomial in $1 + B + R + B_1 + R_1 + T + \ell^{-1} + |k|$. If the physical field is solenoidal, its inverse deformation, evaluated in particle labels, has factorial bounds polynomial in the same parameters, with exponents allowed to depend on d .

Proof. The chain rule for a linear inner map gives the factor $\|J_k/\ell\|^n$, and $\|J_k\| \leq 1 + |k|$. The outer map has norm at most $\ell \leq 1$. This proves the first assertion, including order zero. Apply it to (3)–(5). The first spatial derivative of a flow is the identity plus the derivative of its displacement, so it has the same type of polynomial factorial bound after enlarging the parameters.

Solenoidality gives $\det D\psi = 1$ by the variational equation and Jacobi's formula. Thus $(D\psi)^{-1} = \text{adj}(D\psi)$. Every entry of the adjugate is a sum of $(d-1)!$ products of $d-1$ entries. For a product of r factorially bounded fields, the multinomial Leibniz formula and

$\prod_{i=1}^r n_i! \leq n!$ show that its n th derivative is bounded by $C_d C^r(rR)^n(n!)^2$. Here $r = d - 1$ is fixed. Hence the inverse deformation has a polynomial factorial bound, with no exponential in the physical frequency. For composition with the preceding flow, the finite composition and product formulas combine its factorial bounds with the graph-flow bounds above. Two time derivatives are controlled by the material velocity and acceleration estimates. $\square$

The trace identity needed in this construction is also dimension independent. If a lifted field is tangent to the graph, $b = J_k \pi b$. Differentiating and cyclically commuting the trace gives

$$\mathrm{tr}_{\mathbb{R}^d}(\pi D b J_k) = \mathrm{tr}_{\mathcal{L}}(J_k \pi D b) = \mathrm{tr}_{\mathcal{L}}(D b).$$

Thus lifted divergence zero implies ordinary divergence zero for the graph velocity. Scaling preserves it. The graph flow and the preceding particle map both preserve d -dimensional volume, as does their composition.

4.1 Cylinder and graph bounds in L^2

To derive global L^2 estimates, let b be periodic in its last variable with period P and assume its cylinder divergence is zero. Suppose the input bounds of Lemma 4.2 hold both in supremum norm and in $L^2(\mathbb{R}^d \times \mathbb{T}_P)$, with common amplitudes B, B_1 and radii R, R_1 . The weighted Sobolev estimates for the packet and its time derivative provide these input L^2 bounds.

The flow satisfies $\Psi_t(y, \theta + P) = \Psi_t(y, \theta) + (0, P)$ by uniqueness. It therefore descends to the cylinder. Its variational Jacobian obeys

$$\partial_t \det D \Psi_t = (\mathrm{div} b)(t, \Psi_t) \det D \Psi_t = 0.$$

The inverse flow exists by the same bounded Lipschitz ODE, so change of variables gives

$$\|a \circ \Psi_t\|_{L^2(\mathbb{R}^d \times \mathbb{T}_P)} = \|a\|_2. \quad (6)$$

For the lifted velocity of the corrected packet, $\mathrm{div} b = \kappa \mathrm{div}_y z + \partial_\theta(m_0 \cdot z) = 0$ is the closed solenoidal constraint, which holds pointwise by smoothness.

Set $S_v = \mathcal{R}(R)$ and $S_a = \mathcal{R}(4R + R_1)$, with $\mathcal{R}$ as in Lemma 4.2. Then the fields satisfy

$$\|D^n(b(t) \circ \Psi_t)\|_2 \leq B S_v^n(n!)^2, \quad (7)$$

$$\|D^n((b_t + D b b)(t) \circ \Psi_t)\|_2 \leq (B_1 + 3B^2 R) S_a^n(n!)^2, \quad (8)$$

$$\|D^n(\Psi_t - \mathrm{id})\|_2 \leq B t S_v^n(n!)^2. \quad (9)$$

To prove these assertions, expand the derivative of a composition into its finite Faà di Bruno sum. In each term place the single factor $D^r a \circ \Psi_t$ in L^2 using (6); place all derivatives of the inner flow in L^∞ using Lemma 4.2. The numerical composition convolution is the same as for the supremum bound, so its radius is $\mathcal{R}(S)$ for an outer radius S . For $D b b$, put one factor in L^2 and the other in L^∞ in each Leibniz term. The factorial convolution is bounded by three, giving amplitude $3B^2 R$ at radius $4R$. This proves the first two estimates. Finally integrate the identity $\Psi_t - \mathrm{id} = \int_0^t b(s) \circ \Psi_s ds$ and apply Minkowski's inequality to each derivative. This proves the last one, including order zero, with radius S_v .

The graph trace estimate includes the physical scaling. If a smooth periodic field a has $\|D^n a\|_2 \leq C S^n(n!)^2$, then

$$\|D^n\{\ell \pi a(J_k(\cdot/\ell))\}\|_{L^2(\mathbb{R}^d)} \leq C_P \ell^{1+d/2} C(1+S) \left(\frac{4S(1+|k|)}{\ell} \right)^n (n!)^2. \quad (10)$$

Indeed the one-dimensional periodic estimate

$$\sup_\theta |v(\theta)|^2 \leq \frac{2}{P} \int_{\mathbb{T}_P} |v|^2 + 2P \int_{\mathbb{T}_P} |v_\theta|^2$$

follows from the fundamental theorem of calculus, the triangle inequality and Cauchy–Schwarz. The same constants apply to any finite-dimensional normed target, in particular to the operator norm of $D^n a$, independently of n . Applied at each spatial point it bounds evaluation on any phase graph by $C_P(\|a\|_2 + \|\partial_\theta a\|_2)$; the phase slope does not enter this trace estimate. Differentiation of the affine graph introduces the factor $\|J_k\|^n \leq (1 + |k|)^n$ before applying that trace. The additional angular derivative is bounded using $(n + 1)^2 \leq 4^n$. Finally $x = \ell y$ gives the exact factor $\ell^{1+d/2-n}$. For $0 < \ell \leq 1$ the prefactor $\ell^{1+d/2}$ is bounded above by one and may be omitted. This proves (10). The argument applies to the displacement, material velocity and material acceleration in (7)–(9).

For the parameter induction, use the source polynomial $C_{\text{rel}}(W) \geq 1$ obtained by inserting the packet bounds into the preceding finite composition formulas. Here $W \geq 1$ bounds the source coefficients, the common packet radius, and its profile amplitude. The correction that eliminates the residual has amplitude $\delta_{\text{corr}}(X)$, where X is its expansion parameter. The lifted drift bound is a source polynomial times $k^{-1} + \delta_{\text{corr}}(X)$. Thus, with $T \leq 1$ and the physical scale $\ell = \rho \leq 1$, the same explicit polynomial can be enlarged to give

$$BRT \leq C_{\text{rel}}(W)(k^{-1} + \delta_{\text{corr}}(X)), \quad M_{\text{rel}} \leq C_{\text{rel}}(W), \quad S_{\text{rel}} \leq C_{\text{rel}}(W)\rho^{-1}k.$$

Here $M_{\text{rel}}, S_{\text{rel}} \geq 1$ are common factorial amplitudes and radii for the three relative fields in L^2 , and for the displacement and material velocity also in supremum norm. The latter supremum bounds come directly from the small lifted flow estimates; applying a physical Sobolev embedding here would introduce unnecessary powers of $\rho^{-1}k$ into the amplitude.

The prescribed scales satisfy the smallness conditions

$$16C_{\text{rel}}(W) \leq k^{1/2}, \quad \rho^{-1} \leq k^{1/2}, \quad \delta_{\text{corr}}(X) \leq k^{-1}, \quad k \geq 1.$$

Consequently $BRT \leq 1/8$, $M_{\text{rel}} \leq k$, and $S_{\text{rel}} \leq k^2$. These bounds will be used in the parameter induction. All three fields are restrictions of the same lifted flow. In original physical labels their relative map is $\psi_t(x) = \rho \psi_t^{\text{norm}}(x/\rho)$, and the updated particle map is $X_{\text{parent}}(t, \psi_t(x))$. The composition uses the label bounds of the preceding map in its original physical labels.

4.2 Fixed Sobolev shifts and label renewal

For the evaluation step fix $q = d + 2$, and suppose a smooth Euclidean vector-valued field satisfies $\|D^n a\|_2 \leq CR^n(n!)^2$ for every $n \geq 0$, with $C, R \geq 0$. Define

$$A_d(C, R) = c_d C \sum_{j=0}^q (4R)^j (j!)^2, \quad R_d(R) = 4 \max(1, d)R,$$

where c_d is the fixed Sobolev evaluation constant. Then

$$\|D^n a\|_\infty \leq A_d(C, R) R_d(R)^n (n!)^2. \quad (11)$$

Indeed, apply the fixed H^q evaluation inequality to each coordinate word of length n . Its derivative of order $j \leq q$ has L^2 norm bounded by $\|D^{n+j} a\|_2$. The factorial inequality

$$((n + j)!)^2 \leq 4^{n+j} (n!)^2 (j!)^2$$

gives the amplitude A_d and coordinate-word radius $4R$. The operator norm of an n -linear derivative is at most the sum of its d^n coordinate evaluations, which gives exactly $R_d(R)$. Thus the derivative loss is fixed by d , independently of packet grade or truncation.

The composition with the preceding flow has the following chain-rule expressions. Write $X_{\text{parent}}(t, x) = x + A(t, x)$, with material velocity $U = \partial_t A$ and acceleration $W_0 = \partial_t U$. Write

the relative fields as a, b, c with $\partial_t a = b$, $\partial_t b = c$, and $\psi = x + a$. The updated displacement, velocity, and acceleration are

$$\begin{aligned} A_{\text{new}} &= a + A \circ \psi, \\ U_{\text{new}} &= U \circ \psi + (I + DA \circ \psi)b, \\ W_{\text{new}} &= W_0 \circ \psi + 2(DU \circ \psi)b + (D^2 A \circ \psi)[b, b] + (I + DA \circ \psi)c. \end{aligned}$$

Suppose the three preceding fields have both L^2 and supremum factorial bounds with amplitude and radius $K \leq k$, $K \geq 1$. Let $M = M_{\text{rel}} \leq k$ and $S = S_{\text{rel}} \leq k^2$. Since ψ preserves volume, each factor composed with the preceding flow retains its L^2 norm before applying the derivative composition formula. The relative displacement and velocity provide the needed supremum factors; the relative acceleration is placed in L^2 . Differentiating a factorial bound changes amplitude C and radius R to CR and $4R$. The first and second derivative amplitudes of the preceding fields are consequently K^2 and $4K^3$. The finite product and composition estimates bound all three updated fields in L^2 by the common amplitude and radius

$$C_{\text{new}} = K + M + 9K^2 M + 36K^3 M^2, \quad R_{\text{new}} = (1 + S)((1 + M)16K + 2) + S.$$

For $k \geq 69$ these satisfy $C_{\text{new}} \leq 47k^5 \leq k^6$ and $R_{\text{new}} \leq 69k^4 \leq k^5$.

The evaluation step (11) now applies to these composed fields. Enlarge the fixed lower bound on k to include $4 \max(1, d)$ and $c_d \sum_{j=0}^q (j!)^2$. Since $4R_{\text{new}} \leq k^6$, it gives

$$A_d(C_{\text{new}}, R_{\text{new}}) \leq k^{6q+7}, \quad R_d(R_{\text{new}}) \leq k^6.$$

The sum of one and the amplitude and radius bounds in L^2 and supremum norm is at most $5k^{6q+7} \leq k^{6q+8} \leq k^{10(d+4)}$. Thus $K_{\text{new}} = k^{10(d+4)}$ is a single valid label invariant for all three updated fields in both norms. Differentiating these fields gives the frame coefficients $F = I + DA_{\text{new}}$, $F_1 = DU_{\text{new}}$, and $F_2 = DW_{\text{new}}$, with amplitudes $1 + K_{\text{new}}^2, K_{\text{new}}^2, K_{\text{new}}^2$ and radius $4K_{\text{new}}$. The determinant-one inverse argument above gives the corresponding polynomial bound for the inverse frame of the updated physical particle map.

5 The active three-plane and ambient amplification

All cross products in this section are taken in the oriented Euclidean three-plane E and define an orthonormal basis of E . Write $M = \nabla u(t, 0)$ for the central gradient of the preceding solution and $H = \nabla^2 P(t, 0)$. Oddness fixes the central particle at zero. Reflection implies that M, H are block diagonal on $E \oplus E^\perp$. The active block need not be trace free. Norms of M, H, B in this section are suprema in time at the central particle. Full-space coefficient bounds enter through the polynomial particle-label estimates, and the spatial history inverse uses the global one-sided pressure bound.

The geometric input at a stage is the decomposition

$$M = B + hqp^T + \mathcal{E}, \quad |\mathcal{E}| \leq E_*, \quad h(t_0) = h_*, \quad (12)$$

where p, q are the normalized transported normal and primary velocity of the preceding packet. They lie in E and are orthogonal; put $n = p \times q$. The shear is the preceding packet amplitude, $h = c_0 |m_{\text{old}}| |v_{\text{old}}|$ with constant $c_0 > 0$. Consequently $\dot{h}/h = -B_{pp} - B_{qq}$, by the two transport equations for the preceding normal and velocity. Here B is the gradient of the preceding background solution. On the stage interval $|B| \leq CG_*$ and $|\dot{B}| \leq CG_*^2$. The transported normal and velocity satisfy

$$\dot{m} = -M^T m, \quad \dot{v} = -Mv + 2m \frac{m \cdot Mv}{|m|^2}, \quad m \cdot v = 0. \quad (13)$$

The coefficient 2 follows by differentiating the last constraint. These ambient equations preserve E for central data in E .

At activation put

$$a = p \cdot Bq \in [1/2, 2], \quad \beta = (n \cdot Bq)/a, \quad \frac{1}{2} \leq \beta x_{\text{prev}}^2 \leq 2, \\ \epsilon = \sqrt{a/h_*}, \quad \tau = a(t - t_0)/\epsilon, \quad x = \sqrt{\beta} \tau.$$

The target is $x = x_{\text{tar}} \geq 2$. Write $\tau_{\text{tar}} = x_{\text{tar}}/\sqrt{\beta}$ and use the scaled endpoint $\Theta = \tau_{\text{tar}} + \Delta\tau$. Here $0 < \sqrt{\beta} \leq 1/4$ and $0 < \Delta\tau \leq \Theta^{-60}$, so in particular $\Theta \geq \max\{2, \beta^{-1/2}, \tau_{\text{tar}}\}$. For the next prescribed physical time width w_{next} , set $\Delta\tau = 2\sqrt{ah_*} w_{\text{next}}$, and $(\epsilon/a)\Delta\tau = 2w_{\text{next}}$. The scalar parameter choice provides an envelope $\hat{\Theta} \geq \Theta$ and $\Delta\tau \hat{\Theta}^{60} \leq 1$; monotonicity of reciprocal powers proves the stated bound for Θ . The geometric error parameter is

$$e = C_d(\epsilon\Theta G_*^2 + E_* + \rho K_h^{c_d}/\epsilon), \quad e\Theta^{40} \leq c_d. \quad (14)$$

Here ρ is the physical packet length and K_h bounds the preceding particle map, its inverse deformation and the required derivatives. The small constant on the right is fixed before the scales are chosen.

Lemma 5.1 (Endpoint selection). *On a positive history interval $[0, t_0]$, assume the one-sided pressure bound $H \leq K + I$ and $K + t_0^2/2 \leq 1/2$, provided by the mean reserve. Suppose $\|M\|_\infty \leq C_M h_*$, $\|H\|_\infty \leq C_H h_* h_{\text{old}}$, $1 \leq h_{\text{old}} \leq h_*$, and $t_0^{-1} \leq h_*$. Suppose also $\bar{B} = B(t_0) + \mathcal{E}(t_0)$ has $\bar{B}_{pp} < 0$ and $\|\bar{B}\| \leq \zeta h_*$. For sufficiently small ζ , depending only on C_M, C_H , an endpoint $Y \in E \cap n^\perp$ of size at most C/h_* gives a stationary displacement whose terminal primary velocity satisfies*

$$v_q = 1, \quad -C \leq v_p \leq 0, \quad v_{E^\perp} = 0.$$

The displacement is stationary in the full transverse space.

Proof. Choose $m_0 = F(t_0, 0)^T n / |F(t_0, 0)^T n|$ and transport it by F^{-T} . Let Λ be the endpoint operator of Lemma 3.3. Integration by parts in its stationary equation gives

$$Y \cdot \Lambda Z = \int_0^{t_0} (\eta'_Y \cdot \eta'_Z - H \eta_Y \cdot \eta_Z) dt.$$

The stationary paths start at zero. The inequality $\int |\eta|^2 \leq (t_0^2/2) \int |\eta'|^2$ therefore implies that their action is nonnegative even for a nonzero terminal value. Thus Λ is symmetric and positive semidefinite. To bound it, put $L_0 = t_0^{-1} + \|M\|_\infty + \|H\|_\infty^{1/2} + 1$. A comparison path vanishes before $t_0 - L_0^{-1}$ and then equals $L_0(t - t_0 + L_0^{-1}) \text{proj}_{m(t)^\perp} Y$. Since $\|\partial_t \text{proj}_{m^\perp}\| \leq C\|M\|$, its action is at most $C(L_0 + \|M\|_\infty^2/L_0 + \|H\|_\infty/L_0)|Y|^2 \leq C_0 h_* |Y|^2$. Minimality yields $0 \leq \Lambda \leq C_0 h_* I$ on the full transverse space, and reflection reduces its restriction to $E \cap n^\perp$ to a symmetric two-by-two matrix.

Write $b = \Lambda_{pq}$, $d_0 = \Lambda_{qq}$ and $u_* = \Lambda_{pp} - \bar{B}_{pp} > 0$. The terminal velocity $v = \eta' - M\eta$ has active coordinates

$$\begin{pmatrix} v_p \\ v_q \end{pmatrix} = \begin{pmatrix} u_* & b - \bar{B}_{pq} \\ b - h_* - \bar{B}_{qp} & d_0 - \bar{B}_{qq} \end{pmatrix} \begin{pmatrix} Y_p \\ Y_q \end{pmatrix}.$$

If $b \leq h_*/2$, take $Y = -p$. Then $v_p = -u_* \leq 0$ and $v_q = h_* + \bar{B}_{qp} - b \geq h_*/3$. If $b > h_*/2$, positivity implies $\Lambda_{pp} \geq b^2/d_0 \geq h_*/(4C_0)$. Take $Y = q - (b - \bar{B}_{pq})p/u_*$. Its norm is bounded by a constant depending on C_0 , and $v_p = 0$. Moreover

$$v_q = \left(d_0 - \frac{b^2}{u_*}\right) - \bar{B}_{qq} + \frac{bh_* + b(\bar{B}_{pq} + \bar{B}_{qp}) - h_*\bar{B}_{pq} - \bar{B}_{qp}\bar{B}_{pq}}{u_*}.$$

The first parenthesized term is nonnegative because $u_* \geq \Lambda_{pp}$. The numerator is at least $h_*^2/2 - C(C_0 + 1)\zeta h_*^2$. Since $u_* \leq (C_0 + \zeta)h_*$, choosing ζ small gives $v_q \geq ch_*$. Divide the displacement, endpoint and velocity by this positive terminal value. Both cases give the asserted normalization and endpoint bound. Reflection gives vanishing $E^\perp$ components before and after normalization. At $t_0 = 0$, prescribe $v = q$ directly. $\square$

Lemma 5.2 (The scalar growing solution). *Let $\beta > 0$ be sufficiently small, and set $P_0 = \beta\tau^2$, $Q_0 = -2\beta\tau$ and $D = 1 + P_0^2$. The equation*

$$(DV')' = 2(1 - \beta P_0)V, \quad V_\lambda(0) = 1, \quad V'_\lambda(0) = \lambda \geq 0$$

has a positive solution on every target interval under consideration, and $V_\lambda(x = 1) \geq e^{b/\sqrt{\beta}}$. For $x \geq 1$ put $r_\lambda = -V'_\lambda/V_\lambda$. There is a bounded nonnegative z such that

$$r_\lambda = \frac{\sqrt{\beta}}{x} - \frac{z}{x^2}, \quad z^2 = \frac{2}{1 + x^{-4}} + O(\sqrt{\beta}) \quad (x \geq 2).$$

Writing $g(\tau) = V_0(\tau)$, the two-component propagator for $(-V', V)$ obeys

$$\|\Phi_0(t, s)\| \leq C\Theta^8 g(t)/g(s).$$

The estimates are dimension independent.

Proof. On $x \leq 1$, integrating twice with $D \leq 2$ and $2(1 - \beta P_0) \geq 1$ gives $V_\lambda \geq 1 + \lambda\tau/2 + \frac{1}{2} \int_0^\tau (\tau - s)V_\lambda(s)ds$. Positive iteration bounds it below by $\cosh(\tau/\sqrt{2})$. Its logarithmic derivative l satisfies $0 \leq l$ and $l' \leq 2 - l^2$, whence $l \leq \sqrt{2} \coth(\sqrt{2}\tau)$. For $x \geq 1$, write $\hat{V}(x) = V(x/\sqrt{\beta})$ and set $s = 1 - \rho$, $\rho = x^{-1}$ and $f(\rho) = \hat{V}(1/\rho)/\rho$. Direct substitution gives

$$((1 + \rho^4)f_\rho)_\rho = (2/\beta - 2\rho^2)f.$$

Starting from $f(1) > 0$, $f_\rho(1) < 0$, backward integration preserves both signs. For $z = -\sqrt{\beta}f_\rho/f$ and $\mu = \sqrt{(2 - 2\beta\rho^2)/(1 + \rho^4)}$,

$$\sqrt{\beta}z_s = \mu^2 - z^2 + \sqrt{\beta} \frac{4\rho^3}{1 + \rho^4} z.$$

This gives a uniform upper bound and preserves nonnegativity. Indeed its initial value at $\rho = 1$ is $z(1) = \sqrt{\beta} + V'_\lambda(1/\sqrt{\beta})/V_\lambda(1/\sqrt{\beta})$, which is bounded independently of λ by the preceding logarithmic-derivative estimate. The equation for $z - \mu$ has damping coefficient $z + \mu \geq c > 0$ and a bounded forcing multiplied by $\sqrt{\beta}$. Its integrating factor gives $|z - \mu| \leq Ce^{-c(1-\rho)/\sqrt{\beta}} + C\sqrt{\beta}$, which proves the direction formula for $\rho \leq 1/2$.

Reduction of order gives $V_\lambda = g(1 + \lambda I)$, where $I(\tau) = \int_0^\tau (Dg^2)^{-1}$ and $I(1) \geq c > 0$. Positivity before $x = 1$, and increase of xg afterwards, give $g(s)/g(u) \leq C\Theta$ for $s \leq u$. For arbitrary scalar data at s ,

$$V(t) = \frac{g(t)}{g(s)} \left[V(s) + D(s) \left(V'(s) - \frac{g'(s)}{g(s)} V(s) \right) \int_s^t \frac{(g(s)/g(u))^2}{D(u)} du \right].$$

The integral is at most $C\Theta^3$, $D(s) \leq C\Theta^4$, and $|g'/g| \leq C$. Near $\tau = 0$ the latter bound follows from $g'(0) = 0$ and $l' \leq 2 - l^2$; away from zero it follows from the displayed bounds for l and z . Differentiating this formula introduces at most one additional power of Θ , proving the propagator assertion. This scalar calculation follows [1, §§4.4–4.5]. $\square$

The selected initial conditions at activation are $m(t_0) = s_0 n(t_0)$, $v_q(t_0) = 1$, $v_n(t_0) = 0$ and $-C \leq v_p(t_0) \leq 0$. At the central label express $m = s_0(Pp + \epsilon Qq + Nn)$ and $v = \epsilon Up + Vq + \epsilon Wn$. The moving-frame connection has $(S_f)_{12} = B_{pq}$, $(S_f)_{13} = B_{pn}$ and $(S_f)_{32} = -B_{nq}$; skew symmetry determines its other entries. Equations (13) then give

$$(P, Q, N) = (\beta\tau^2, -2\beta\tau, 1) + O(\epsilon\Theta^5), \quad W = -(PU + QV)/N, \quad N \geq 1/2.$$

The reference ray system is $P' = -Q$, $Q' = -2\beta N$, $N' = 0$. Its triangular propagator is bounded by $C\Theta^2$. The error integral is at most $C\epsilon\Theta^5 + C\epsilon\Theta^3 \sup |\text{error}|$, and the last term is absorbed by (14). Substitution in the velocity equation gives a coefficient error at most $C\epsilon\Theta^{13}$ from the two-component ideal system

$$U' = -2V + \frac{2P_0((P_0 + \beta)V + Q_0U)}{1 + P_0^2}, \quad V' = -U.$$

Indeed the only unsuppressed contributions to $J = (m \cdot Mv)/(s_0a)$ are $PB_{pq}V/a$, $h\epsilon^2QU/a$ and $NB_{nq}V/a$. The identity $\dot{h}/h = -B_{pp} - B_{qq}$ gives $h/h_* = 1 + O(\epsilon\Theta G_*)$ on this interval, so $h\epsilon^2/a = 1 + O(\epsilon\Theta G_*)$. The remaining contributions contain either ϵG_* or a term controlled by (14). Transversality bounds $|W| \leq C\Theta^2(|U| + |V|)$; the denominator $|m|^2/s_0^2$ is at least $1/4$. These estimates bound every rational coefficient independently of the trace of the active block. The full matrix estimate, including all transverse coordinates, is derived in Section 8.

Apply Duhamel's formula with weight $g(s)/g(t)$. Lemma 5.2 bounds its error operator by $C\epsilon\Theta^{22}$. A geometric series therefore gives the active propagator bound $2C\Theta^8 g(t)/g(s)$. For the selected data, $\lambda = -v_p(t_0)/\epsilon \geq 0$,

$$|U - U_\lambda| + |V - V_\lambda| \leq C\epsilon\Theta^{30}(1 + \lambda)g.$$

For $\tau \geq 1$, $V_\lambda \geq c(1 + \lambda)g$, so this is a relative error. In particular

$$J/V = P_0 + \beta + Q_0 r_\lambda + O(\epsilon\Theta^{34}) > 0. \quad (15)$$

For $x \leq 1$ its leading term is at least β , since $Q_0, r_\lambda \leq 0$. For $x \geq 1$ it equals $x^2 - \beta + 2\sqrt{\beta}z/x$. For $x \leq 1$, the error is at most $Cc_d\Theta^{-6} \leq Cc_d\beta^3$, hence less than $\beta/2$ after fixing c_d sufficiently small. For $x \geq 1$ the leading term is bounded below by $1 - \beta$, and the same error bound suffices.

Lemma 5.3 (Central frame renewal). *At the target set $p_2 = m/|m|$, $q_2 = v/|v|$, $n_2 = p_2 \times q_2$, $a_2 = p_2 \cdot Mq_2$ and $\beta_2 = (n_2 \cdot Mq_2)/a_2$. Then*

$$\begin{aligned} a_2/a &= 1 + O(x_{\text{tar}}^{-4} + \beta x_{\text{tar}}^{-2} + \sqrt{\beta} x_{\text{tar}}^{-3} + \epsilon\Theta^{40}), \\ \beta_2 x_{\text{tar}}^2 &= 1 + O(\sqrt{\beta} + x_{\text{tar}}^{-4} + \epsilon\Theta^{40}), \\ p_2 \cdot Mp_2 &\leq -\frac{c\sqrt{h_*}}{x_{\text{prev}} x_{\text{tar}}} + C(G_* + E_*). \end{aligned}$$

Moreover $A_{\text{tar}} = |m||v|$ at the target satisfies $A_{\text{tar}} \geq K_h^{-c}\Theta^{-c}e^{bx_{\text{prev}}}$.

Proof. Put $r = U/V$, $w = W/V$, $D_a = |m|^2/s_0^2$ and $E_v = 1 + \epsilon^2(r^2 + w^2)$. Normalize the two vectors directly:

$$a_2/a = \frac{J/V}{\sqrt{D_a}\sqrt{E_v}}, \quad \beta_2 = \frac{[(P, \epsilon Q, N) \times (\epsilon r, 1, \epsilon w)] \cdot (M(v/V)/a)}{(J/V)\sqrt{E_v}}.$$

In the second numerator substitute $P = x^2$, $Q = -2\sqrt{\beta}x$, $r = \sqrt{\beta}/x - z/x^2$. The terms of size βx^2 and $\sqrt{\beta}xz$ cancel, leaving

$$-1 + (1 + x^{-4})z^2 + \beta x^{-2} - 2\sqrt{\beta}zx^{-3} = 1 + O(\sqrt{\beta}).$$

The perturbation contributes at most $Ce\Theta^{36}$. Also $J/V = x^2 - \beta + 2\sqrt{\beta}z/x + O(e\Theta^{34})$, $D_a = 1 + x^4 + O(e\Theta^7)$ and $E_v = 1 + O(\epsilon^2\Theta^4)$. These identities prove the first two estimates. The rank-one part of (12) gives $p_2 \cdot Mp_2 = h\epsilon QP/D_a + O(G_* + E_*)$. Since $QP \leq -c\sqrt{\beta}x^3$, $D_a \leq Cx^4$ and $h\epsilon \asymp \sqrt{h_*}$, this proves compression. Finally $V_\lambda(x_{\text{tar}}/\sqrt{\beta}) \geq x_{\text{tar}}^{-1}V_\lambda(1/\sqrt{\beta}) \geq x_{\text{tar}}^{-1}e^{b/\sqrt{\beta}}$ and $s_0^{-1} \leq K_h^c$; the relative-error estimates give the lower bound for A_{tar} . $\square$

Let $\mathcal{P} \geq 1$ bound the source quantities $K_h, \Theta, \epsilon^{-1}, h_{\text{child}}, \delta^{-1}$ and the fixed coefficient bounds and inverse estimates. These quantities are estimated in Section 8 before the frequency is selected. The choice $\alpha = \delta h_{\text{child}}/A_{\text{tar}}$ therefore satisfies $\alpha \leq \mathcal{P}^c e^{-bx_{\text{prev}}}$. At the central target the cutoff equals one, the phase is zero and $f'_\delta(0) = \delta^{-1}$, so the leading gradient is exactly $\alpha\delta^{-1}v \otimes m = h_{\text{child}}q_2p_2^T$. Section 8 proves the corresponding full transverse propagator and pressure-sign estimates in a neighborhood of the central label. The following consequence of that comparison gives a bound uniform over the stages of the iteration.

Lemma 5.4 (Uniform target-size comparison). *For the selected primary, on every label in the quantitative packet neighborhood, the full ambient relative estimates of Section 8 give*

$$\frac{|m(t, y)||v(t, y)|}{A_{\text{tar}}} \leq C_d \quad (1 \leq \tau \leq \tau_{\text{end}}).$$

On the history interval and on $0 \leq \tau < 1$ the same ratio is at most $K_h^{c_d}\Theta^{c_d}\epsilon^{-c_d}e^{-bx_{\text{prev}}}$. For $g(\tau) = V_0(\tau)$ and $\alpha = \delta h_{\text{child}}/A_{\text{tar}}$, one also has $\sup \alpha g \leq \mathcal{P}^{c_d}$ and $\alpha \leq \mathcal{P}^{c_d}e^{-bx_{\text{prev}}}$.

Proof. The full relative comparison, including the $E^\perp$ components, and $\epsilon\Theta^2 \ll 1$ imply, for $\tau \geq 1$,

$$|m(t, y)||v(t, y)| \asymp_d s_0 \sqrt{1 + x^4} V_\lambda(x/\sqrt{\beta}).$$

The small perturbations of the initial label and of s_0 are included in the same constants. At the central target this equivalence bounds A_{tar} from below by a fixed multiple of the right side. These constants do not grow with Θ or the stage.

For $0 \leq x \leq 1$, V_λ is increasing. For $x \geq 1$ the reciprocal-variable proof of Lemma 5.2 shows that $xV_\lambda(x/\sqrt{\beta})$ is increasing. Hence for $1 \leq x \leq x_{\text{tar}}$,

$$(1 + x^2)V_\lambda(x/\sqrt{\beta}) \leq (x + x^{-1})x_{\text{tar}}V_\lambda(x_{\text{tar}}/\sqrt{\beta}) \leq 2x_{\text{tar}}^2V_\lambda(x_{\text{tar}}/\sqrt{\beta}).$$

For $x \leq 1$ the same conclusion follows by comparison first with $V_\lambda(1/\sqrt{\beta})$ and then with the target. Since $\sqrt{1 + x^4} \asymp 1 + x^2$, this proves the constant size comparison up to the target. Afterwards the logarithmic derivative in τ of $\sqrt{1 + x^4}V_\lambda$ is

$$\frac{2\sqrt{\beta}x^3}{1 + x^4} - \frac{\sqrt{\beta}}{x} + \frac{z}{x^2},$$

which is bounded for $x \geq x_{\text{tar}} \geq 2$. The retained scaled interval has length at most Θ^{-60} , so the additional factor is at most two. The full relative errors preserve this fixed bound.

The stationary history inverse retains the affine endpoint contribution. If C_{hist} is its polynomial norm from continuous forcing to continuous high velocity, the affine-plus-forced construction gives

$$\sup_{0 \leq t \leq t_0} |b_Y(t, y)| \leq (2t_0^{-1}\|F_t\|_\infty C_{\text{hist}} + t_0^{-1}\|F\|_\infty)|Y|.$$

Here Y is the fixed transverse coordinate, so $|Y| \leq \|F(t_0, 0)^{-1}\| |Y_{\text{phys}}| \leq CK_h/h_*$. The inverse norm is polynomial in the known frame quantities and t_0, t_0^{-1} . Using $t_0 \leq 1$, $t_0^{-1} \leq h_*$ and $h_* = a\epsilon^{-2} \leq 2\epsilon^{-2}$ bounds this expression by $K_h^{c_d}\epsilon^{-c_d}$ after enlarging the fixed exponent. This also accounts for the generally nonzero initial high velocity. On $\tau \leq 1$, the scaled system

has bounded coefficients and initial norm at most $C(1 + \lambda)$, with $\lambda \leq C/\epsilon$; the coordinate transformation is therefore bounded by the displayed polynomial in the preceding-stage quantities. Divide these estimates by the exponential target lower bound from Lemma 5.3 to obtain the early and history ratio. Finally $V_0 \leq V_\lambda$. The preceding monotonicity argument gives $\sup g \leq Cx_{\text{tar}}V_\lambda(x_{\text{tar}}/\sqrt{\beta})$, whereas $A_{\text{tar}} \geq c_d s_0 x_{\text{tar}}^2 V_\lambda(x_{\text{tar}}/\sqrt{\beta})$. Their growing factors cancel in αg . The remaining factors $\delta, h_{\text{child}}, s_0^{-1}, x_{\text{tar}}$ have the fixed polynomial source bounds. The target exponential lower bound similarly proves the assertion for α itself. $\square$

6 The finite packet and its exact Euler correction

Fix a smooth reference solution, called the parent, on the closed interval $[0, T]$. Constants may depend on this solution, on d , and on the prescribed seed. The higher-order high-frequency coefficients are constructed by the variational history and forward-tail method of [1, §3]. All source constants are fixed before the packet frequency is chosen. No uniform bound on these constants over an infinite sequence of parents is assumed.

Let U be the parent, let P_U be its Newton pressure, and let Φ_t be its volume-preserving flow, with $\Phi_0 = \text{id}$. In particle labels put

$$F = D\Phi_t, \quad M = (DU) \circ \Phi_t = F_t F^{-1}, \quad H = (D^2 P_U) \circ \Phi_t = -F_{tt} F^{-1}. \quad (16)$$

We assume that these flow and coefficient fields have bounded smooth spatial derivatives and continuous coefficient paths on $[0, T]$. For the quantitative assertions we assume a Gevrey bound on the known coefficients F, F^{-1}, F_t, F_{tt} :

$$\sup_{t,y} \|D_y^j C(t, y)\| \leq C_C R_C^j (j!)^2, \quad C \in \{F, F^{-1}, F_t, F_{tt}\}, \quad j \geq 0. \quad (17)$$

These conditions concern the parent. The lower singular-value bound of F follows from its inverse. All derivative norms in this section are operator norms unless specified otherwise.

Fix a unit vector $m_0 \in \mathbb{R}^d$ and set

$$m(t, y) = F(t, y)^{-\top} m_0.$$

Then $m \neq 0$ and $m_t = -M^\top m$. The phase variable θ belongs to $\mathbb{T}_P = \mathbb{R}/P\mathbb{Z}$, with $P = 2\pi$ when a specific periodic seed is used. Every vector field below has d components, and the transverse space $m^\perp$ has dimension $d - 1$.

For physical Sobolev persistence one may use $s_d = \lfloor d/2 \rfloor + 3 > d/2 + 1$. Packet estimates on the cylinder $\mathbb{R}^d \times \mathbb{T}_P$ use the fixed base order

$$b_d = 2(d + 2) + 1 = 2d + 5. \quad (18)$$

The cylinder embedding, product, and commutator estimates at this order follow by Fourier series in θ and the ordinary Sobolev estimates in y . All their constants depend on d and P , not on the number of additional derivatives.

6.1 The mean inverse

Let $\mathcal{H} = L_\sigma^2(\mathbb{R}^d; \mathbb{R}^d)$ be the closed solenoidal subspace. Choose a nonnegative even smooth cutoff χ , equal to one near zero and supported in the unit ball, and put $\chi_\ell(y) = \chi(\ell y)$. There is a bounded map from the homogeneous first-order tensor space to L^2 defined first on compact smooth tensors by

$$(T_{\chi_\ell} Q)_j = \sum_{i=1}^d \partial_i \{ \chi_\ell (Q_{ij} - Q_{ji}) \}, \quad A_{\chi_\ell} = T_{\chi_\ell} T_{\chi_\ell}^*.$$

The tensor space here is the completion for the square sum of all $\partial_k Q_{ij}$, with $1 \leq i, j, k \leq d$. Hardy's inequality for $d \geq 3$ controls the term $(\partial_i \chi_\ell) Q_{ij}$ and extends T continuously. Its bound is uniform for $0 < \ell \leq 1$ after dilation. Consequently A_{χ_ℓ} is positive and symmetric, its range is solenoidal, and every output vanishes off $\text{supp } \chi_\ell$. The construction requires only boundedness and spatial support of the outputs.

We record the localization inequality used below. There are constants $C_{1,d}, C_{2,d}$ and a fixed $a_d > 0$ such that, if

$$\begin{aligned} \langle M(0, y)z, z \rangle &\geq -B_e |z|^2 & \text{for } |y| \geq a_d r / \ell, \\ \langle M(0, y)z, z \rangle &\geq -B_c |z|^2 & \text{for } |y| < a_d r / \ell, \end{aligned}$$

where $0 \leq r \leq 1/4$, then, for $z \in \mathcal{H}$ and $\mathcal{L} \geq C_{1,d} B_c$,

$$\langle M(0)z, z \rangle_{L^2} + \mathcal{L} \langle A_{\chi_\ell} z, z \rangle_{L^2} \geq -(B_e + C_{2,d} B_c r^d) \|z\|_2^2. \quad (19)$$

Here and below constants absorb the fixed normalization of the cutoff plateau. To obtain (19), set $Q = T_{\chi_\ell}^* z$. On the plateau, subtract one half of the uncut antisymmetric divergence of Q from z . The difference is weakly harmonic, by the defining Riesz identity for Q . Its L^2 mass on the concentric ball of relative radius r is bounded by $C_d r^d$ times its mass on the plateau ball. This follows from the interior harmonic estimate, obtained by nested cutoff energy estimates and Sobolev embedding, and then by dilation. The other part is controlled by $\|Q\|_{\dot{H}^1}^2 = \langle A_{\chi_\ell} z, z \rangle$. Combining this local bound with the exterior lower bound proves the inequality in every dimension $d \geq 3$.

Assume $H(t, y) \leq KI$ in quadratic-form order and

$$K \frac{T^2}{2} + (B_e + C_{2,d} B_c r^d) T \leq \frac{1}{2}. \quad (20)$$

For $v \in L^2(0, T; \mathcal{H})$ define

$$Iv(t) = - \int_t^T v(s) ds, \quad X_v = FIv, \quad D_F v = \partial_t X_v = F_t Iv + Fv.$$

The mean form is

$$\begin{aligned} \mathfrak{a}(v, w) &= \int_0^T \langle D_F v, D_F w \rangle dt - \int_0^T \langle H X_v, X_w \rangle dt \\ &\quad + \langle (M(0) + \mathcal{L} A_{\chi_\ell}) X_v(0), X_w(0) \rangle. \end{aligned} \quad (21)$$

Spatial integrations are included in each inner product. The form need not be symmetric because $M(0)$ need not be symmetric. It is bounded and coercive. Indeed $X_v(T) = 0$, hence

$$\|X_v\|_{L_t^2 L_y^2}^2 \leq \frac{T^2}{2} \|D_F v\|_{L_t^2 L_y^2}^2, \quad \|X_v(0)\|_2^2 \leq T \|D_F v\|_{L_t^2 L_y^2}^2.$$

Moreover $X_v(0) = Iv(0) \in \mathcal{H}$, because $F(0) = I$. Equations (19) and (20) give $\mathfrak{a}(v, v) \geq \frac{1}{2} \|D_F v\|_2^2$. The inverse frame gives a polynomial norm comparison. If $I = \|F^{-1}\|_\infty$ and $C_1 = \|F_t\|_\infty$, then $v = F^{-1} D_F v - F^{-1} F_t F^{-1} X_v$, so $\|v\|_2 \leq I(1 + C_1 IT) \|D_F v\|_2$. Thus the Hilbert-space coercive inverse constructs the unique v with

$$\mathfrak{a}(v, w) = - \int_0^T \langle f, X_w \rangle dt \quad \text{for every } w \in L^2(0, T; \mathcal{H}). \quad (22)$$

Set $B = Fv$.

Lemma 6.1 (Strong mean and its pressure). *For a forcing f with continuous L^2 spatial derivatives of every order, the preceding construction gives continuous paths B and B_t in every spatial Sobolev space and a spatially smooth scalar q , whose spatial derivatives depend continuously on time locally in space, normalized by $q(t, 0) = 0$, such that*

$$B_t + MB + F^{-\top} \nabla_y q = f, \quad F^{-1} B \in \mathcal{H}. \quad (23)$$

These identities hold pointwise, with one-sided time derivatives at both endpoints. In addition, there exists $z_0 \in \mathcal{H}$ such that

$$B(0) = \mathcal{L}A_{\chi_\ell} z_0, \quad \text{supp } B(0) \subset \overline{B}(0, \ell^{-1}). \quad (24)$$

Proof. Translate the coefficients and forcing in the spatial variable. The coercive inverse is differentiable under each translation; differentiating its inverse identity expresses each new derivative using previously constructed derivatives and the known coefficient derivatives. This constructs the spatial derivatives in L^2 . The same argument applies to the translated cutoff operator on the fixed homogeneous tensor space. The argument yields derivatives of every finite order.

Let $Q_t : \mathcal{H} \rightarrow L^2$ be $Q_t z = F(t)z$. Its Gram operator $Q_t^* Q_t$ has a positive lower bound from F^{-1} . The weak equation, $F_{tt} = -HF$, and integration by parts give

$$v_t = (Q_t^* Q_t)^{-1} Q_t^* \{f - 2F_t v\}.$$

First this is an L^2 time identity. The H^1 trace construction and continuity of the coefficients then give continuous representatives of v and v_t , with the same spatial derivatives. The residual $r = f - 2F_t v - Fv_t$ satisfies $Q_t^* r = 0$; equivalently, $F^\top r$ lies in the closed L^2 gradient space. Its smooth representative is curl-free; hence

$$q(t, y) = \int_0^1 \langle F(t, sy)^\top r(t, sy), y \rangle ds$$

satisfies $q(t, 0) = 0$ and $\nabla q = F^\top r$. This proves (23). The pointwise equations follow from equality of continuous representatives of the same L^2 field. The Gram construction and closedness give $v(t) \in \mathcal{H}$ at every time in the closed interval. In particular, $B(0) = v(0) \in \mathcal{H}$. With $z_0 = Iv(0)$, the boundary identity in (22) is

$$P_\sigma \{-F_t(0)z_0 - B(0) + (M(0) + \mathcal{L}A_{\chi_\ell})z_0\} = 0.$$

Since $F_t(0) = M(0)$ and both $B(0)$ and $A_{\chi_\ell} z_0$ are solenoidal, this yields (24) as an L^2 identity. Spatial continuity gives the asserted pointwise support. $\square$

For a transverse endpoint problem the map Q_t is replaced by $F(t)R_\perp$, where $R_\perp : \mathbb{R}^{d-1} \rightarrow m_0^\perp$ is a linear isometry onto the whole orthogonal complement. Its Gram lower bound is again supplied by F^{-1} . The energy, trace, and coercive-inverse arguments are unchanged; there are $d - 1$ transverse variables throughout.

6.2 The joined high inverse and the full tensor corrector

Fix an activation time $0 < t_0 < T$. Choose an isometry $R_\perp : \mathbb{R}^{d-1} \rightarrow m_0^\perp$ onto the whole orthogonal complement and put $Q = FR_\perp$. Its range is $m(t, y)^\perp$, and $Q_t = MQ$, $Q_{tt} = -HQ$. The fixed coordinate space $\mathbb{R}^{d-1}$ is used at every label and time.

For an angular-mean-zero forcing $f \in C([0, T]; H^q(\mathbb{R}^d \times \mathbb{T}_P; \mathbb{R}^d))$ for every finite q , first solve on $[0, t_0]$ the following displacement problem. Find $z \in H_0^1(0, t_0; L^2(\mathbb{R}^d \times \mathbb{T}_P; \mathbb{R}^{d-1}))$ such that

$$\int_0^{t_0} [\langle (Qz)_t, (Q\zeta)_t \rangle - \langle HQz, Q\zeta \rangle] dt = - \int_0^{t_0} \langle f, Q\zeta \rangle dt \quad (\zeta \in H_0^1). \quad (25)$$

Both displacement traces are zero. This condition does not prescribe the initial high velocity. Set $A = Qz_t$ on the history interval.

Lemma 6.2 (The bounded joined high inverse). *Under the known frame bounds and $Kt_0^2/2 \leq 1/2$, problem (25) has a unique solution. Its velocity A , its time derivative, and their spatial derivatives are continuous on $[0, t_0]$. At t_0 its velocity trace can be continued by the forward high equation. This gives a unique joined high operator on $[0, T]$ with the specified history boundary conditions. The history inverse and trace bounds are polynomial in the known coefficient norms, inverse lower bound, t_0 and t_0^{-1} . No exponential forward estimate on $[0, t_0]$ is used.*

Proof. Write $X = Qz$. The one-sided Poincaré estimate and the curvature upper bound give

$$\int_0^{t_0} (|X_t|^2 - \langle HX, X \rangle) dt \geq \frac{1}{2} \|X_t\|_{L^2}^2.$$

If $Q^*Q \geq cI$, then $z = (Q^*Q)^{-1}Q^*X$, and direct differentiation, using $X_t = Q_tz + Qz_t$, gives the polynomial estimate

$$\|z_t\|_2 \leq c^{-1/2}(1 + t_0c^{-1/2}\|Q_t\|_\infty)\|X_t\|_2.$$

Thus the form is coercive on the fixed Hilbert space of coordinate velocities z_t with zero time integral. Its bounded coercive inverse constructs z . Integration by parts and $Q_{tt} = -HQ$ give

$$(Q^*Q)z_{tt} = Q^*(f - 2Q_tz_t).$$

The positive Gram inverse therefore gives z_{tt} and the continuous H^1 traces. Repeated differentiation under spatial translation is justified on this same fixed Hilbert space, exactly as for the mean inverse. It gives all the stated derivatives and preserves both homogeneous displacement boundary conditions.

At order zero, a physical-space estimate is available directly. Put $M_* = \sup \|M\|$. Testing the form with z gives $\|X_t\|_2 \leq 2t_0\|f\|_2$. Since $A = X_t - MX$, it follows that

$$\|A\|_2 \leq 2t_0(1 + M_*t_0)\|f\|_2.$$

The velocity equation derived below implies $\|A_t\|_2 \leq 3M_*\|A\|_2 + \|f\|_2$. The elementary Hilbert-valued trace estimate consequently gives

$$\|A\|_{C_tL^2} \leq [2t_0(1 + M_*t_0)(t_0^{-1/2} + 3M_*\sqrt{t_0}) + \sqrt{t_0}]\|f\|_{L_t^2L^2}. \quad (26)$$

This also bounds the initial datum for the forward segment. $\square$

The Gram equation and tangency show that the same A satisfies

$$A_t = \left(-M + 2\frac{m \otimes M^\top m}{|m|^2}\right)A + \left(I - \frac{m \otimes m}{|m|^2}\right)f. \quad (27)$$

Here $(a \otimes b)v = a\langle b, v \rangle$. Indeed the residual $f - A_t - MA$ is perpendicular to the whole range of Q , hence is a multiple of m . Differentiating $m \cdot A = 0$ gives its coefficient. For $t \geq t_0$ use the fundamental evolution of (27) and Duhamel's formula with datum $A(t_0)$. The velocities and their derivatives match at t_0 by the common right-hand side, so the joined field is strongly differentiable there as well as at the two closed endpoints.

Let g be a positive continuous function with $g(t_0) = 1$, and assume the supported forward bound

$$\|\mathcal{U}(t, s; y)\| \leq C_f g(t)/g(s), \quad t_0 \leq s \leq t \leq T, \quad y \text{ in the fixed support neighborhood.} \quad (28)$$

This hypothesis concerns the known parent evolution on the activation tail. The polynomial history estimate is supplied separately by (26).

Define the normalized periodic primitive ∂_θ^{-1} and put

$$\pi = \partial_\theta^{-1} \frac{\langle m, f \rangle - 2\langle m, MA \rangle}{|m|^2}. \quad (29)$$

Then $A_t + MA + m\partial_\theta\pi = f$. The inverse, trace, and forward maps commute with the angular integral. They therefore preserve zero angular mean. They also commute with multiplication by a fixed spatial cutoff, since these linear high equations contain no spatial derivative. Their uniqueness proves support preservation and covariance. Thus the joined high field and its pressure are smooth in (y, θ) , periodic, and supported where the forcing is supported. Their spatial derivatives are continuous in time; the high velocity also has the continuous time derivative given by (27).

The primary field uses a nonzero terminal displacement. To distinguish its physical and coordinate forms, let the selected physical endpoint be $Y_{\text{phys}} \in m(t_0, 0)^\perp$, and fix

$$Y = R_\perp^* F(t_0, 0)^{-1} Y_{\text{phys}} \in \mathbb{R}^{d-1}.$$

Then $Q(t_0, 0)Y = Y_{\text{phys}}$ and $|Y| \leq \|F(t_0, 0)^{-1}\| |Y_{\text{phys}}|$. This same coordinate vector is used at every label.

The cylinder problem has terminal datum $\alpha\chi_{\text{in}}(y)f_\delta(\theta)Y$, which belongs to every cylinder Sobolev space, and initial displacement zero. Subtract the affine lift $\alpha(t/t_0)\chi_{\text{in}}f_\delta Y$. Using $Q_{tt} = -HQ$, its remaining zero-trace problem is exactly (25) with forcing $-2\alpha t_0^{-1}\chi_{\text{in}}f_\delta Q_t Y$. This forcing is continuous in every cylinder Sobolev space and has the prescribed compact spatial support.

Equivalently, at each fixed label solve the finite-dimensional homogeneous displacement problem with endpoints $0, Y$, and write $b_Y(t, y) = Qz_t$ for its velocity, continued homogeneously on the tail. The uniform frame bounds justify this pointwise solution and its label derivatives. Linearity and uniqueness identify the cylinder velocity with the compact periodic primary

$$A_1(t, y, \theta) = \alpha\chi_{\text{in}}(y)f_\delta(\theta)b_Y(t, y). \quad (30)$$

Its initial value generally depends on y . The first stage, whose activation time is zero, instead uses a prescribed homogeneous forward datum. The same recursion applies in both cases. For higher grades the displacement endpoints in (25) are both zero, but their initial velocities are retained.

For every such A define all d^2 entries of the potential

$$Q_{ij} = \partial_\theta^{-1} \frac{m_i A_j - m_j A_i}{|m|^2}, \quad Q_{ij} = -Q_{ji}, \quad C_j = \sum_{i=1}^d D_y Q_{ij} [F^{-1} e_i]. \quad (31)$$

If $\kappa = k^{-1}$ and $\psi = \Phi_t^{-1}$, the chain rule gives

$$A_j(\psi(x), km_0 \cdot \psi(x)) + \kappa C_j(\psi(x), km_0 \cdot \psi(x)) = \kappa \sum_{i=1}^d \partial_{x_i} Q_{ij}(\psi(x), km_0 \cdot \psi(x)). \quad (32)$$

Indeed the phase part on the right is $\sum_i m_i \partial_\theta Q_{ij} = A_j$, by tangency. Taking divergence and interchanging the two ordinary derivatives cancels all terms by antisymmetry. This identity holds for every pair (i, j) . Applying the volume-preserving Piola transformation shows that $F^{-1}(A + \kappa C)$ has zero divergence on every phase-shifted label graph. Integration over the shift gives its membership in the cylinder's closed divergence-free L^2 space. The potentials are smooth and have the same compact spatial support as A , so the integrations by parts are legitimate. The time derivative of C is obtained by differentiating (31): it includes the derivatives of F^{-1} , m , $|m|^{-2}$, the primitive, and A . The same formula gives the one-sided time derivatives at the endpoints.

6.3 Recursive construction from preceding grades

For vector fields a, b on the time-label-angle cylinder write

$$S(a, b) = D_y b[F^{-1}a], \quad T(a, b) = \langle m, a \rangle \partial_\theta b, \quad La = \partial_t a + Ma.$$

Let A_1 be the homogeneous joined primary (30) (or the first-stage forward primary). Put $B_1 = q_1 = 0$, form C_1 by (31), and set every grade-zero field to zero. Suppose grades below $p \geq 2$ have been constructed. Set

$$V_i = A_i + B_i + C_{i-1} \quad (1 \leq i < p), \quad \widehat{V}_p = C_{p-1},$$

and extend this known sequence by zero above p . The forcing determined by the preceding grades is

$$K_p = -LC_{p-1} - F^{-\top} \nabla_y \pi_{p-1} - \sum_{i+j=p} S(\widehat{V}_i, \widehat{V}_j) - \sum_{i+j=p+1} T(\widehat{V}_i, \widehat{V}_j). \quad (33)$$

Every term uses only the strict prefix. Let an overline denote normalized angular averaging. First solve the mean problem with forcing $\overline{K}_p$, obtaining B_p, q_p . Then apply the joined high inverse of Lemma 6.2 with forcing

$$f_p = K_p - \overline{K}_p - \langle m, B_p \rangle \partial_\theta A_1, \quad (34)$$

obtaining A_p, π_p , and form C_p from A_p . The final term of (34) has zero angular mean. Terms in K_p containing only means are angularly constant and disappear from $K_p - \overline{K}_p$. Every remaining term contains a high field or corrector, so it has the same compact spatial support as the seed. The mean forcing has all continuous L^2 spatial derivatives by the Sobolev product estimates. The high forcing is smooth, periodic, mean zero, and spatially supported. These facts inductively verify the domains of the two inverse operators.

In the quadratic coefficient, replacing $\widehat{V}_p$ by $\widehat{V}_p + A_p + B_p$ adds only $\langle m, B_p \rangle \partial_\theta A_1$. The other potentially new fast terms vanish because A_p, A_1 are tangent and B_p is angularly constant. This proves both strict-prefix coherence and the grade- p momentum equation. In particular, families constructed for different finite cutoffs are restrictions of one family.

6.4 Finite truncation, divergence, and the residual identity

For $N \geq 1$ define

$$\begin{aligned} V^{[N]} &= \sum_{i=1}^N \kappa^i (A_i + B_i + \kappa C_i), \\ p^{[N]} &= \sum_{i=1}^N \kappa^i (q_i + \kappa \pi_i). \end{aligned} \quad (35)$$

Equivalently the velocity grades are $V_i = A_i + B_i + C_{i-1}$ for $1 \leq i \leq N$, followed by the terminal grade $V_{N+1} = C_N$. The pressure grades are $p_i = q_i + \pi_{i-1}$ for $1 \leq i \leq N$, followed by $p_{N+1} = \pi_N$. Grades outside this range are zero. In particular the terminal grade does not include unsolved A_{N+1} or B_{N+1} .

Each high-corrector pair in (35) is exactly solenoidal after graph evaluation by (32). Each mean has $F^{-1}B_i \in \mathcal{H}$ by Lemma 6.1; its constant-angle embedding therefore belongs to the same cylinder solenoidal space. Thus the complete finite sum satisfies the closed divergence constraint. The constant-angle embedding of an ordinary L^2 gradient lies in the cylinder's closed gradient space: approximate it by gradients of compactly supported smooth scalars and use $\nabla_\kappa \varphi = \kappa \nabla_y \varphi$ for an angle-independent scalar. This applies to the mean pressure

gradients. The compact periodic high scalar pressures themselves give gradient generators. Their finite sum therefore belongs to the required closed gradient space.

For any V, p define the label momentum residual by

$$\mathcal{R}_\kappa(V, p) = LV + F^{-\top} \nabla_y p + \kappa^{-1} m \partial_\theta p + S(V, V) + \kappa^{-1} T(V, V). \quad (36)$$

For the finite grade sequences above, let

$$R_n = LV_n + F^{-\top} \nabla_y p_n + m \partial_\theta p_{n+1} + \sum_{i+j=n} S(V_i, V_j) + \sum_{i+j=n+1} T(V_i, V_j).$$

The mean and high equations give $R_n = 0$ for $0 \leq n \leq N$. Expanding the finite sums with their derivatives, including the one-sided time derivatives, consequently gives the exact identity

$$\mathcal{R}_\kappa(V^{[N]}, p^{[N]}) = \sum_{n=N+1}^{2N+2} \kappa^n R_n. \quad (37)$$

This finite identity involves no time derivative of p . In particular, R_{N+1} retains LC_N and $F^{-\top} \nabla_y \pi_N$.

6.5 Quantitative bounds at one known parent

We give the estimates in a form that makes the derivative counting explicit. For an integer $a \geq 0$ set $\mathcal{G}_a(j) = R^{j+a}((j+a)!)^2$. Use ordered cylinder derivatives with alphabet $\{1, \dots, d+1\}$. More precisely, the block of order j is

$$\text{Block}_j(u) = \sum_{|w|=j} \sum_{l=0}^{b_d} \sum_{|v|=l} \|\partial^v \partial^w u\|_{C([0,T];L^2)}.$$

At fixed time the corresponding word Sobolev norm is equivalent to the usual H^{b_d} norm, with fixed constants depending only on d, b_d . A positive time weight is inserted by dividing the field by that weight before taking spatial or angular derivatives. For one sufficiently large R , depending only on the known coefficients, seed, and inverse bounds, the preceding recursion gives

$$\begin{aligned} \text{Block}_j(B_p, B_{p,t}, \nabla_y q_p) &\leq H_0^{2p-2} \mathcal{G}_{100p-140}(j), & p \geq 2, \\ \text{Block}_j(A_p, A_{p,t}, C_p, C_{p,t}, \nabla_y \pi_p) &\leq \gamma(t) H_0^{2p-2} \mathcal{G}_{100p-80}(j), & p \geq 1. \end{aligned} \quad (38)$$

Here each entry has the stated bound separately. We use $\gamma(t) = \alpha$ for $0 \leq t \leq t_0$ and $\gamma(t) = \alpha g(t)$ on the tail, with $H_0 \geq \max(1, \sup \gamma)$. The notation with $\gamma(t)$ means the uniform-time block of the field divided by γ . The supported forward constant in (28), the history trace bound, and the terminal datum are included among the known source constants controlling R . In particular the higher initial high fields are bounded with the factor α , not set to zero. This is the family used below and in Section 9.

For completeness, the derivative estimates underlying (38) are as follows. The word Leibniz rule is bounded by a binomial convolution. Its Gevrey convolution closes because

$$\binom{j}{l} (l!)^2 ((j-l)!)^2 = \frac{(j!)^2}{(j)!}, \quad \sum_{l=0}^j \binom{j}{l}^{-1} \leq 3.$$

Fixed-order Sobolev products add a constant depending on b_d, d . On the history interval, differentiate the fixed-space coercive inverse and then use the Gram equation and its trace. On the tail, differentiate the Duhamel equation and use (28). The history terminal trace is the

datum in this Volterra recurrence. The joined high velocity and its time derivative require shifts of at most three grades. Differentiating the coercive inverse, followed by its Gram reconstruction and the H^1 time trace, requires shifts of at most three grades for the mean, its time derivative, and its pressure gradient. These are estimates on word blocks with the full alphabet. The angular mean is a bounded L^2 map with norm $P^{-1/2}$, and the constant-angle embedding has norm $P^{1/2}$; the factors cancel. Thus converting mean input to mean output does not multiply the unknown word radius at each grade. Only the known coefficient radius is enlarged once. Also the mean time weight is the constant H_0^{2p-2} , so no commutation of a time-dependent weight with the mean inverse is claimed. The history estimate is applied to f/α and the tail estimate to $f/(\alpha g)$; linearity preserves the common factor α . The tail estimate uses the weighted Duhamel operator without differentiating g . The tensor potential and its time derivative require one spatial derivative and a fixed d -dependent multiplier bound; the pressure source (29) and one slow derivative require shifts of at most two additional grades. Enlarging R once absorbs these fixed constants. The grade counting can be made explicit. The mean force has shift at most $100p - 150$, and the high force has shift at most $100p - 90$. For example a slow high-high term with $i + j = p$ has shift $100p - 159$. In a fast term with $i + j = p + 1$, the first high factor is annihilated by $m \cdot A_i = 0$. A first corrector factor has shift $100i - 180$, so the resulting high term has shift at most $100p - 159$; a first mean factor gives shift $100p - 119$, and has zero angular average. The new term $(m \cdot B_p) \partial_\theta A_1$ has the latter shift as well. These shifts satisfy the stated forcing bounds. The profile powers close because $\gamma \leq H_0$; in particular a mean-high fast product has the required factor γH_0^{2p-2} . The at most quadratic number of summands is absorbed by the additional factorial shifts: for $a \geq 0$, $\mathcal{G}_{a+2}(j)/\mathcal{G}_a(j) = R^2(j+a+1)^2(j+a+2)^2$. The low grades satisfy the bounds after one enlargement of the known radius. The mean inverse and the joined high inverse, followed by the corrector and pressure estimates, require fewer than ten additional shifts. Strong induction proves (38) for this joined family using only previously constructed fields.

For the correction use the truncated weighted norm

$$\|u\|_{r,b_d;J} = \sum_{j=0}^J \frac{r^j}{(j!)^2} \sum_{|w|=j} \sum_{l=0}^{b_d} \sum_{|v|=l} \|\partial^v \partial^w u\|_2.$$

A block bound $C\mathcal{G}_a(j)$ gives a weighted norm bound of at most $2C(4R)^a(a!)^2$ when $4Rr \leq 1/2$. This conversion is performed after the word recursion, not at every recursive step. In particular, at every derivative cutoff,

$$\|V_i\|_{r,b_d;J} \leq 6H_0^{2i}(4R)^{100i}((100i)!)^2 \quad (i \geq 1),$$

with analogous bounds for the time derivatives and pressure forces at a smaller fixed radius. To quantify the dependence on the expansion order, choose a known-source constant $C_0 \geq 1$ and put $S_N = C_0(N+1)^{200}$. The inequality $(100i)! \leq (100(N+1))^{100i}$ for $i \leq N+1$ bounds each displayed grade by S_N^i after enlarging C_0 . The time derivatives and pressure terms obey the same bound; one slow or angular derivative is controlled by one fixed radius reduction, whose reciprocal factor is included in C_0 . A slow product with $i + j = n$ is bounded by CS_N^n , and a fast product with $i + j = n + 1$ by CS_N^{n+1} . There are at most $(N+2)^2$ terms. The terminal linear terms have the same bound, including the C_N time derivative and the slow derivative of π_N . Thus $C_1(N+2)^2 S_N^{n+1}$ bounds the entire residual grade. Taking $B_N = C_2(N+1)^{202}$ with a sufficiently large fixed C_2 absorbs this factor for every $n \geq N+1$. Thus the fixed tail degree is 202, and

$$B_N \leq C_*(N+1)^{202}, \quad \|R_n\|_{r,b_d;J} \leq B_N^{n+1} \quad (N+1 \leq n \leq 2N+2). \quad (39)$$

The constant C_* is fixed by the current parent and seed, and is independent of N, k, J . Spatial derivatives of the high pressure, the time derivative of C_N , and the terminal linear terms are all included in this estimate.

Put

$$Z = kF^{-1}V^{[N]}, \quad Q^a = k^2p^{[N]}, \quad \nabla_\kappa = \kappa\nabla_y + m_0\partial_\theta, \quad G = F^{-1}F^{-\top}. \quad (40)$$

Then the coordinate residual is $\mathcal{E} = kF^{-1}\mathcal{R}_\kappa(V^{[N]}, p^{[N]})$. If $B_N \leq k^{1/100}$ and the known inverse multiplier bound is at most $k^{1/100}$, the geometric sum in (37) yields, for $N \geq X - 1$, $X \geq 6$,

$$\|\mathcal{E}\|_{r,b_d;J} \leq \exp(-\frac{7}{10}X \log k). \quad (41)$$

The field Z and its first cylinder derivatives have fixed weighted bounds B_0, B_1 . More sharply, its cylinder transport vector is

$$\mathcal{V}_\kappa Z = (\kappa Z, m_0 \cdot Z), \quad \|\mathcal{V}_\kappa Z\|_{r,b_d;J} \leq D_*/k.$$

The order-one high field is tangent, so its fast transport component vanishes exactly. All other contributions carry at least one inverse frequency. The radius estimate uses this small transport norm rather than the full $O(1)$ norm of Z .

6.6 An exact correction on the cylinder

The coordinate equation is

$$W_t + (W \cdot \nabla_\kappa)W + 2F^{-1}F_tW + \kappa \sum_{i=1}^d W_i F^{-1}(\partial_i F)W + G\mathbf{p} = 0, \quad \nabla_\kappa \cdot W = 0. \quad (42)$$

Here $\mathbf{p}$ is a vector pressure variable in $\mathcal{G}_\kappa = \overline{\{\nabla_\kappa \varphi : \varphi \text{ smooth, periodic, compact in } y\}}^{L^2}$. The construction requires only membership in this closed space, without a global periodic scalar potential or a closed-range property for the degenerate d -component operator on the $(d+1)$ -dimensional cylinder. The equation contains all d quadratic coefficient matrices. Substituting Z and the approximate pressure vector $\mathbf{p}^a = \nabla_\kappa Q^a$ gives exactly the residual $\mathcal{E}$ above. The correction pressure is solved in this closed subspace of vector-valued cylinder L^2 . The positive matrix G gives a coercive variational pressure inverse on that subspace. Its derivatives are controlled by the coefficient jets and commutators.

Seek $W = Z + E$ with $E(0) = 0$. Use the inverse metric $G^{-1} = F^\top F$ in the finite weighted Sobolev energy. The top pressure term cancels since $\langle G^{-1}E, G\mathbf{p}^e \rangle = \langle E, \mathbf{p}^e \rangle = 0$ for $\mathbf{p}^e \in \mathcal{G}_\kappa$. The top transport term is integrated by parts using the closed solenoidal constraint. Differentiated metric terms and lower products have a known coefficient bound. To specify the energy convention, put $\|u\|_{F^\top F} = (\int \langle F^\top F u, u \rangle)^{1/2}$ and set

$$\mathcal{A}_J = \sum_{j=0}^J \frac{r^j}{(j!)^2} \sum_{|w|=j} \sum_{l=0}^{b_d} \sum_{|v|=l} \|\partial^v \partial^w E\|_{F^\top F}.$$

The quantity $\mathcal{A}_J$ is a sum of metric norms and is uniformly equivalent to the preceding weighted norm by the known frame bounds. Let $\mathcal{L}_J$ be the same sum with the additional factor j in every summand. Derivatives of norms at zero can be justified by first replacing each norm by its quadratic regularization and then letting the regularization parameter decrease to zero.

The orthogonal gradient projection commutes with every cylinder derivative. In the differentiated pressure equation, each nontrivial commutator with G lowers the derivative order on the pressure by at least one. This compensates for the one derivative in the transport source. After the fixed-base Sobolev pressure estimate, its remaining coefficient-word convolutions are absorbed by choosing the known radius small enough. Thus no unknown derivative of order

$J + b_d + 1$ is left in the energy bound. The word product convolution and the metric transport cancellation consequently give a single radius-loss factor. In fact, in a transport commutator with $1 \leq j \leq n$, put $l = n - j + 1$. The ratio of Gevrey weights, including its Leibniz coefficient, is

$$\frac{r^n}{(n!)^2} \binom{n}{j} \frac{(j!)^2 (l!)^2}{r^j r^l} = \frac{l^2}{r \binom{n}{j}} \leq \frac{l}{r},$$

since $\binom{n}{j} \geq l$. The excluded term $j = 0$ is precisely the cancelled principal transport term. Thus the loss is the factor l in $\mathcal{L}_J$, with no additional derivative beyond the cutoff. We obtain

$$\mathcal{A}'_J \leq C\{\mathcal{A}_J + \mathcal{A}_J^2 + \varepsilon\} + \left\{ \frac{r'}{r} + C(r^{-1} + R_c)(D_*/k + \mathcal{A}_J) \right\} \mathcal{L}_J. \quad (43)$$

The same C, R_c work for every external derivative cutoff J . Here $\varepsilon = 2 \exp(-7X \log k/10)$ includes the metric conversion of the residual after a fixed enlargement of constants.

Choose a positive r_0 from the known coefficient and packet radii so that all pressure absorptions hold, in particular $r_0 R_c \leq 1$, and set

$$\eta = \exp(-\sqrt{X}), \quad r(t) = r_0 - 2C(D_*/k + \eta)t.$$

If

$$2\varepsilon e^{3CT} \leq \eta/2, \quad 2C(D_*/k + \eta)T \leq r_0/2, \quad (44)$$

then $r_0/2 \leq r(t) \leq r_0$. In the bootstrap range $\mathcal{A}_J \leq \eta \leq 1$ the loss coefficient in (43) is nonpositive. The scalar integral inequality gives the strict bound $\mathcal{A}_J \leq \eta/2$ and closes the bootstrap.

For existence, we follow the pressure projection and vanishing-viscosity argument of [1, §3.6]. Let $\mathcal{G}_\kappa$ be the closed gradient space above and let $P_{\mathcal{G}}$ be its orthogonal projection in cylinder L^2 . On $\mathcal{G}_\kappa$ the map $g \mapsto P_{\mathcal{G}}(G(t)g)$ is coercive, since $\langle g, G(t)g \rangle \geq c\|g\|_2^2$ with $c > 0$ supplied by F . Set

$$\mathcal{T}_t = (P_{\mathcal{G}}G(t)|_{\mathcal{G}_\kappa})^{-1}P_{\mathcal{G}}, \quad \mathcal{P}_t = I - G(t)\mathcal{T}_t.$$

Then $P_{\mathcal{G}}\mathcal{P}_t = 0$, $\mathcal{P}_t u = u$ for $u \in \mathcal{G}_\kappa^\perp$, and $\mathcal{P}_t G(t)g = 0$ for $g \in \mathcal{G}_\kappa$. Differentiating the coercive pressure equation gives the Sobolev bounds for $\mathcal{T}_t$ and $\mathcal{P}_t$ at every fixed order. Their constants may depend on that fixed order; the previously derived weighted energy constants do not depend on the external derivative cutoff.

Write the unprojected correction source as

$$\mathcal{N}(E) = b_E \cdot \nabla_{y,\theta} E + \mathcal{Z}(E) + \mathcal{E}, \quad b_E = \mathcal{V}_\kappa(Z + E),$$

where $\mathcal{Z}(E)$ contains the terms $E \cdot \nabla_\kappa Z$, $2F^{-1}F_t E$ and the differences of the two quadratic coefficient terms. It has no derivative of E . For $0 < \nu \leq 1$, solve

$$E_t^\nu - \nu \Delta_{y,\theta} E^\nu = -\mathcal{P}_t \mathcal{N}(E^\nu), \quad E^\nu(0) = 0. \quad (45)$$

The right side is locally Lipschitz $H^q \rightarrow H^{q-1}$ for $q \geq b_d$, and the Fourier heat estimate

$$\|e^{\nu t \Delta} f\|_{H^q} \leq C_q(1 + (\nu t)^{-1/2})\|f\|_{H^{q-1}}$$

gives a local mild solution by contraction. Every component of ∇_κ has constant coefficients and commutes with the full cylinder Laplacian. The heat semigroup therefore preserves the fixed space $\mathcal{G}_\kappa^\perp$. Since the non-heat term in (45) takes values in this space, the mild solution is exactly divergence free. No time derivative of $\mathcal{P}_t$ and no commutation of $\mathcal{P}_t$ with the heat semigroup is required.

The pressure gradient is $-\mathcal{T}_t \mathcal{N}(E^\nu)$. The viscosity contributes no pressure, because ΔE^ν is still divergence free. Heat maximal regularity justifies testing differentiated equations by $F^\top F \partial^w E^\nu$: on each compact existence interval, $E^\nu \in L_t^2 H^{q+1}$ and $E_t^\nu \in L_t^2 H^{q-1}$. The differentiated metric-viscosity term satisfies

$$\nu \int \langle F^\top F u, \Delta u \rangle \leq -\frac{c\nu}{2} \|\nabla u\|_2^2 + C\nu \|u\|_2^2.$$

This follows by integration by parts and Young's inequality, using the known first derivatives of $F^\top F$. Thus it adds only a bounded multiple of the lower energy to (43), uniformly for $\nu \leq 1$. In the choice of C above we include this known metric-viscosity constant before selecting the radius or frequency; it depends only on the fixed source and not on J or ν . The same norm regularization justifies summing this estimate for $\mathcal{A}_J$. The bootstrap extends every viscous solution to T . Uniqueness in the low Sobolev class identifies the solutions constructed at different higher orders. In particular they have uniform bounds in $C_t H^q$ for each finite q .

The viscosity limit follows from a Cauchy estimate in the global Sobolev spaces. For two viscosities, subtract their equations and use the same metric L^2 difference energy. The transport and pressure leading terms cancel as before. The remaining viscous defect is $(\nu - \mu)\Delta E^\mu$, uniformly bounded in L^2 by a constant of the fixed source times $|\nu - \mu|$. With zero initial difference, Gronwall gives

$$\|E^\nu - E^\mu\|_{C_t L^2} \leq C_* |\nu - \mu|.$$

The dissipative term $\nu\Delta(E^\nu - E^\mu)$ is handled by the same integration by parts. Interpolation with the uniform higher-order bounds now gives a strong limit in $C_t H^q$ for every finite q . The nonlinear source and the pressure inverse pass to this limit one order lower; $\nu\Delta E^\nu \rightarrow 0$ there. The gradient and solenoidal constraints are closed, the finite weighted energies pass to the limit, and the initial trace remains zero. The limiting equation supplies the continuous strong time derivative at every Sobolev order, including the within-interval endpoint derivatives.

These steps use the heat Fourier multiplier on $\mathbb{R}^d \times \mathbb{T}_P$, a closed Hilbert gradient space, and the Sobolev product threshold (18). In the three-dimensional proof the corresponding cylinder has dimension four; here the only changes are the $d + 1$ Fourier variables, all d quadratic coefficient matrices, and the dimension-dependent Sobolev constants. The argument uses the full transverse space and tensor coefficients.

There is an explicit finite frequency choice. With

$$X = k^{1/203000}, \quad N = \lfloor X \rfloor,$$

(39) meets $B_N \leq k^{1/100}$ at sufficiently large k . Denoting by I_* the inverse-multiplier bound, the packet and correction smallness conditions hold if

$$k \geq \left(64 + e + 2^{202} C_* + I_* + 12CT + \frac{8CTD_*}{r_0} + \left(\frac{8CT}{r_0} \right)^2 \right)^{203000}. \quad (46)$$

Indeed the 203000th root controls X , the polynomial tail, $\log k$, and the two terms in the radius loss; the negative exponential in (41) dominates e^{3CT} and $e^{-\sqrt{X}}$. The constants in (46) depend only on the fixed source; no bound in terms of a subsequent parent's parameters is assumed.

6.7 The physical child and the signed pressure term

Define the perturbed solution, called the child, by

$$U^+(t, x) = U(t, x) + \kappa F(t, y) W(t, y, km_0 \cdot y), \quad y = \Phi_t^{-1}(x). \quad (47)$$

The chain rule, $\Phi'_t = U \circ \Phi_t$, and (16) transform (42) into the physical momentum equation with pressure force

$$\nabla P_U(t, x) + \kappa F(t, y)^{-\top} \mathbf{p}(t, y, km_0 \cdot y).$$

This force is a gradient on physical space. Indeed, closed-gradient membership implies

$$D_{\kappa, i} \mathbf{p}_j - D_{\kappa, j} \mathbf{p}_i = 0$$

in distributions: it holds for each compact smooth gradient generator and is preserved by the L^2 limit. All spatial derivatives of the constructed pressure vector are continuous, so the same identity holds pointwise. The pullback chain rule for a one-form gives

$$\partial_{x_i}(\kappa F^{-\top} \mathbf{p})_j - \partial_{x_j}(\kappa F^{-\top} \mathbf{p})_i = \sum_{a, b} (F^{-1})_{bi} (F^{-1})_{aj} (D_{\kappa, b} \mathbf{p}_a - D_{\kappa, a} \mathbf{p}_b) = 0$$

after graph evaluation. The terms differentiating $F^{-1} = D\Phi_t^{-1}$ cancel by the symmetry of its ordinary second derivatives.

Graph evaluation uses the smooth periodic representatives and one extra angular Sobolev trace estimate; the parent change of variables is volume preserving. It gives all physical L^2 spatial derivatives and their continuous time paths, both for the velocity and for this pressure force. A curl-free L^2 vector field on $\mathbb{R}^d$ lies in the ordinary closed gradient space, by its Fourier characterization. The Piola identity also transforms the closed cylinder solenoidal constraint into $\text{div } U^+ = 0$. Taking the ordinary divergence of the physical momentum equation and applying the Helmholtz decomposition identifies its pressure force with the gradient of the Newton pressure in Lemma 2.2. We choose that normalized scalar pressure. The smooth Sobolev local theory then identifies the constructed field with the Euler evolution on the whole interval. The mean and finite angular scalar pressures provide normalized representatives, while the correction is constructed directly in the closed gradient space.

The same field has the decomposition

$$U^+ = U + V^{[N]}(t, y, km_0 \cdot y) + \kappa F(t, y) E(t, y, km_0 \cdot y).$$

Thus its initial increment is exactly the finite packet, and the correction initially vanishes. The correction estimates at all derivative orders give physical velocity-gradient and pressure-Hessian errors bounded by fixed current-source constants times η . One fast derivative of the velocity correction is paired with its prefactor κ . The physical pressure force of the correction is $\kappa F^{-\top} \mathbf{p}^e$ evaluated on the graph. Its one physical derivative, which is the pressure Hessian, has the same cancellation of κ against one fast derivative. For the finite scalar primary pressure the corresponding factor is its prefactor κ^2 against two fast derivatives. All remaining powers of κ are nonnegative. A fixed number of the uniform weighted estimates controls the required coefficient and angular derivatives; the stated constants are independent of k .

For both the joined and initial forward packets, set $P = 2\pi$ and use the explicit profile

$$f_\delta(\theta) = \arctan \frac{\sin \theta}{1 + \delta - \cos \theta}, \quad 0 < \delta \leq 1.$$

Its first denominator is at least δ , so the profile is smooth, odd and 2π -periodic. Oddness gives zero mean over $[-\pi, \pi]$, and periodicity gives zero mean over any full period. Also $|f_\delta| \leq \pi/2$. Direct differentiation gives

$$f'_\delta(\theta) = \frac{N_\delta(\theta)}{D_\delta(\theta)}, \quad N_\delta = (1 + \delta) \cos \theta - 1, \quad D_\delta = (1 + \delta)^2 - 2(1 + \delta) \cos \theta + 1 \geq \delta^2.$$

At zero the quotient equals δ^{-1} . The identities

$$D_\delta + N_\delta = (1 + \delta)(1 + \delta - \cos \theta) > 0, \quad D_\delta - \delta N_\delta = (1 + \delta)(2 + \delta)(1 - \cos \theta) \geq 0$$

give the bounds used in the pressure estimate:

$$f'_\delta(0) = \delta^{-1}, \quad -1 \leq f'_\delta \leq \delta^{-1}.$$

The dependence of the Gevrey bounds on the width is explicit. All derivatives of N_δ have magnitude at most 3, and the positive-order derivatives of D_δ have magnitude at most 4. Differentiate $D_\delta D_\delta^{-1} = 1$ and use the Gevrey binomial convolution to obtain, by induction,

$$\|(D_\delta^{-1})^{(n)}\|_\infty \leq 10\delta^{-2}(40\delta^{-2})^n(n!)^2.$$

Multiplying by N_δ in the formula for f'_δ uses the same radius; the convolution factor is at most 3. Since $40\delta^{-2} \geq 1$, the resulting bound at order $n+1$, together with the bound $|f_\delta| \leq \pi/2$ at order zero, yields

$$\|f_\delta^{(n)}\|_\infty \leq 100\delta^{-2}(40\delta^{-2})^n(n!)^2 \quad (n \geq 0).$$

Both the amplitude constant and radius are polynomial in δ^{-1} .

Use this profile for the joined primary (30). The uncut physical high velocity is b_Y , and the amplitude is α . At $y = \Phi_t^{-1}(x)$ and $\vartheta = km_0 \cdot y$, the leading velocity and pressure Hessian tensors are respectively

$$\begin{aligned} \mathcal{S} &= \alpha \chi_{\text{in}}(y) f'_\delta(\vartheta) b_Y \otimes m, \\ \mathcal{Q} &= -2\alpha \chi_{\text{in}}(y) \frac{\langle m, Mb_Y \rangle}{|m|^2} f'_\delta(\vartheta) m \otimes m. \end{aligned} \quad (48)$$

Direct differentiation of the primary velocity gives

$$D_x \{k^{-1} \alpha \chi_{\text{in}}(y) f_\delta(\vartheta) b_Y(t, y)\} = \mathcal{S} + k^{-1} \alpha f_\delta(\vartheta) D_y(\chi_{\text{in}} b_Y)(t, y) F^{-1}(t, y).$$

The slow coefficient therefore contributes a constant times k^{-1} . The corresponding pressure-force coefficient is $-2\alpha \chi_{\text{in}} \langle m, Mb_Y \rangle m / |m|^2$; its derivative is the other slow primary remainder. To bound these derivatives, it suffices to bound the carrier and its first label derivative on the ball of radius $1/2$. Both coefficients vanish off the ball of radius $1/4$, so their derivatives vanish locally outside that support. The history inverse and the supported tail estimates above supply these carrier bounds. This gives uniform spatial remainder constants without any sharp shear estimate away from the central particle. The finite higher grades, derivatives of slow coefficients, and exact correction give

$$\|DU^+ - DU - \mathcal{S}\| \leq C_v/k + C'_v \eta, \quad \|D^2 P_{U^+} - D^2 P_U - \mathcal{Q}\| \leq C_p/k + C'_p \eta, \quad (49)$$

uniformly on the closed time interval and physical space. The pressure identity follows first for the finite scalar pressure and then for the correction pressure; equality of their gradient with the canonical Euler pressure gradient transfers the identity to P_{U^+} .

If $\langle m, Mb_Y \rangle \geq 0$, the lower slope bound gives the one-sided estimate

$$\langle \mathcal{Q}z, z \rangle \leq 2\alpha \chi_{\text{in}} \langle m, Mb_Y \rangle |z|^2. \quad (50)$$

Without this sign condition, the absolute estimate is

$$\|\mathcal{Q}\| \leq 2\alpha \|M\| \chi_{\text{in}} |m| |b_Y| / \delta.$$

The one-sided estimate retains the gain from the narrow profile in the mean curvature bound at the next stage. All tensors in (48) act on the entire $\mathbb{R}^d$, and the norm identities used here hold in every dimension.

For spatial localization the preceding construction is applied to the rescaled parent

$$U^\rho(t, x) = \rho^{-1}U(t, \rho x), \quad P_{U^\rho}(t, x) = \rho^{-2}P_U(t, \rho x), \quad 0 < \rho \leq 1.$$

Direct substitution proves the Euler equations on the same time interval. Its flow is $\rho^{-1}\Phi_t(\rho y)$, so its frame, strain and pressure Hessian in labels are exactly $F(t, \rho y)$, $M(t, \rho y)$ and $H(t, \rho y)$. Undoing the scaling after constructing the child gives the physical field $\rho(U^\rho)^+(t, x/\rho)$. Velocity gradients and pressure Hessians are unchanged in size by these two inverse scalings, and the support radius of a compact oscillatory coefficient is multiplied by ρ . Thus the low error bounds and leading tensors above transfer to the original parent with no factor ρ^{-1} in those two derivative orders. Higher-order initial derivatives retain their scaling factors as recorded in Section 9.

6.8 Physical localization of the mean boundary condition

The oscillatory localization scale and the mean cutoff scale are chosen separately. With a fixed physical cutoff parameter ℓ_* and physical unscaling $x = \rho y$, use the normalized cutoff parameter $\ell = \rho\ell_*$. Then $\chi(\ell y) = \chi(\ell_*x)$, and (24) places each physical mean initial increment in $\overline{B}(0, \ell_*^{-1})$. If $\text{supp } \chi_{\text{in}} \subset \overline{B}(0, 1/4)$, the high and corrector initial increments are supported in $\overline{B}(0, \rho/4)$. The exact correction has initial value zero. Lemma 7.1 proves propagation of the mean reserve with this fixed physical cutoff, and Lemma 7.2 gives the resulting common compact support through the iteration.

6.9 The packet construction for one parent

The following lemma summarizes the construction for use in the iteration. Its assumptions concern only the fixed source.

Lemma 6.3 (Exact joined packet child). *Fix $d \geq 3$ and a smooth Sobolev parent on $[0, T]$ with flow (16), the known coefficient bounds (17), and the mean reserve (20). Fix its transverse terminal datum, compact Gevrey cutoffs, and the positive forward profile with (28). There is a finite constant determined by these data such that every sufficiently large k gives the child (47) from the joined strict-prefix family. It solves the full d -dimensional Euler equation with its Newton pressure on the whole closed interval and belongs to $C_t H^m \cap C_t^1 H^{m-1}$ for every integer $m \geq 1$.*

Its leading velocity-gradient and pressure-Hessian increments are exactly the two tensors in (48). After an explicit enlargement of the known-source frequency threshold, each error in (49) is at most $k^{-1/4}$, uniformly in time and space. Its initial increment is the entire finite sum (35) at time zero, including every higher-order high-frequency initial velocity and the terminal corrector C_N ; the exact correction has initial value zero.

If the parent is odd and the cutoffs are even, then the child is odd. If in addition a fixed orthogonal coordinate reflection $\mathcal{R}$ obeys

$$U(t, \mathcal{R}x) = \mathcal{R}U(t, x), \quad \mathcal{R}m_0 = m_0, \quad \mathcal{R}R_\perp Y = R_\perp Y,$$

and preserves the cutoffs, then

$$U^+(t, \mathcal{R}x) = \mathcal{R}U^+(t, x), \quad DU^+(t, \mathcal{R}x) = \mathcal{R}DU^+(t, x)\mathcal{R}^{-1}.$$

For the fixed physical mean cutoff described above, every initial increment is supported in $\overline{B}(0, \max(1/4, \ell_^{-1}))$. Thus the child's initial support is contained in the union of this ball and the parent's initial support.*

Proof. The preceding constructions give existence, the Euler equation, the normalized pressure, and all Sobolev orders. The residual identity includes the terminal corrector and pressure.

To obtain the stated error bound, let $\beta = 203000$ and let $C_{\text{err}} \geq 1$ dominate the four constants in (49). Increase the lower bound for $X = k^{1/\beta}$ to include $(2\beta)^4$ and $(4C_{\text{err}})^2$, as well as the quantity inside parentheses in (46). For $X \geq 1$, $\log X \leq 4X^{1/4}$, and hence

$$\frac{1}{4} \log k \leq \frac{1}{2} \sqrt{X}, \quad \log(4C_{\text{err}}) \leq \frac{1}{2} \sqrt{X}.$$

It follows that $C_{\text{err}} e^{-\sqrt{X}} \leq k^{-1/4}/4$ and $C_{\text{err}}/k \leq k^{-1/4}/4$. The sum satisfies the asserted error bound. The enlarged threshold remains a fixed power of a sum of known source constants.

For symmetry, uniqueness of the parent's flow first gives $\Phi_t(-y) = -\Phi_t(y)$ and the corresponding reflection covariance. The mean form commutes with the induced orthogonal action, since its cutoff tensor operator is constructed using all tensor indices and the cutoff is invariant. Uniqueness of its coercive inverse gives the mean parities. For the joined high operator the action preserves the fixed zero-trace displacement space; for the primary it also preserves the terminal datum. Uniqueness of the history inverse and then of the forward initial-value equation proves the high parities. The normalized angular primitive, the full tensor corrector, and each product in the strict-prefix force commute with the same actions. Induction therefore gives odd vector fields and even scalar pressures for the whole finite family, together with the stated reflection covariance.

The projected coordinate equation and its closed solenoidal and gradient spaces are also invariant under these actions. Its unique correction with zero initial value has the same parities. Transforming back gives the child symmetry; differentiating this identity gives the claimed full derivative covariance. Finally, the initial support follows from (24), support preservation of the joined high inverse and the tensor potential, and $E(0) = 0$. $\square$

7 A fixed compact support and the full mean reserve

The initial mean boundary cutoff and the oscillatory packet have different spatial scales. In physical coordinates fix two parameters $0 < \ell_* \leq 1$ and $0 \leq r_* \leq 1/4$. Write $c_d = C_{2,d}$ for the nonnegative volume-error constant in the localized mean boundary inequality, and $C_{1,d}$ for its boundary-operator constant. These dimensional constants are denoted separately from the stage-dependent bound on the second frame coefficient. Let $s_{\text{cut},d} > 0$ be the fixed dimensional cutoff scale and put $a_d = (3s_{\text{cut},d})^{-1}$. For the current solution horizon $L > 0$, define the mean reserve by

$$\mathfrak{M}(L) = \frac{1}{2}KL^2 + (B_e + c_d B_e r_*^d)L. \quad (51)$$

The parameters K, B_c, B_e are nonnegative. Here K bounds the pressure Hessian above, while B_c and B_e bound the negative part of the initial symmetric gradient inside and outside the fixed ball of radius $a_d r_*/\ell_*$, respectively. The boundary penalty $\mathcal{L}$ is chosen to be at least $C_{1,d}B_c$.

Lemma 7.1 (Fixed-cutoff propagation). *For a constructed child let $e_p \geq 0$ be its upper pressure-Hessian increment and $e_i \geq 0$ its uniform initial-gradient increment. Retain the same physical ℓ_*, r_* , replace*

$$K \mapsto K + e_p, \quad B_e \mapsto B_e + e_i, \quad B_c \mapsto B_c + e_i, \quad \mathcal{L} \mapsto C_{1,d}(B_c + e_i).$$

All corresponding mean inequalities hold whenever the resulting $\mathfrak{M}(L)$ is less than $1/2$, and

$$\mathfrak{M}_{\text{child}}(L) - \mathfrak{M}_{\text{parent}}(L) = \frac{1}{2}e_p L^2 + (1 + c_d r_*^d)e_i L. \quad (52)$$

Proof. The pressure upper bound adds e_p . If the difference of the initial gradients has operator norm at most e_i , its quadratic form is at least $-e_i|z|^2$. The old interior and exterior inequalities therefore persist on exactly the same regions with the indicated new constants. The new boundary penalty meets the same coercivity requirement. Expanding (51) proves (52). On any shorter horizon the reserve only decreases. The updated parameters are used for the subsequent mean problem; the child's flow, inverse, time derivatives, and pressure remain those obtained from the packet construction. $\square$

The comparisons in this lemma are made at the parent's horizon L . Passing to a shorter horizon decreases the reserve. Each nonnegative increment may be estimated using a common upper horizon, while the preceding reserve is evaluated on the current solution's horizon.

The spatial normalization preserves this full functional exactly. At physical localization length ρ , the normalized mean scale is $\rho\ell_*$ and its splitting parameter is still r_* . Unscaling changes the scale back to ℓ_* and leaves K, B_c, B_e, r_* unchanged. Since $\ell_* \leq 1$, the final cutoff normalization makes no further change. Thus the physical interior radius remains $a_d r_*/\ell_*$, and both the normalized and physical mean reserves equal (51). Changing auxiliary source bounds does not alter these mean parameters.

The two initialization packets permit distinct initial increments e_c, e_e on the interior and exterior regions. The same proof then gives

$$\mathfrak{M}_{\text{child}}(L) - \mathfrak{M}_{\text{parent}}(L) = \frac{1}{2}e_p L^2 + (e_e + c_d r_*^d e_c)L.$$

For the seed the fixed physical choice is $\ell_* = 1$, $r_* = X^{-1000}$. Its source and frequency bounds make the full child reserve at most $1/8$ by increasing X after J is fixed. The first normal forward packet contributes at most $1/16$ to the pressure-time term and $1/16$ to the initial-time term. Consequently the first positive-history state has mean reserve at most $1/4$. These estimates include both the interior contribution and the nonzero exterior mean fields with the same fixed cutoff.

For subsequent stages, the one-sided pressure estimate is as follows. Let w be the transported primary carrier, including its selected amplitude α , let m be the transported normal, and let M be the parent gradient in particle labels. On the packet support, $\|M\| \leq M_0$. Write $b = \chi|m||w|$, with the nonnegative spatial cutoff χ . The principal pressure Hessian is

$$Q = -2\chi \frac{\langle m, Mw \rangle}{|m|^2} f'_\delta m \otimes m.$$

For the chosen periodic profile $f'_\delta \geq -1$ and $|f'_\delta| \leq \delta^{-1}$ when $0 < \delta \leq 1$. On the good-time set the flux is nonnegative; hence $Q \leq 2\chi \langle m, Mw \rangle I \leq 2M_0 b I$ as quadratic forms. The supported localization estimates give $b \leq \delta G$ there. On the complementary times they give $b \leq \delta E$, and the absolute estimate yields $\|Q\| \leq 2M_0 E$. Outside the cutoff $Q = 0$. The child's canonical pressure Hessian therefore has the global upper increment

$$e_p \leq 2M_0(\delta G + E) + \frac{C_p(\mathcal{W})}{k} + C_{\text{corr}}(\mathcal{W})\delta_{\text{corr}}(X_{\text{exp}}).$$

The remainder polynomials follow from the carrier and coefficient derivative estimates, including the correction that eliminates the residual. Every carrier norm is controlled by these estimates. The supported bounds on the selected profile give $G \leq C_d h_{\text{child}}$ and $E \leq C_d(\mathcal{P})h_{\text{child}}e^{-x_n/8}$, where $\mathcal{P}$ is the source scale determined by the parent. The coefficient of δh_{child} is controlled by this scale, which excludes the angular factor δ^{-1} .

The initial-gradient increment satisfies $e_i \leq C_i(\mathcal{W})(\alpha + k^{-2})$. Retain the factor $1 + c_d r_*^d$ in (52), and bound it above in the new error terms by the fixed constant $B_{\text{weight}} = 1 + C_{2,d}$. There are five error contributions: the principal pressure terms on the good-time set and

its complement, the initial term proportional to α , the pressure remainder, and the mean initial term proportional to k^{-2} . The prescribed expansion satisfies $\delta_{\text{corr}}(X_{\text{exp}}) \leq k^{-1}$, so both pressure remainders are included in the same frequency-dependent contribution. The two initial contributions include B_{weight} . The preceding reserve retains the factor r_*^d .

The coefficient and damping estimates in Section 8 and Section 10 control these five contributions at the prescribed frequencies. In particular, $\delta h_{\text{child}} \leq e^{-x_n/(2(J+n)^3)}$ and $h_{\text{child}} e^{-x_n/8} \leq e^{-x_n/16}$, while the selected α and the frequency remainders have their own packet-source bounds. The finite parameter selection bounds the sum of each of these five contributions by $1/64$. At every finite positive-history stage the mean reserve is bounded by the partial sum

$$\frac{1}{4} + \sum_{i < n} \sum_{a=1}^5 \text{cost}_{a,i} \leq \frac{1}{4} + \frac{5}{64} = \frac{21}{64} < \frac{1}{2}.$$

Including the nonnegative index-zero contribution in this partial sum preserves the upper bound, since the two initialization reserves have already been estimated separately. The dimensional multiplier and the seven additional initialization contributions belong to the finite list fixed before choosing J . The remaining lower bounds are met by enlarging X before constructing any packet. This changes the frequency values through their fixed formulas, while all construction exponents, error contributions, and the chosen J remain fixed.

Lemma 7.2 (One compact set for all initial increments). *With this fixed-cutoff construction every initial increment is supported in $\overline{B}(0, \max(1/4, \ell_*^{-1}))$, irrespective of its frequency, localization length, or finite truncation order.*

Proof. At a stage with physical localization length $0 < \rho \leq 1$, normalized labels are $y = x/\rho$, and the mean cutoff is $\chi((\rho\ell_*)y)$. The mean boundary condition expresses its initial velocity as the compactly supported solenoidal boundary operator with this cutoff, multiplied by its scalar penalty. Hence the normalized mean initial field vanishes for $|y| > (\rho\ell_*)^{-1}$. Physical unscaling replaces y by x/ρ , so it vanishes for $|x| > \ell_*^{-1}$. This holds for every mean grade. The oscillatory profiles and their full tensor correctors are supported in $|y| \leq 1/4$, hence after unscaling in $|x| \leq \rho/4 \leq 1/4$. The exact correction has zero initial value. Summing the finite family, including its terminal corrector, proves the assertion. Adding the fixed compact support of the base datum gives one compact set for all finite-stage initial velocities and their pointwise limit. $\square$

8 Joined source estimates and a uniform parameter envelope

We fix the dimension throughout this section. The Sobolev base order $q = 2d+5$ is fixed before any stage or truncation is chosen. Increasing this order changes constants and polynomial degrees only.

8.1 The variational history inverse

For every forced high grade, the inverse consists of a Dirichlet problem before activation and a forward evolution after activation. The Dirichlet coercivity estimate gives polynomial dependence on the parent coefficients on the history interval. On the forward interval this dependence follows from the relative propagator estimate. Compact-interval existence of a forward fundamental matrix alone does not provide either quantitative bound.

The quantitative estimate is established in the transverse space $U = m_0^\perp$, of dimension $d-1$. Let

$$Q(t, x) = F(t, x)|_U, \quad Q_t = Q_1, \quad Q_{tt} = -\mathcal{H}Q, \quad \|Qv\|^2 \geq c\|v\|^2, \quad c > 0.$$

The symmetric matrix $\mathcal{H}$ is the pressure Hessian along the preceding particle map. For a positive activation time $0 < t_0 \leq T$, suppose on $[0, t_0]$

$$\langle \mathcal{H}v, v \rangle \leq K\|v\|^2, \quad 0 \leq K, \quad Kt_0^2/2 \leq 1/2.$$

These are the inherited global one-sided pressure bounds. The absolute coefficient norms below are obtained from the particle-label fields: $F_1 = D_x(\partial_t \Phi)$, $F_2 = D_x(\partial_t^2 \Phi)$, $\mathcal{H} = -F_2 F^{-1}$, and $\det F = 1$ expresses F^{-1} by cofactors. The fixed Sobolev shift and factorial product bounds therefore make all these norms polynomial in the label amplitude and radius. The sharper central bounds of the preceding section are used for central endpoint selection and renewal. All norms below can also be taken in spatial-angular L^2 ; the multiplication coefficients have uniform pointwise bounds.

For a forcing f , solve for $z \in H_0^1((0, t_0); U)$:

$$\int_0^{t_0} \langle (Qz)', (Q\zeta)' \rangle - \langle \mathcal{H}Qz, Q\zeta \rangle dt = - \int_0^{t_0} \langle f, Q\zeta \rangle dt \quad (\zeta \in H_0^1((0, t_0); U)). \quad (53)$$

Its physical Jacobi displacement is $y = Qz$; its physical high velocity is

$$b = Qz', \quad (54)$$

which is generally different from y' . In particular, $z(0) = z(t_0) = 0$ does not assert that $b(0)$ or $b(t_0)$ vanishes.

Lemma 8.1 (Polynomial history inverse). *The solution of (53), its high velocity, its time derivative and its terminal high velocity have bounds polynomial in*

$$1 + t_0 + t_0^{-1} + c^{-1} + \|Q\|_\infty + \|Q_1\|_\infty + \|\mathcal{H}\|_\infty.$$

Their fixed-order spatial-angular Sobolev bounds are also polynomial in the corresponding fixed-order coefficient bounds. The assertion holds in every finite transverse dimension.

Proof. Put $C_0 = \|Q\|_\infty$, $C_1 = \|Q_1\|_\infty$ and $L = (Q^*Q)^{-1}Q^*$. Then $LQ = I$ and

$$\|L\| \leq c^{-1}C_0, \quad \|L'\| \leq c^{-1}C_1 + 2c^{-2}C_0^2C_1.$$

Consequently, with

$$A = 1 + c^{-1}C_0 + t_0(c^{-1}C_1 + 2c^{-2}C_0^2C_1),$$

the identity $z = Ly$ and the elementary inequality $\|y\|_{L^2} \leq t_0\|y'\|_{L^2}$ give $\|z'\|_{L^2} \leq A\|y'\|_{L^2}$. The sharper primitive bound $\|y\|_{L^2}^2 \leq (t_0^2/2)\|y'\|_{L^2}^2$ gives

$$\int_0^{t_0} (|y'|^2 - \langle \mathcal{H}y, y \rangle) \geq \frac{1}{2}\|y'\|_{L^2}^2 \geq (2A^2)^{-1}\|z'\|_{L^2}^2.$$

Thus the coercive variational theorem on the fixed Hilbert space H_0^1 constructs the inverse. Testing with z also gives

$$\|y'\|_{L^2} \leq 2t_0\|f\|_{L^2}, \quad \|z'\|_{L^2} \leq 2At_0\|f\|_{L^2}.$$

Both bounds are polynomial in C_0 , C_1 and $\|\mathcal{H}\|$.

Integration by parts and $Q'' = -\mathcal{H}Q$ give

$$Q^*(Qz'' + 2Q_1z') = Q^*f.$$

Therefore the acceleration is

$$z'' = (Q^*Q)^{-1}\{Q^*f - 2Q^*Q_1z'\}, \quad (55)$$

and

$$\|z''\|_{L^2} \leq c^{-1}C_0(\|f\|_{L^2} + 2C_1\|z'\|_{L^2}).$$

The time trace inequality

$$\|z'\|_{C_t} \leq t_0^{-1/2}\|z'\|_{L^2} + t_0^{1/2}\|z''\|_{L^2}$$

bounds both endpoints polynomially. For continuous f , (55) gives continuous z'' , and $b' = Q_1 z' + Q z''$ is the derivative on the closed interval. In particular the forward initial datum $b(t_0)$ is a bounded trace of the solution of (53).

Spatial derivatives are obtained by differentiating the operator on the fixed space $H_0^1((0, t_0); U)$. Every commutator contains only coefficient derivatives and a lower derivative of the same solution. Applying the same coercive inverse proves the fixed-order estimates by finite induction. For example, a sufficient bound for the inverse is

$$I_0 = I, \quad I_{r+1} = I + I_r + 2^r B I_r^2,$$

where I bounds the base inverse and B the coefficient block. This is a polynomial after the fixed number q of steps. The Gram inverse in (55) has the same positive-order commutator proof. These arguments use adjoints, norms and Hilbert-space coercivity in arbitrary transverse dimension. $\square$

On $[t_0, T]$ solve the forward transverse equation with initial datum $b(t_0)$ obtained above. Its value and derivative match those of the history, by the same equation and forcing at the junction. The algebraic normal pressure is reconstructed from this joined field. The profile is $\gamma(t) = \alpha$ on the history interval and $\gamma(t) = \alpha g(t)$ afterwards, where $g(t_0) = 1$ and $\alpha > 0$. Normalization by γ in spatial word estimates never differentiates γ in time.

8.2 The full transverse relative estimate

We derive the estimate for the full transverse space. At activation let p, q be the preceding unit normal and unit velocity, and put

$$a = \langle Bq, p \rangle > 0, \quad \beta = \frac{\|\Pi_{\{p,q\}^\perp} Bq\|}{a}, \quad \sigma = \sqrt{\beta}, \quad \varepsilon = \sqrt{a/h}.$$

Here h is the preceding shear height. Assume $0 < \sigma \leq 1/4$, $1/2 \leq a \leq 2$ and $0 < \varepsilon \leq 1$. Choose the third unit vector n in the direction of $\Pi_{\{p,q\}^\perp} Bq$; the positive tilt makes it nonzero. Write $E = \text{span}\{p, q, n\}$ and $\mathcal{E} = E^\perp$. The center reflection fixes E and reverses $\mathcal{E}$. The actual center coefficient therefore preserves this splitting. Neighboring coefficients need not preserve it.

Use the inherited bounds $G \geq 1$, $\|B\| \leq G$ and $\|B_t\| \leq G^2$. Transport the entire orthonormal frame by its skew connection S . In this frame, for the preceding background matrix B ,

$$S = p \otimes B^* p - B^* p \otimes p + q \otimes \Pi_{\{p,q\}^\perp} Bq - \Pi_{\{p,q\}^\perp} Bq \otimes q.$$

Consequently $\|S\| \leq 4G$ if $\|B\| \leq G$, and the exact column identity is

$$(B + S)q = 2\langle Bq, p \rangle p + \langle Bq, q \rangle q. \quad (56)$$

The normal-normal connection is zero. Differentiating the transported background gives norm at most $G^2 + 2(4G)G = 9G^2$. At activation its column outside the q diagonal is $a(p + \beta n)$. At later physical time t its defect is therefore at most $9G^2|t - t_0|$. These statements concern the complete vector column, including all additional components.

Use fast time $\tau = a(t - t_0)/\varepsilon$. The estimate is restricted to $0 \leq \tau \leq \tau_{\text{end}}$; assume $\tau_{\text{end}} \leq \Theta$, where $\Theta \geq 1$ is the numerical horizon. The estimate does not require extending the coefficients. At a neighboring label the actual gradient in this same frame is

$$M = B + h(t)q \otimes p + \mathcal{R}, \quad \|\mathcal{R}\| \leq e_{\text{old}} + \rho L_M.$$

The rank-one term uses the center pair; the neighboring discrepancy is included in $\mathcal{R}$.

Define the diagonal maps

$$D_r = \text{diag}(1, \varepsilon, 1, I_{\mathcal{E}}), \quad D_v = \text{diag}(\varepsilon, 1, \varepsilon, \varepsilon I_{\mathcal{E}}).$$

They satisfy $D_r D_v = \varepsilon I$ and both have norm at most one. The actual ray and high velocity, including every extra coordinate, are

$$m = s_0 O(t) D_r y, \quad y = (P, Q, N, r), \quad v = O(t) D_v z, \quad z = (U, V, W, Z), \quad r, Z \in \mathcal{E}. \quad (57)$$

The primary solution uses one fixed coordinate endpoint at every label. More precisely, if the endpoint lemma selects the physical vector $Y_{\text{phys}} \in m(t_0, 0)^\perp$, set

$$Y_{\text{coord}} = R_\perp^* F(t_0, 0)^{-1} Y_{\text{phys}}.$$

At the center, $F(t_0, 0) R_\perp Y_{\text{coord}} = Y_{\text{phys}}$. At every other label the history has the same coordinate endpoint Y_{coord} . Its norm is at most $\|F(t_0, 0)^{-1}\| \|Y_{\text{phys}}\|$. Thus Y_* in the history Lipschitz estimates below includes this inverse-frame factor.

For $t_0 > 0$, the unlocalized history is solved pointwise in the finite-dimensional transverse space and has uniform spatial derivative bounds. Its cylinder L^2 realization is obtained after multiplication by the compact seed. To verify this directly, put $Z(y, \theta) = \chi_{\text{in}}(y) f_\delta(\theta) Y_{\text{coord}}$ and $z_{\text{lin}}(t, y, \theta) = t Z(y, \theta) / t_0$. The zero-trace remainder $w = z - z_{\text{lin}}$ solves (53) with forcing

$$f_{\text{lin}} = -(2/t_0) Q_1 Z.$$

This identity follows from $Q_{tt} = -\mathcal{H}Q$ and $z_{\text{lin}, tt} = 0$. The forcing is a continuous path in every cylinder Sobolev space, is angular-mean zero, and has compact spatial support. Thus the Hilbert-space inverse applies. Its uniqueness identifies the result with the localized pointwise history. All cylinder estimates concern this localized field. For $t_0 = 0$ the prescribed forward primary is used instead of the history construction.

Here $s_0 > 0$ and $O(t)$ is the transported orthogonal frame. The velocity v is the unamplified selected high solution. The packet uses αv , with $\alpha > 0$ determined by its target size below. Set

$$A = a^{-1} D_r M D_v, \quad C = a^{-1} D_r (M + S) D_v.$$

Direct substitution into the ray and high equations gives

$$y' = (C^* - 2A^*)y, \quad z' = \mathcal{B}_\varepsilon(y)z, \quad \mathcal{B}_\varepsilon(y) = -C + 2 \frac{D_r^2 y}{|D_r y|^2} \otimes A^* y. \quad (58)$$

For example, the pressure numerator transforms by $\langle D_r y, M D_v z \rangle = a \langle y, A z \rangle$. Thus (58) is the rescaled equation for all transverse components.

Let

$$A_0 = (p + \beta n) \otimes q + q \otimes p, \quad C_0 = 2p \otimes q + q \otimes p.$$

The large shear term scales to $(\varepsilon^2 h(t)/a) q \otimes p$. For any matrix L , separating its q input column gives

$$a^{-1} D_r L D_v = (\varepsilon/a) D_r L + ((1 - \varepsilon)/a) (D_r L q) \otimes q.$$

Together with (56), this proves

$$\max(\|A - A_0\|, \|C - C_0\|) \leq \left| \frac{\varepsilon^2 h(t)}{a} - 1 \right| + \frac{11\varepsilon G + 18G^2|t - t_0| + e_{\text{old}} + \rho L_M}{a}. \quad (59)$$

One obtains the slightly better coefficient 3 in place of 11 for A ; the displayed common bound suffices. The norm of an additional input column has the small factor ε : the shear is zero on $\{p, q\}^\perp$. Both diagonal maps have norm at most one, so matrix scaling introduces no factor ε^{-1} .

The logarithmic derivative of $h(t)$ is bounded by $2G$. For $G\varepsilon\Theta/a$ small, its contribution to (59) is $O(G\varepsilon\Theta/a)$. Choose a single error parameter $e \geq \varepsilon$ large enough that the right side of (59) is at most $2e$, and that the initial ray error is at most e . It is bounded by a fixed constant times

$$\varepsilon\Theta G^2 + e_{\text{old}} + \rho \left(L_M + \frac{3L_m}{s_0\varepsilon} + \frac{2L_{\text{hist}}Y_*}{\varepsilon} \right). \quad (60)$$

The lower bound on a was used here. The constant L_m bounds the spatial Lipschitz norm of the normal at activation. The constant L_{hist} bounds, in operator norm, the spatial derivative of the linear map $Y_{\text{coord}} \mapsto b_{Y_{\text{coord}}}(t_0, y)$. Finally Y_* bounds the norm of the selected coordinate endpoint. Consequently the spatial variation of its high velocity is at most $\rho L_{\text{hist}}Y_*$.

The reference ray is

$$y_0(\tau) = (\beta\tau^2, -2\beta\tau, 1, 0).$$

Its full generator $C_0^* - 2A_0^* = -p \otimes q - 2\beta q \otimes n$ is nilpotent of order three. Its evolution is exactly

$$I - (\tau - s)p \otimes q - 2\beta(\tau - s)q \otimes n + \beta(\tau - s)^2 p \otimes n$$

and has norm at most $5\Theta^2$. Duhamel's formula and absorption in the supremum norm give, if $60e\Theta^3 \leq 1$,

$$\|y - y_0\| \leq 310e\Theta^5. \quad (61)$$

Indeed the difference generator has norm at most $6e$; the resulting bound is $10\Theta^2(e + 30e\Theta^3) \leq 310e\Theta^5$. Under the stronger smallness condition used below, this is at most $1/2$. Thus

$$N \geq \frac{1}{2}, \quad |D_r y|^2 \geq \frac{1}{4}, \quad \|y\| \leq 6\Theta^2.$$

In particular, (61) controls the extra ray components r .

Tangency is $y \cdot z = 0$, so the eliminated coordinate is exactly

$$W = -\frac{PU + QV + r \cdot Z}{N}. \quad (62)$$

Let $\zeta = (U, V, Z)$, let Π omit the n coordinate, and let $E_y \zeta$ insert the value (62). Then $\|E_y\| \leq 1 + 2\|y\| \leq 13\Theta^2$, and the actual reduced equation on the full $d - 1$ dimensional transverse space is $\zeta' = \Pi \mathcal{B}_\varepsilon(y) E_y \zeta$. It retains all couplings to Z .

The pressure denominator admits the following operator-norm estimate on the full ambient space. If $r_* = \|y - y_0\| \leq 1/2$, $R_* \geq \max(\|y\|, \|y_0\|)$ and $\eta = \max(\|A - A_0\|, \|C - C_0\|) \leq 1$, then

$$\begin{aligned} \|\mathcal{B}_\varepsilon(y) - \mathcal{B}_0(y_0)\| &\leq \eta + 8R_*(4 + 8R_*^2)(r_* + \varepsilon R_*) \\ &\quad + 2R_*(\eta R_* + 3r_*). \end{aligned} \quad (63)$$

To prove it, put $u_\varepsilon = D_r^2 y / |D_r y|^2$. The preceding lower denominator bounds and $\|D_r y - D_0 y_0\| \leq r_* + \varepsilon R_*$ give $\|u_\varepsilon - u_0\| \leq (4 + 8R_*^2)(r_* + \varepsilon R_*)$. Also $\|A^* y - A_0^* y_0\| \leq \eta R_* + 3r_*$, $\|A^* y\| \leq 4R_*$ and $\|u_0\| \leq R_*$. Insert these estimates in the difference of the two rank-one operators in (58). All estimates are Euclidean operator estimates on the complete space.

Using (61), $\eta \leq 2e$, $R_* \leq 6\Theta^2$, and $\varepsilon \leq e$, the right side of (63) is at most $5 \cdot 10^6 e \Theta^{11}$. The ideal generator has zero n input column. Therefore $\Pi \mathcal{B}_0(y_0)(E_y - E_{y_0}) = 0$, and the actual reduced generator differs from the ideal reduced generator by at most

$$65 \cdot 10^6 e \Theta^{13}. \quad (64)$$

The ideal reduced generator is explicitly

$$\begin{pmatrix} \frac{2P_0 Q_0}{1+P_0^2} & -2 + \frac{2P_0(P_0 + \beta)}{1+P_0^2} \\ -1 & 0 \end{pmatrix} \oplus 0_\varepsilon, \quad P_0 = \beta\tau^2, \quad Q_0 = -2\beta\tau.$$

Hence its active equation is

$$((1 + \beta^2\tau^4)V')' = 2(1 - \beta^2\tau^2)V, \quad U = -V',$$

and every ideal extra coordinate is constant.

Let $g = V_0$ be the positive scalar solution with $g(0) = 1$, $g'(0) = 0$. The scalar relative propagator and the ratio estimate $g(s)/g(t) \leq C\Theta$ bound the complete ideal reduced evolution by

$$K_d \frac{g(t)}{g(s)}, \quad K_d = C_d \Theta^8.$$

The extra identity block is included using this ratio estimate; g need not be nondecreasing. Whenever the reduced coefficient gap is Δ , the weighted Duhamel estimate gives

$$\frac{g(s)}{g(t)} \|U(t, s)\| \leq K_d + K_d \Delta \int_s^t \frac{g(s)}{g(r)} \|U(r, s)\| dr.$$

Thus $K_d \Theta \Delta \leq 1/2$ implies $\|U(t, s)\| \leq 2K_d g(t)/g(s)$. This estimate covers arbitrary transverse initial data and forcing, including additional modes.

It also bounds the joined high evolution in the fixed transverse coordinates used in the history problem. At each endpoint the actual physical field and reduced coordinates are related by

$$b = OD_v E_y \zeta, \quad \zeta = \Pi D_v^{-1} O^* b.$$

Their operator norms are at most $13\Theta^2$ and ε^{-1} , respectively. If $R_\perp : U \rightarrow m_0^\perp$ is an isometric identification and $b = FR_\perp a_{\text{hist}}$, then

$$a_{\text{hist}} = R_\perp^* F^{-1} b, \quad b = FR_\perp a_{\text{hist}}.$$

These two endpoint transformations have norms at most $\|F^{-1}\|$ and $\|F\|$. Thus the evolution in the actual a_{hist} coordinates is bounded by $26K_d \Theta^2 \varepsilon^{-1} \sup \|F^{-1}\| \sup \|F\| g(t)/g(s)$. Every factor has a known polynomial bound in the source parameters. Applying the same endpoint maps to the variation-of-constants formula gives the forced tail bound with these factors. This is the full $d - 1$ dimensional forward operator joined to the history trace, including its action on all additional transverse components.

For the selected primary let $\lambda \geq 0$ be its actual normalized initial slope and let $V_\lambda(0) = 1$, $V'_\lambda(0) = \lambda$. The ideal reduced initial vector is $(-\lambda, 1, 0)$. At a neighboring label the discrepancy of (U, V, Z) is at most $\rho L_{\text{hist}} Y_*/\varepsilon$, hence at most e after the choice (60). In particular the extra primary initial component is bounded by the same history Lipschitz estimate. The remaining initial component is determined by (62).

The weighted Duhamel estimate applied to the difference from $\zeta_\lambda = (-V'_\lambda, V_\lambda, 0)$ yields

$$\|\zeta - \zeta_\lambda\| \leq C_d e \Theta^{30} (1 + \lambda) g(\tau). \quad (65)$$

Indeed the initial term is $K_d e g(\tau)$ and the integral term is at most $2K_d^2 \Theta \Delta(1 + \lambda + e)g(\tau)$. The exponent is $2 \cdot 8 + 1 + 13 = 30$. Restoring the eliminated coordinate introduces at most two more powers of Θ , by (62) and (61).

The late-interval comparison is uniform in λ . The second fundamental solution satisfies

$$\frac{V_1(\tau) - g(\tau)}{g(\tau)} = \int_0^\tau \frac{ds}{(1 + \beta^2 s^4)g(s)^2}.$$

On $[0, 1]$ the scalar equation has uniformly bounded coefficients, so this integral at one has a fixed positive lower bound. Consequently, for $\tau \geq 1$, $V_\lambda(\tau) \geq c(1 + \lambda)g(\tau)$. The actual normalized flux is $J = \langle y, Az \rangle$; the physical flux for the unamplified field equals $s_0 a J$; the amplified field has flux $\alpha s_0 a J$. The ideal flux is $J_0 = Q_0(-V'_\lambda) + (P_0 + \beta)V_\lambda$. Equations (61)–(65) give

$$|J - J_0|/V_\lambda \leq C_d e \Theta^{34} \quad (1 \leq \tau \leq \tau_{\text{end}}). \quad (66)$$

For example, expand its difference into $\langle y, (A - A_0)z \rangle + \langle y - y_0, A_0 z \rangle + \langle y_0, A_0(z - z_\lambda) \rangle$. The final term introduces two powers of Θ beyond the restored-state bound; the other two terms are smaller.

The scalar formula gives $J_0/V_\lambda \geq \beta = \sigma^2$. Below the threshold $x = \sigma\tau = 1$, both the $Q_0(-V'_\lambda/V_\lambda)$ contribution and P_0 are nonnegative. Above the threshold $x = 1$, its equivalent expression is $x^2 - \beta + 2\sqrt{\beta}z/x \geq 1 - \beta \geq 3/4$, with the nonnegative scalar Riccati variable z . Let $x_{\text{tar}} \geq 2$ be the target coordinate and assume $\tau_{\text{tar}} = x_{\text{tar}}/\sigma \leq \Theta$. Then $\beta \geq 4/\Theta^2$. The geometric smallness condition is

$$C_d e \Theta^{40} \leq 1. \quad (67)$$

Indeed the flux error is then at most a fixed small multiple of Θ^{-6} , which is at most $\beta/2$ after fixing the dimensional constant. This gives a strictly positive lower bound for the late flux. The ray, denominator, reduced-propagator and relative-state estimates above involve smaller powers and follow from the same smallness condition. Enlarge the fixed C_d to cover their finitely many numerical constants, including the full transverse dimensional perturbation bound. The power 60 in the separate short-extension estimate concerns the numerical time envelope; it is not an additional geometric-error hypothesis.

The same condition transfers the ideal uniform-size estimate to the actual neighbor. Indeed the unamplified size is $s_0 \|D_r y\| \|D_v z\|$; the packet size is α times this quantity. Subtract the corresponding central reference expression for the packet size and apply the two state bounds. At the target the actual q velocity component is at least $V_\lambda(\tau_{\text{tar}})/2$ and the normal component N is at least $1/2$. Hence the unamplified target size is at least $s_0 V_\lambda(\tau_{\text{tar}})/4$. The actual normalization $\alpha \text{size}_{\text{target}} = \delta h_{\text{child}}$ and the scalar ratio $g(\tau)/g(\tau_{\text{tar}}) \leq C\Theta$ make this size error at most $C_d e \Theta^{40} \delta h_{\text{child}}$, hence at most δh_{child} . On $0 \leq \tau \leq 1$, the reference states are at most $C(1 + \lambda)$ and the perturbation bound adds at most the same size. The scalar target growth gives $\alpha s_0(1 + \lambda) \leq C\Theta \delta h_{\text{child}} e^{-b/\sigma}$. Here $b > 0$ is a fixed constant from the scalar growth lemma, decreased if needed. Thus the early size retains its exponentially small factor and a fixed polynomial in Θ (the larger Θ^5 bound suffices). All these statements concern the actual short interval ending at τ_{end} ; the ideal scalar equation alone is defined globally.

The bounded profile height. The preceding normalization also establishes the polynomial bound for the profile constant H_0 through cancellation of the scalar growth against the terminal normalization. The forward scalar profile is $g = V_0$. In physical time this means $\gamma(t) = \alpha$ for $t \leq t_0$ and $\gamma(t) = \alpha V_0(a(t - t_0)/\varepsilon)$ for $t \geq t_0$. Since $V_0 \leq V_\lambda$, the scalar ratio estimate on the retained interval through the target and its stipulated short extension gives $g(\tau)/V_\lambda(\tau_{\text{tar}}) \leq C\Theta$. Together with the actual target-size lower bound this yields

$$H_0 = \max(1, \sup \gamma) \leq 1 + C\Theta \delta h_{\text{child}} s_0^{-1}. \quad (68)$$

The constant s_0 is that of the actual unit initial phase: $s_0^{-1} = \|F(t_0)^* n_{\text{new}}\| \leq \|F(t_0)\|$. It therefore has a bound in terms of the preceding particle-label parameters. Likewise the scalar target lower bound gives

$$\alpha \leq C\Theta \delta h_{\text{child}} s_0^{-1} (1 + \lambda)^{-1} e^{-b/\sigma}.$$

Thus both the polynomial height and the small history amplitude come from normalization at the same target; the resulting source majorant has polynomial dependence on the parent parameters.

Finally, (60) explains how to choose a quantitative spatial neighborhood. Only preceding unnormalized coefficient and history Lipschitz bounds occur. Their positive-word inverse and endpoint estimates are polynomial in the preceding particle-label data. These bounds depend only on the parent geometry, its selected history and ε^{-1} . In particular, they are independent of the new angular width δ_n . The prescribed ρ_n below satisfies the resulting neighborhood conditions.

8.3 A common radius and polynomial source bounds

For an ordered-word Gevrey block put $\mathbf{m}_{R,b}(n) = R^{n+b}((n+b)!)^2$. The fixed-base coefficient constant is bounded by the explicit polynomial

$$S_{d,q}(R_c, C) = 2^q C \sum_{r=0}^q (4(d+1)R_c)^r (r!)^2.$$

Here R_c is the known coefficient radius. The full word alphabet is $\{1, \dots, d+1\}$.

Indeed, at external order n and base order $0 \leq r \leq q$,

$$(n+r)!^2 \leq 4^{n+r} (n!)^2 (r!)^2.$$

The $(d+1)^{n+r}$ ordered coordinate choices therefore give a coefficient-block bound

$$S_{d,q}(R_c, C) [4(d+1)R_c]^n (n!)^2.$$

Thus the known coefficient radius is replaced once by $4(d+1)R_c$. In the commutator argument below R_c denotes this enlarged known radius. This conversion is applied only to coefficient tensor bounds; the forcing is already measured by its full ordered-word block.

For a positive continuous profile γ and a fixed base order q , the normalized block used here is

$$\mathcal{B}_n^\gamma(f) = \sum_{I \in \{1, \dots, d+1\}^n} \|\gamma^{-1} \partial_I f\|_{C([0,T]; H^q)}.$$

The finite sum is outside the time norm. At each fixed time the block dominates the corresponding sum of Sobolev norms and hence the weighted norms used in the correction estimates. Equivalent choices of the base H^q norm change only constants depending on d, q .

Lemma 8.2 (Same-radius joined bounds). *There is a threshold $R_* \geq 1$, polynomial in the known source coefficients, their reciprocal coercivity bounds, inverse time lengths and the relative propagator constant of Section 8.2, such that for every $R \geq R_*$ the joined high inverse maps a forcing satisfying $\mathcal{B}_n^\gamma(f) \leq A \mathbf{m}_{R,b}(n)$ to velocity and time derivative bounds*

$$\mathcal{B}_n^\gamma(h_f) \leq C_v A \mathbf{m}_{R,b+3}(n), \quad \mathcal{B}_n^\gamma(\partial_t h_f) \leq C_t A \mathbf{m}_{R,b+3}(n).$$

Here h_f denotes the joined high field. The constants and radius threshold are independent of A, b and $R \geq R_*$. The mean inverse, pressure inverse and tensor corrector have analogous fixed-shift bounds at this same radius.

Proof. The history base estimate is Lemma 8.1. Its terminal trace supplies the actual forward datum. The forward base estimate is the relative estimate in Section 8.2, on the fixed packet support. Products with the known frame turn coordinate velocities into physical velocities, and the Gram equation controls the true time derivative.

We prove the all-order estimate by induction at the same radius. Differentiating the actual equation produces positive-order coefficient commutators. The elementary inequality

$$\binom{n}{j} (j!)^2 ((n-j+b)!)^2 \leq ((n+b)!)^2$$

holds for every $b \geq 0$. If M is the fixed-base inverse or forward constant and $BR_c^j(j!)^2$ is the known coefficient block, the normalized positive-order sum is bounded by

$$MB \sum_{j \geq 1} (R_c/R)^j \leq \frac{1}{2} \quad \text{when} \quad R \geq R_c(1 + 2MB).$$

Thus a factor two closes the induction at all orders, without changing the unknown forcing radius. The subsets of an ordered word give exactly the binomial convolution above. A dimensional factor is permitted when converting the known coefficient tensor bounds into word bounds; it is absorbed once into R_c .

The fixed-base inverse is the finite polynomial recursion in Lemma 8.1. The mean form and pressure equation have the same commutator identity and use their actual coercive inverses. The corrector uses known coefficient products, one angular primitive, and one spatial derivative. The primitive has operator norm bounded by the fixed period and the coordinate sums are finite ($\leq d^3$ terms). Taking the sum of these finitely many required radii gives a polynomial common radius. Linearity, applied after dividing the forcing and datum by A , proves that the radius is independent of A ; uniqueness gives the zero-amplitude case. The time profile itself is never differentiated. $\square$

These estimates define an explicit polynomial source majorant. Replace known primitive bounds by $W \geq 1$, coefficient blocks by $S_{d,q}$, fixed inverses by their q -step polynomial recursion, and maxima by sums. Reciprocal coercivity, ray and Gram constants are first replaced by the bounds proved above. Every remaining operation in the source construction is addition, multiplication, a fixed power or one of these fixed recursions. The resulting positive polynomial $\Phi_d(W)$ gives the bound

$$\Phi_d(W) \leq A_d W^{e_d}, \quad e_d = \deg \Phi_d, \quad A_d = 1 + \sum_j |[W^j] \Phi_d|.$$

The exponent is therefore determined by a fixed finite recursion. The mean and high grade recursion uses this common radius at every grade.

8.4 The explicit nonlinear threshold

Let $p = 202$ be the tail-polynomial degree proved in (39), and put $\vartheta = 1/(1000(p+1)) = 1/203000$. Write $Z = k^\vartheta$, $N = \lfloor Z \rfloor$. Let C, I, G, D, T, r_a^{-1} be the tail, inverse, correction-growth, drift, time and reciprocal radius constants of that packet, with $C, I, G, D, T \geq 0$. Here $r_a > 0$ is its analytic radius; it is distinct from the physical localization length ρ and from the fixed mean cutoff parameter ℓ_* . Define

$$H = 64 + \exp(1) + 2^p C + I + 12GT + \frac{8GTD}{r_a} + \left(\frac{8GT}{r_a} \right)^2, \quad k_* = H^{1000(p+1)}. \quad (69)$$

For $k \geq k_*$ one has $H \leq Z$, $Z \geq 64$ and $\log k \geq 1$. In particular

$$C(N+1)^p \leq C(2Z)^p \leq Z^{p+1} = k^{1/1000}.$$

The residual and correction tolerance obey

$$2[2e^{-(7/10)Z \log k}]e^{3GT} \leq \frac{1}{2}e^{-\sqrt{Z}}, \quad 2G(D/k + e^{-\sqrt{Z}})T \leq r_a/2.$$

Indeed $3GT \leq Z/4$, $\sqrt{Z} \leq Z/8$, $\log 8 \leq Z/8$ prove the first inequality. For the second, use $8GTD \leq r_a k$ and $8GT/r_a \leq \sqrt{Z} \leq e^{\sqrt{Z}}$. This gives a polynomial frequency threshold for the exponential correction-growth condition.

If these constants are bounded by $Q \geq 1$, the explicit bound is

$$k_* \leq (2^p + 200)^{1000(p+1)} Q^{6000(p+1)}. \quad (70)$$

For $p = 202$ the power of Q is 1 218 000. This exponent is independent of the stage and the truncation.

8.5 The next particle-label bound

Use the lifted and physical fields constructed in Lemmas 4.1–4.3 and (10). If u is the corrected coordinate packet, its lifted velocity is

$$\mathcal{V} = (k^{-1}u, \langle m_0, u \rangle), \quad \|m_0\| = 1,$$

and its material acceleration is the field $(\mathcal{V}_t + D\mathcal{V}\mathcal{V}) \circ \Psi$. The cylinder time-derivative estimate and the finite product and composition bounds define the source polynomial $C_{\text{rel}}(W)$ of the lifted-flow construction. At the prescribed scales its smallness conditions are

$$16C_{\text{rel}}(W) \leq k^{1/2}, \quad \rho^{-1} \leq k^{1/2}, \quad \delta_{\text{corr}}(k^\vartheta) \leq k^{-1}.$$

Here δ_{corr} is the correction tolerance, distinct from the profile width. The drift cancellation gives $BRT \leq C_{\text{rel}}(W)(k^{-1} + \delta_{\text{corr}}) \leq 1/8$. The graph trace and physical scaling therefore give common relative-field amplitude $M \leq k$ and radius $S \leq k^2$. These bounds hold for displacement, material velocity and material acceleration in L^2 , and for displacement and material velocity also in supremum norm. All fields come from the same lifted flow. The preceding conditions require only the prescribed source smallness exponents.

Write $X_{\text{parent}} = \text{id} + D_0$, with material velocity V_0 and acceleration W_0 . In original physical labels the relative map is $Y = \text{id} + a = \rho\psi^{\text{norm}}(\cdot/\rho)$, with time derivatives b, c . The full particle map is $X_{\text{parent}} \circ Y$, so its three fields are

$$\begin{aligned} D_0^+ &= D_0 \circ Y + a, \\ V_0^+ &= V_0 \circ Y + b + (DD_0 \circ Y)b, \\ W_0^+ &= W_0 \circ Y + 2(DV_0 \circ Y)b + (D^2D_0 \circ Y)[b, b] + c + (DD_0 \circ Y)c. \end{aligned}$$

The parent fields are estimated in the original physical labels, with common L^2 and supremum amplitude and radius $K \leq k$. Volume preservation of Y retains the parent L^2 factors in each composition term. The product and composition estimates proved in the lifted-flow section give common child L^2 amplitude and radius

$$C^+ = K + M + 9K^2M + 36K^3M^2, \quad R^+ = (1 + S)((1 + M)16K + 2) + S.$$

For the fixed lower frequency bound $k \geq 69$, these satisfy $C^+ \leq 47k^5 \leq k^6$ and $R^+ \leq 69k^4 \leq k^5$. The fixed Sobolev evaluation shift in (11) has order $q_{\text{ev}} = d + 2$, separate from the packet base order $q = 2d + 5$. After absorbing its fixed dimensional constants at the prescribed frequency, it yields supremum amplitude at most $k^{6q_{\text{ev}}+7}$ and radius at most k^6 . Thus the single bound

$$K^+ = k^{10(d+4)}$$

controls all three child fields in both norms, since the sum of one, the two L^2 bounds and the two supremum bounds is at most $5k^{6q_{\text{ev}}+7} \leq k^{6q_{\text{ev}}+8} \leq k^{10(d+4)}$. In particular, the acceleration has the supremum jet bounds needed for the next F_2^+ source. These are the fixed source-polynomial and frequency conditions included in the single scale choice.

The full inverse deformation is polynomial as well. Since $\det F^+ = 1$, its inverse is the adjugate. If $\|D^n F^+\| \leq AR^n(n!)^2$, expansion of each cofactor into $(d-1)!$ products and the Gevrey product constant 3 give

$$\|D^n(F^{+,-1})\| \leq d(d-1)! 3^{d-2} A^{d-1} R^n(n!)^2.$$

The factor d bounds operator norm by the largest entry. The time derivative satisfies $(F^{+,-1})_t = -F^{+,-1} F_1^+ F^{+,-1}$. Also $F_1^+ = D_x V_0^+$ and $F_2^+ = D_x W_0^+$. Thus the source coefficients have fixed polynomial bounds in the new particle-label parameter. Fix, for instance,

$$D_d = 10(d+4)$$

for the particle-label invariant $K^+ \leq k^{D_d}$. The inverse and coefficient bounds are fixed polynomials in K^+ , possibly with exponents different from D_d .

8.6 Uniform scales and a single frequency choice

Set $s_n = J + n$ and

$$x_0 = X, \quad x_{n+1} = s_n^2 x_n, \quad k_n = e^{x_n/s_n^2}, \quad h_n = e^{x_n/s_n^5}, \quad \delta_n = e^{-x_n/s_n^3}, \quad \rho_n = e^{-x_n/s_n^{7/2}}.$$

Here $n = j - J$ and $x_n = x_{j-1}$ in the notation of Section 10; ρ_n is the physical localization, not the fixed mean cutoff ℓ_* . For $n \geq 1$ the identities

$$\log k_{n-1} = \frac{x_n}{(s_n - 1)^4}, \quad \log h_{n-1} = \frac{x_n}{(s_n - 1)^7}, \quad \log \rho_{n-1}^{-1} = \frac{x_n}{(s_n - 1)^{11/2}}$$

are exact. If $K_n \leq k_{n-1}^{D_d}$, every preceding geometric constant in (60) is bounded by $C s_n^a x_n^b e^{cx_n/(s_n-1)^4}$. For sufficiently large fixed J and then X ,

$$\rho_n C s_n^a x_n^b e^{cx_n/(s_n-1)^4} \leq e^{-x_n/(2s_n^{7/2})}.$$

The additional factors in the ambient smallness condition have the same inherited scale: $\Theta_n \leq C(1+s_n^2 x_n^2)$, $s_0^{-1} \leq \|F(t_0, 0)\|$, and $\varepsilon^{-1} \leq C\sqrt{h_{n-1}}$. They therefore contribute only polynomial factors in s_n, x_n or an exponential with denominator $(s_n - 1)^7$, which is already absorbed by the displayed denominator $(s_n - 1)^4$. The terms $\varepsilon \Theta G^2$ and e_{old} in (60) are controlled by the separate shear separation and preceding frequency-error comparisons in Section 10; they are not made small by localization. At $n = 0$ the activation time is zero, and the forward high construction is used. This case contains no reciprocal history-length factor. The seed predecessor has polynomial bounds in X . The normal-stage factor $e^{-X/J^{7/2}}$ dominates those bounds for the same final large choice of X .

Consequently all neighborhood conditions hold at the prescribed localization length ρ_n .

The full source list additionally includes δ_n^{-1} and the new scalar height. Its positive-polynomial bounds therefore have the form

$$\mathcal{P}_n = C s_n^a x_n^b \exp\left(\frac{cx_n}{(s_n - 1)^3}\right). \quad (71)$$

In particular the radius, H_0 , reciprocal analytic radii, history and mean inverse constants, and the constants in (69) are bounded by fixed powers of (71).

Lemma 8.3 (Uniform domination). *For a finite list of fixed $A, C, c, a, b, N > 0$ and $\gamma > 0$, one common choice of J and then X ensures*

$$A\mathcal{P}_n^N \leq k_n^\gamma \quad (n \geq 0)$$

for every entry of the list.

Proof. Choose J so that $Nc/(s-1)^3 \leq \gamma/(2s^2)$ for $s \geq J$; $J \geq 16Nc/\gamma + 2$ suffices. Increase X until

$$\log(AC^N) + Na \log s_n + Nb \log x_n \leq \frac{\gamma x_n}{2s_n^2}$$

uniformly in n . To see the uniformity, use $x_n = X \prod_{r < n} (J + r)^2$. The sequences $s_n^2 \log s_n / \prod_{r < n} (J + r)^2$ and $s_n^2 \log(\prod_{r < n} (J + r)^2) / \prod_{r < n} (J + r)^2$ are bounded, since the denominator eventually dominates every polynomial in n while its logarithm grows at most $2n \log(J + n)$. The resulting bound is $O_J((1 + \log X)/X)$ and tends to zero. Adding the two inequalities proves the assertion. Take the maximum of the finitely many choices of J, X . $\square$

Apply this lemma simultaneously to the source constants, the neighborhood conditions, (70), the lifted-flow smallness conditions and their numerical constants. It gives the actual nonlinear correction at the preselected k_n and then $K_{n+1} \leq k_n^{D_d}$. The exponent D_d is fixed first; the source polynomials and their degrees are then fixed; J, X are chosen once for the resulting finite list. The same exponents and starting index apply to every subsequent child.

The zero-history initialization adds seven fixed constants to this same finite list: the actual forward family, weighted carrier localization, support geometry, gradient and pressure errors, relative particle fields, and the initial amplitude ratio. Their exponents and constants are fixed before the single choice of J . The finitely many remaining seed and first forward-stage thresholds depend on d, J and are met by increasing X alone. The seed cutoff normalization, the two initial mean reserves, and the preserved core difference are specified in Section 10. Thus the first positive-history state and every successor use the same J, X and the same prescribed frequencies. The induction then applies to the joined variational-history family and the quantitative neighborhood constructed in this section.

9 Initial increments at every Sobolev order

We use the joined displacement-history and forward inverse of Lemma 6.2. When the activation time is positive, the forced high coefficients generally have nonzero initial velocities; all of them enter the estimates below.

Fix the dimension d and an admissible integer base Sobolev order s for the packet estimates. Constants below may depend on d and on a fixed derivative order, but the frequency condition will not depend on that derivative order. Write $\mathcal{P} \geq 1$ for a bound on the source primitives, including the coefficient and inverse constants, the terminal datum, the reciprocal phase scale, the fixed mean cutoff constant, and the common Gevrey radius. Section 8 bounds these primitives for the constructed sources, uniformly along the prescribed scale sequence.

9.1 The initial coefficient sums

Put $\kappa = k^{-1}$ and let N be the chosen truncation. At time zero, the parent flow is the identity and the nonlinear correction vanishes. The initial increment is

$$u_{\text{osc},0}(a) = \rho \sum_{p=1}^N [\kappa^p A_p + \kappa^{p+1} C_p](0, a/\rho, km_0 \cdot a/\rho), \quad (72)$$

$$u_{\text{mean},0}(a) = \rho \sum_{p=2}^N \kappa^p B_p(0, a/\rho), \quad \Delta u_0 = u_{\text{osc},0} + u_{\text{mean},0}. \quad (73)$$

Here C_p is the full ambient antisymmetric-potential corrector and B_p is independent of the angle. The estimates include every high coefficient, its tensor corrector, and every mean coefficient. Formula (72) extends [1, §3.9, (3.60)] to the full ambient tensor construction.

Let $\alpha > 0$ be the selected primary amplitude. The joined profile is

$$\gamma(t) = \alpha \quad (0 \leq t \leq t_0), \quad \gamma(t) = \alpha g(t) \quad (t_0 \leq t \leq T), \quad g(t_0) = 1,$$

and put $H_0 = \max(1, \alpha, \sup \alpha g)$. The coefficient induction assigns the oscillatory profiles γH_0^{2p-2} and the mean profiles H_0^{2p-2} .

The factor α on the history interval follows from linearity. The primary terminal displacement is α times the normalized compact datum, and the displacement inverse is linear. In every higher zero-mean force, terms containing only means cancel under angle-mean subtraction. Every remaining term contains a previous oscillatory factor or a derivative of one. The term involving the newly solved mean is $-(m \cdot B_p) \partial_\theta A_1$, which also contains this factor. Extra oscillatory factors are bounded by the indicated powers of H_0 . The history inverse and the angular tensor primitive therefore retain α at time zero. This factor is preserved at each step of the graded construction by linearity of its high solver.

Suppose first that $t_0 > 0$. In fixed transverse coordinates, every forced history solves $\xi(0) = \xi(t_0) = 0$ and has velocity $A = FR_\perp \xi_t$. The primary instead has $\xi(0) = 0$ and $\xi(t_0) = \alpha \chi_{\text{in}}(y) f_\delta(\theta) \xi_T$, where ξ_T is the selected terminal coordinate. It is obtained from the physical central endpoint by the inverse frame; its norm is included in $\mathcal{P}$. Subtracting the affine extension $(t/t_0) \alpha \chi_{\text{in}} f_\delta \xi_T$ reduces this primary problem to the same zero-endpoint inverse. All spatial and phase words of the forced problem still have zero endpoint values. For the primary, differentiating this explicit affine extension supplies exactly the differentiated boundary datum; the differentiated boundary traces are therefore explicit. The forward solution starts with the trace $FR_\perp \xi_t(t_0)$. It agrees there with the history solution, and the common velocity equation then gives equality of the time derivatives at the junction. These are the boundary conditions of Lemma 6.2. When $t_0 = 0$, the primary instead has its prescribed forward datum and the forced high fields have zero initial velocity. The forward estimates give the same bounds below, without a history inverse or reciprocal factor t_0^{-1} . This applies both to the exceptional seed and to the first normal stage $j = J$; positive history begins at $j = J + 1$.

The all-word estimate uses the same inverse. After translation in the label and angle, write the equation as $L(a)u(a) = f(a)$. At a fixed admissible base Sobolev order let its inverse constant be M , and let the known coefficient derivative bounds be $BR_c^j(j!)^2$. Differentiating a word of length n gives the exact Leibniz identity

$$L\partial^\omega u = \partial^\omega f - \sum_{\emptyset \neq S \subset \{1, \dots, n\}} (\partial^{\omega_S} L) \partial^{\omega_{S^c}} u.$$

The homogeneous boundary space is unchanged, so the same inverse applies to each term. In the block norm this yields a binomial convolution. For $b \geq 0$,

$$\binom{n}{j} (j!)^2 ((n-j+b)!)^2 \leq ((n+b)!)^2.$$

Consequently a forcing bound $FR^{n+b}((n+b)!)^2$ gives $2MFR^{n+b}((n+b)!)^2$ for the solution as soon as $R \geq R_c(1 + 2MB)$: the positive-order convolution contributes at most $MB \sum_{j \geq 1} (R_c/R)^j \leq 1/2$. The initial finite base inverse is obtained by the finite Sobolev commutator recursion. Its order is fixed by d , not by the packet grade. Time traces follow from the H^1 equation and have a polynomial bound in the reciprocal history length and known frame bounds. The same estimate on the forward interval uses its relative propagator and the history trace. This proves preservation of the unknown forcing radius; known coefficient conversion enlarges R_c only once.

The displacement argument uses the full transverse space of dimension $d - 1$. If $R_\perp$ has orthonormal columns, $\eta = FR_\perp\xi$, and $\eta(t_0) = 0$, then

$$\int_0^{t_0} (|\eta_t|^2 - \langle H\eta, \eta \rangle) dt \geq \left(1 - \frac{K+t_0^2}{2}\right) \int_0^{t_0} |\eta_t|^2 dt \geq \frac{1}{2} \int_0^{t_0} |\eta_t|^2 dt.$$

The one-sided Poincaré inequality here follows by integrating $\eta(t) = -\int_t^{t_0} \eta_t(r) dr$. Conversion to ξ uses only the known bounds for F^{-1} and its derivative. Thus the inverse constant is polynomial in those bounds in every transverse dimension.

9.2 Separating derivative order from expansion order

For a fixed integer $m \geq 0$, the coefficient estimates need derivatives only through $m + s + 1$. Their shift is at most $100p$; replacing this fixed coefficient by a larger dimensional constant makes no change to the argument. For $b = m + s + 1$,

$$(b + 100p)! \leq 2^{b+100p} b! (100p)!, \quad ((b + 100p)!)^2 \leq C_b (C_p)^{200p}.$$

Also $R^{b+100p} = R^b (R^{100})^p$. Consequently all dependence on m can be put in a prefactor $C_{d,m} \mathcal{P}^{c_{d,m}}$, while the expansion-order bound has the form

$$B_N^{p+1}, \quad B_N = \mathcal{P}^{c_d} (C_d N)^{L_d}. \quad (74)$$

with one finite exponent L_d independent of m . For the displayed shifts one may take $L_d = 300$ after enlarging constants. Choose $N = \lfloor k^\vartheta \rfloor$ with $L_d \vartheta < 1/100$. A single fixed source-to-frequency condition then ensures $B_N \leq k^{1/100}$.

At time zero the cylinder Sobolev norms of A_p and C_p are bounded by

$$\alpha C_{d,m} \mathcal{P}^{c_{d,m}} B_N^{p+1}.$$

The mean has the same bound without α . For $k \geq 4$,

$$\sum_{p=1}^N k^{-p} B_N^{p+1} \leq C, \quad \sum_{p=1}^N k^{-p-1} B_N^{p+1} \leq C, \quad \sum_{p=3}^N k^{-p} B_N^{p+1} \leq C k^{-2}.$$

These follow by summing the geometric series with ratio $k^{-99/100}$. In the last sum its first power is $k^{-3+4/100} \leq k^{-2}$. The grade-two mean is estimated separately at its fixed order, giving $C_{d,m} \mathcal{P}^{c_{d,m}} k^{-2}$ as well. This fixed-order estimate preserves the power k^{-2} in the mean bound.

For $T_k f(y) = f(y, km_0 \cdot y)$ and $|m_0| = 1$, the chain rule and $H^1(\mathbb{T}) \hookrightarrow L^\infty(\mathbb{T})$ give

$$\|T_k f\|_{H^m(\mathbb{R}^d)} \leq C_{d,m} (1+k)^m \|f\|_{H^{m+1}(\mathbb{R}^d \times \mathbb{T})}.$$

Indeed each spatial derivative is a product of the commuting operators $\partial_{y_i} + km_{0,i} \partial_\theta$. Apply the periodic one-dimensional estimate to each resulting derivative at fixed y , then integrate in y . Exactly one additional angle derivative is needed. For the mean, angle independence eliminates the additional graph derivative. Finally, physical scaling multiplies the L^2 norm of an order- r derivative by $\rho^{1+d/2-r} \leq \rho^{-m}$ for $r \leq m$ and $0 < \rho \leq 1$. We obtain the estimates

$$\|u_{\text{osc},0}\|_{H^m} \leq C_{d,m} \rho^{-m} k^m \alpha \mathcal{P}^{c_{d,m}}, \quad (75)$$

$$\|u_{\text{mean},0}\|_{H^m} \leq C_{d,m} \rho^{-m} k^{-2} \mathcal{P}^{c_{d,m}}. \quad (76)$$

The selected central amplitude satisfies

$$\alpha \leq \mathcal{P}^{c_d} e^{-bx}$$

for a fixed $b > 0$, by the central target growth, with the normalized terminal and reciprocal phase constants included in $\mathcal{P}$. In the scalar indexing below, $n = j - J$ corresponds to normal stage j of Section 10: $a_n = j$, $x_n = x_{j-1}$, $k_n = k_j$, and $\rho_n = \rho_j$. The inherited geometric parameter is therefore x_n , while the new target is x_{n+1} . The amplitude factor is e^{-bx_n} . This exponential factor applies to all initial high coefficients, including those produced by positive history intervals.

9.3 One scale sequence gives all orders of summability

Lemma 9.1 (Scalar summability). *Fix $J \geq 2$, $X \geq 1$, and define*

$$a_n = J + n, \quad x_0 = X, \quad x_{n+1} = a_n^2 x_n, \quad k_n = e^{x_n/a_n^2}, \quad \rho_n = e^{-x_n/a_n^{7/2}}.$$

Here ρ_n is the physical packet length; it is distinct from the fixed physical mean-cutoff parameter ℓ_ and from the Gevrey weight radius used in the cylinder estimates. Let*

$$\mathcal{P}_n = C a_n^p x_n^q \exp\left(\frac{c x_n}{(a_n - 1)^3}\right)$$

with fixed $C \geq 1$, $c \geq 0$, $p, q \in \mathbb{N}$ and $b > 0$. For every fixed $m, \nu \in \mathbb{N}$, both series

$$\sum_n \rho_n^{-m} \mathcal{P}_n^\nu k_n^m e^{-b x_n}, \quad \sum_n \rho_n^{-m} \mathcal{P}_n^\nu k_n^{-2}$$

converge.

Proof. We have $x_n \geq X 4^n$ and $\log x_n \leq \log X + 2n \log(J + n)$. Therefore, for every fixed $r \geq 0$,

$$\frac{a_n^r (1 + \log a_n + \log x_n)}{x_n} \rightarrow 0.$$

The logarithm of the first summand, divided by x_n , tends to $-b$: its other exponential terms are m/a_n^2 , $m/a_n^{7/2}$ and $c\nu/(a_n - 1)^3$, and its logarithmic terms vanish. Eventually the summand is at most $e^{-b x_n/2}$, a summable sequence.

For the second summand, divide its logarithm by x_n/a_n^2 . The result tends to -2 , since

$$m a_n^{-3/2} \rightarrow 0, \quad c\nu \frac{a_n^2}{(a_n - 1)^3} \rightarrow 0,$$

and the logarithmic terms again vanish. Eventually it is at most $e^{-x_n/a_n^2} \leq e^{-n}$, proving convergence. $\square$

The choices of J, X in this lemma are independent of m . Only the starting index of the convergence comparison depends on m ; the construction itself uses the same scales for every Sobolev order. The same proof permits a localization scale whose reciprocal is bounded by a polynomial times $\exp(c_0 x_n/a_n^\beta)$ for any fixed $\beta > 2$. The quantitative localization estimate is therefore part of the source construction.

To verify that the prescribed frequency satisfies all construction conditions, consider any finite list $A\mathcal{P}_n^\nu \leq k_n^\theta$ with $\theta > 0$. Choose J so that $c\nu a^2/(a - 1)^3 \leq \theta/2$ for all $a \geq J$ and every condition. Writing $x_n = X b_n$ with $b_n \geq J^{2n}$, the sequences a_n^2/b_n , $a_n^2 \log a_n/b_n$ and $a_n^2 \log b_n/b_n$ are bounded. Hence a single increase of X absorbs all the remaining logarithmic terms, uniformly in n . This proves the conditions at the prescribed k_n . Only finitely many construction conditions enter at fixed d ; the fixed- m prefactors in (75)–(76) do not introduce further construction conditions.

9.4 The smooth compact limit

Section 8 bounds the stage- n source primitives by $\mathcal{P}_n$ and verifies the prescribed localization scale for the joined construction. The preceding estimates consequently give

$$\sum_n \|\Delta u_n(0)\|_{H^m} < \infty \quad \text{for every integer } m \geq 0.$$

For each spatial order r , take $m > r + d/2$. The Sobolev embedding in dimension d gives uniform convergence of the derivative series through order r . Include the base solution, the seed increment and the first normal zero-history forward increment in $u_{\text{seed},0}$. The latter two are fixed finite packet increments, including their mean fields and terminal tensor correctors. The remaining sum starts with the first positive-history joined increment; this finite index shift changes neither the convergence proof nor the already fixed J, X . Consequently the single sum $u_0 = u_{\text{seed},0} + \sum_{n \geq 1} \Delta u_n(0)$ is smooth, has the termwise derivatives, and is divergence free. The fixed physical mean-cutoff construction puts the supports of all these initial increments in one compact ball. Enlarging it to contain the base support proves $u_0 \in C_{c,\sigma}^\infty(\mathbb{R}^d)$.

10 The scale choice and the finite-stage induction

The construction uses one prescribed scale sequence. The quantitative source estimate supplies a fixed exponent D_d for the particle-map invariant and a finite list of polynomial construction conditions. All choices in this section are made after those exponents have been fixed.

10.1 The base flow and the order of choices

Choose an oriented orthonormal basis (p, q, n) of E and let the full matrix L_B vanish on $E^\perp$ and satisfy

$$L_B q = p + X^{-2} n, \quad L_B p = L_B n = 0.$$

It is trace free. Lemma 3.1, with a fixed radial cutoff, gives a compact, odd, reflection-equivariant solenoidal datum equal to $L_B x$ near zero. Its Sobolev and Gevrey bounds are uniform for $X \geq 1$. Lemma 2.1 gives its actual Euler solution u_B on a uniform short interval. Its Gevrey bounds also persist on a uniform shorter interval: with fixed base order $q_d > (d+1)/2 + 2$ and radius $r(t)$, set

$$\mathcal{X}_r = \sum_{n \geq 0} \frac{r^n}{(n!)^2} \sum_{|I|=n} \|\partial_I u_B\|_{H^{q_d}}, \quad \mathcal{Y}_r = \sum_{n \geq 0} \frac{nr^n}{(n!)^2} \sum_{|I|=n} \|\partial_I u_B\|_{H^{q_d}}.$$

At each finite cutoff the differentiated transport cancellation and factorial product convolution give $\mathcal{X}'_r \leq C_d \mathcal{X}_r^2 + (r'/r + C_d \mathcal{X}_r/r) \mathcal{Y}_r$. Choose r' a sufficiently negative fixed constant. A short-time bootstrap bounds $\mathcal{X}_r$ and keeps $r > 0$ uniformly in the cutoff. Passing to the increasing sums proves the claim. The projected Euler equation controls the time derivative and material acceleration at a smaller radius. The integral flow equation then bounds the actual particle map and its first two time derivatives in the same manner. This proof uses the Sobolev embedding at an order appropriate to the fixed dimension. Write K_B for a uniform upper bound on its pressure Hessian.

First fix d , the packet base order and all finite inverse, coefficient, flow and source exponents, including D_d . Fix C_M, C_H large enough for the central gradient and Hessian induction below. The global coefficient bounds are separately supplied by the actual particle-label invariant; the global pressure bound is one-sided. These fix the endpoint constant and then its small threshold ζ . Choose D_0 sufficiently large for the first geometric comparison; $D_0 = 1000$ suffices for the displayed exponent 60. All seed mean-cutoff and boundary bounds are fixed

powers of X . Choose $D_k \geq 1000$ larger than the resulting seed polynomial exponent divided by each of the finitely many frequency margins, including the seed gradient, pressure and particle-label estimates. Fix also $C_{\text{init}} = 2C_{\text{lit}}$, where $C_{\text{lit}} = 1 + \pi C_{\chi,d}$ and $C_{\chi,d}$ is the fixed seed cutoff derivative bound. This bounds the combined contribution of the two initializations to $B_c - B_e$.

The same finite scale choice includes the twelve later-stage estimates and seven additional estimates for the first normal forward stage: its finite family and correction, weighted carrier localization, support geometry, gradient error, pressure error, relative particle fields, and initial amplitude ratio. All seven are fixed source polynomials multiplied by the prescribed frequency or exponential damping. In particular, the last condition is

$$[1 + c_d(1 + \pi C_{\chi,d})] \alpha_J / \delta_J \leq 1/64.$$

Choose the starting index J once for this combined list, and then increase X to meet the seed, short-time, support and reserve thresholds. These thresholds depend only on d, J and constants already fixed before J . No second choice of J , frequency or exponent is made in either initialization or the later induction.

Put $h_{J-2} = 1$, $x_{J-1} = X$, and $x_j = j^2 x_{j-1}$ for $j \geq J$. The auxiliary value h_{J-2} records the uniform scale of the base solution. Prescribe

$$h_{J-1} = X^{D_0}, \quad k_{J-1} = X^{D_k}, \quad \rho_{J-1} = s_{\text{cut},d}^{-1} X^{-D_0}, \quad \delta_{J-1} = X^{-(D_0+10)}, \quad (77)$$

$$h_j = e^{x_{j-1}/j^5}, \quad k_j = e^{x_{j-1}/j^2}, \quad \delta_j = e^{-x_{j-1}/j^3}, \quad \rho_j = e^{-x_{j-1}/j^{7/2}} \quad (j \geq J). \quad (78)$$

Here $s_{\text{cut},d} > 0$ is the fixed dimensional cutoff scale, normalized so that $a_d = 1/(3s_{\text{cut},d})$. The seed frequency is fixed by its polynomial source bound in X ; the source-envelope lemma establishes the required estimates at the prescribed normal frequencies. Use the explicit physical mean-cutoff parameters

$$\ell_* = 1, \quad r_* = X^{-D_0}.$$

For large X , $r_* \leq 1/4$. The splitting ball of radius $a_d r_* / \ell_* = \rho_{J-1}/3$ contains the oscillatory seed support of radius $\rho_{J-1}/4$. The normalized seed mean has parameters $\ell = \rho_{J-1}$ and $r = r_*$, hence splitting radius $1/3$. These parameters are fixed functions of the seed value before the final choice of X ; their physical values are retained throughout the iteration and are not replaced by the shrinking ρ_j . Their reciprocals and all subsequent boundary penalties are bounded by fixed powers of X . The multiplier in the full mean reserve is at most $1 + c_d$, a dimensional constant fixed before the scale choice. The first normal support condition follows from $\rho_J s_{\text{cut},d} X^{D_0} \leq 1$: for fixed d, J , the left side tends to zero as $X \rightarrow \infty$. Thus this condition only enlarges the final lower bound on the seed parameter.

10.2 Time and geometric comparisons

Let

$$W_j = \frac{3x_j x_{j-1}}{\sqrt{h_{j-1}}}, \quad S_{J-1} := S_{\text{base}} = 2W_J = 6J^2 X^{2-D_0/2}, \quad t_{J-1} = 0.$$

At stage j , the preceding solution determines a_j, β_j . Define

$$t_j = t_{j-1} + \frac{x_j}{\sqrt{\beta_j a_j h_{j-1}}}, \quad S_j = t_j + 2W_{j+1}. \quad (79)$$

When $a_j, \beta_j x_{j-1}^2 \in [1/2, 2]$, $W_j/6 \leq t_j - t_{j-1} \leq 2W_j/3$. For $j > J$ the exact identities

$$\log k_{j-1} = \frac{x_{j-1}}{(j-1)^4}, \quad \log h_{j-1} = \frac{x_{j-1}}{(j-1)^7}$$

show that W_{j+1}/W_j is arbitrarily small uniformly after J : its logarithm is

$$2 \log j + 2 \log(j+1) - \frac{x_{j-1}}{2j^5} + \frac{x_{j-1}}{2(j-1)^7}.$$

At $j = J$ the last term is instead $(D_0/2) \log X$. The negative term dominates in both cases. Hence $S_j \leq S_{j-1}$, and with the numerical envelope $\hat{\Theta}_j = C(1 + x_j x_{j-1})$ the extra scaled interval obeys $\Delta\tau_j = 2\sqrt{a_j h_{j-1}} W_{j+1} \leq \hat{\Theta}_j^{-60}$. The actual central parameter is $\Theta_j = H_j^{\text{sc}} = x_j/\sqrt{\beta_j} + \Delta\tau_j$ and satisfies $\Theta_j \leq \hat{\Theta}_j$. In particular $\Delta\tau_j \leq \Theta_j^{-60}$. The superscript distinguishes this scaled horizon from a pressure Hessian. Also $t_J \geq S_{\text{base}}/12$, so at positive-history stages $t_{j-1}^{-1} \leq 12S_{\text{base}}^{-1} \leq h_{j-1}$.

Use $E_* = k_{j-1}^{-1/4}$, $G_* = 1 + h_{j-2}$ and $K_h = k_{j-1}^{D_d}$. Each part of (14) satisfies its smallness condition at the same scale choice. For the spatial term the logarithm is at most

$$-\frac{x_{j-1}}{j^{7/2}} + \frac{c_d D_d x_{j-1}}{(j-1)^4} + \frac{C x_{j-1}}{(j-1)^7} + C \log x_{j-1} \leq -\frac{x_{j-1}}{2j^{7/2}}.$$

This estimate uses the inherited geometric bound, which excludes the new δ_j^{-1} ; using the complete source list here would not imply the required smallness. The old frequency error has logarithm $-x_{j-1}/(4(j-1)^4)$. For $j \geq J+2$,

$$\log(\epsilon G_*^2) \leq -\frac{x_{j-1}}{2(j-1)^7} + \frac{2x_{j-1}}{(j-1)^2(j-2)^7} + C \leq -\frac{2x_{j-1}}{5(j-1)^7} + C.$$

For $j = J$, $\epsilon \hat{\Theta}_j^{61} \leq C(J) X^{122-D_0/2}$; for $j = J+1$ the new exponential dominates the preceding polynomial shear. These cover the two exceptional stages. Multiplying by any of the fixed powers of x_{j-1} , $\hat{\Theta}_j$ still gives the required smallness, including the actual $e\Theta_j^{40}$ condition. The same comparisons give

$$h_j/h_{j-1}^2 \gg 1, \quad \frac{\sqrt{h_{j-1}}}{x_{j-1}x_j} \gg G_* + 1, \quad \rho_j < r_{\text{core}}. \quad (80)$$

Here $r_{\text{core}} > 0$ is the fixed physical seed core radius. Each use of $\gg$ in this proof denotes a fixed finite inequality whose constant was chosen before J, X .

10.3 Summable error bounds

For positive-history stages let $\mathcal{P}_j^{\text{par}} \geq 1$ bound the parent coefficient and geometric bounds, including C_1, C_I , but excluding the new reciprocal phase scale. The particle-label invariant and determinant-one inverse formula give

$$\mathcal{P}_j^{\text{par}} \leq C j^a x_{j-1}^b \exp\left(\frac{c x_{j-1}}{(j-1)^4}\right).$$

At the first normal stage its predecessor bounds are fixed polynomials in X . The full packet envelope $\mathcal{P}_j$ also includes δ_j^{-1} and has denominator $(j-1)^3$ as in (71).

The relative size estimate for the selected packet bounds the history, early-time and initial primary errors by a fixed polynomial times $h_j e^{-bx_{j-1}}$. After enlarging that polynomial to cover all the actual initial fields, a common bound is

$$b_j = C_d \mathcal{P}_j^{c_d} h_j e^{-bx_{j-1}} \leq e^{-bx_{j-1}/2}.$$

The logarithm of the prefactor divided by x_{j-1} is at most $C_d/(j-1)^3 + o(1)$; the seed predecessor contributes only a polynomial in X . This comparison is ensured by the same finite choice of J, X .

The late upper-pressure estimate uses the actual global parent strain bound $M_{\text{all}} \leq 3C_1C_I \leq 3(\mathcal{P}_j^{\text{par}})^2$. The pressure sign (15) on the packet support and $f'_\delta \geq -1$ therefore bound its late contribution by $C_d(\mathcal{P}_j^{\text{par}})^2\delta_j h_j$. The large positive part of f'_δ gives a nonpositive leading Hessian contribution; only its bounded negative part enters this upper bound. Outside the support the leading tensor is zero. In particular, no global sharp bound of size h_{j-1} on the parent gradient is needed. Since

$$\delta_j h_j \leq e^{-x_{j-1}/(2j^3)},$$

the parent-only polynomial factor is absorbed by the stronger damping with denominator j^3 . For large fixed J and then X , the resulting contribution is at most $e^{-x_{j-1}/(4j^3)}$. This absorption would not be valid with an envelope containing the new δ_j^{-1} .

The exact packet gradient and Hessian remainders are bounded by $k_j^{-1/4}$. Thus nonnegative bounds for the increments are

$$e_{p,j} = C_d(b_j + (\mathcal{P}_j^{\text{par}})^2\delta_j h_j + k_j^{-1/4}), \quad e_{i,j} = C_d(b_j + k_j^{-1/4}). \quad (81)$$

The latter applies to the complete initial finite family, including all means and terminal correctors. For each fixed $A > 0$, x_{j-1}/j^A increases at least geometrically for large J . The damped error terms and fixed negative powers of x_{j-1} consequently have arbitrarily small sums after increasing X . The five terms in the full mean-reserve increment are bounded by $1/64$ each; their total is at most $5/64$.

The initial core term in that reserve is retained. With the fixed cutoff above and $B_c = O_d(X^{D_0})$, it satisfies

$$c_d B_c r_*^d S_{\text{base}} = O_d(J^2 X^{2+(1/2-d)D_0}) \longrightarrow 0.$$

The remaining seed reserve terms also tend to zero. The seed solution is chosen with full reserve at most $1/8$, and the first normal forward step contributes at most another $1/8$. Later finite prefixes are therefore at most $1/4 + 5/64 = 21/64 < 1/2$, on their actual horizons.

Boundary penalties remain polynomial by an exact difference invariant, not by a global sharp initial-gradient assertion. After the two initializations,

$$B_c - B_e \leq C_{\text{init}} X^{D_0}.$$

Every subsequent joined update adds the same e_i to B_c, B_e , so this inequality persists. On a positive-history horizon T the full reserve implies $B_e T \leq 1/2$, while $T \geq S_{\text{base}}/12$. For the already chosen large X these give

$$B_e \leq 2X^{D_0}, \quad B_c \leq (C_{\text{init}} + 2)X^{D_0}, \quad |\mathcal{L}| \leq C_{1,d}(C_{\text{init}} + 2)X^{D_0}.$$

These are the boundary bounds used in the next source envelope.

10.4 Seed and induction step

Set $u_{J-2} = u_B$. Apply Lemma 6.3 on $[0, S_{\text{base}}]$ with the seed scales, $m_0 = p$, $v(0) = q$ and $\alpha = \delta_{J-1} h_{J-1} = X^{-10}$. The base mean parameters have $B_c = B_e = C_{B,d}$, $K = K_{B,d}$ and $\mathcal{L} = C_{1,d} C_{B,d}$, with fixed nonnegative dimensional constants. Its positive core radius and cutoff are exactly the normalized parameters specified above. Their full mean reserve is small on the short base interval; no boundary term is set to zero.

At time zero $m \cdot M_B v = 1$ throughout the oscillatory support. The uniformly bounded base equations preserve $m \cdot M_B v \geq 1/2$ on the chosen short interval. The seed upper-pressure increment is therefore at most $C_d X^{-10} + k_{J-1}^{-1/4}$. The forced high initial data vanish in this zero-time forward construction, but the actual mean initial fields need not vanish. Retain

every finite mean and full tensor corrector. The correction removing the residual has zero initial value, so the initial increment is precisely this complete finite sum.

The estimates for that sum give interior and exterior gradient bounds

$$e_i = \frac{\alpha}{\delta_{J-1}} + \frac{\pi C_{\chi,d}\alpha + C_{\nabla}}{k_{J-1}}, \quad e_o = \frac{C_{\nabla}}{k_{J-1}}.$$

Here C_{∇} is the higher-grade source bound, controlled at the prescribed seed frequency. Update B_c by e_i and B_e by e_o ; retain the physical cutoff and set the new penalty to $C_{1,d}(B_c + e_i)$. The difference obeys $B_c^+ - B_e^+ \leq C_{\text{lit}}X^{D_0}$. The nonoscillatory initial terms outside the small core are bounded by e_o and remain in the common fixed compact support. The full reserve of this seed child is at most $1/8$ by the explicit short-time and frequency estimates above.

The seed expansion is, at the fixed origin,

$$M_{J-1} = M_B + h(t)q(t)p(t)^T + \mathcal{E}(t), \quad |\mathcal{E}| \leq k_{J-1}^{-1/4}, \quad h(0) = h_{J-1}.$$

The normalized frame is transported by the actual base gradient. Its activation parameters are exactly $a_J = 1$ and $\beta_J = X^{-2}$. The actual lifted flow estimate gives its polynomial particle-map bound at k_{J-1} . Thus the complete input state for stage J consists of two exact solutions, the actual frame and shear decomposition, the central gradient and Hessian bounds, the global one-sided pressure bound, the strict fixed-cutoff reserve and the polynomial label bounds. The first activation time is zero, so stage J uses the forward primary with its actual weighted growth and initial amplitude. Its seven estimates were included in the sole scale choice. Its initial interior and exterior updates can differ; their difference adds at most $C_{\text{lit}}X^{D_0}$. Thus $C_{\text{init}} = 2C_{\text{lit}}$ covers both initializations. Its pressure-time and initial-time reserve increments are each at most $1/16$, leaving the first positive-history state with full mean reserve at most $1/4$. Its actual activation time is $t_J > 0$; all later steps use the joined history construction.

Assume that state at stage j . The actual parent is available on $[0, S_{j-1}]$, and $S_{j-1} - t_{j-1} = 2W_j$ contains the target and short retained extension. Restrict it to $[0, S_j]$ when forming the actual central source and packet. The central horizon is precisely

$$H_j^{\text{sc}} = a_j(S_j - t_{j-1})/\epsilon_j = \Theta_j;$$

it is bounded by the numerical source envelope $\hat{\Theta}_j$. Coefficient bounds inherited from the longer parent remain valid under restriction. The source envelope bounds the error contributions; it does not extend any field past its actual time domain. Duhamel and energy estimates and the relative target-size comparison are used only on the actual retained interval, whose scaled extension past the target is $\Delta\tau_j$. Along the older central solution, $\dot{B} = -B^2 - H_{j-2}$, which gives the central bounds on $B, \dot{B}$. Lemma 5.1 supplies the actual stationary history for $j > J$, and the zero-time prescription supplies it for $j = J$. The ambient comparison and source-envelope argument verify the quantitative neighborhood, full transverse propagator and every coefficient bound at the already prescribed ρ_j, k_j . Take

$$\alpha_j = \frac{\delta_j h_j}{|m(t_j, 0)| |v(t_j, 0)|}.$$

The joined construction in Lemma 6.3 then gives an exact smooth odd reflection-equivariant solution u_j on $[0, S_j]$. Its full ambient tensor correctors, all means and the residual correction are part of this same solution. The graph-flow identities give the next actual particle-map invariant $K_{h,\text{next}} \leq k_j^{D_d}$.

At the fixed central particle the expansion gives

$$\sup_{t \leq S_j} |M_j(t, 0) - M_{j-1}(t, 0)| \leq C_d h_j + k_j^{-1/4}, \quad \sup_{t \leq S_j} |H_j(t, 0) - H_{j-1}(t, 0)| \leq C_d h_j h_{j-1} + k_j^{-1/4}.$$

Here the leading Hessian uses the sharp central parent strain bound. Together with (80) these preserve

$$\sup_{t \leq S_j} |M_j(t, 0)| \leq C_M h_j, \quad \sup_{t \leq S_j} |H_j(t, 0)| \leq C_H h_j h_{j-1}.$$

These are the central gradient and Hessian bounds needed by the central endpoint and renewal equations. They are not asserted as full-space sharp norms. Full-space coefficient and spatial derivative bounds instead follow from the constructed particle-label fields and the determinant-one polynomial inverse. The one-sided bounds (81) preserve the global pressure upper bound, the initial interior and exterior quadratic-form bounds, and the full fixed-cutoff reserve. The pressure is not assumed local: its signed leading expansion and its residual error estimate hold globally, with the leading tensor vanishing off the packet support.

For $t \in [t_j, S_j]$, the new normalized pair p_2, q_2 satisfies

$$M_j = M_{j-1} + h(t) q_2 p_2^T + \mathcal{E}_j, \quad |\mathcal{E}_j| \leq k_j^{-1/4}, \quad h(t_j) = h_j.$$

Its frame equations have older field M_{j-1} , by the exact ray and velocity equations, and $h'/h = -(M_{j-1})_{p_2 p_2} - (M_{j-1})_{q_2 q_2}$. Lemma 5.3 bounds each relative change in a_j by $C_d x_{j-1}^{-4}$, whose sum is arbitrarily small. Since $a_J = 1$, the product of the successive relative factors remains in $[1/2, 2]$. The same lemma gives directly $\beta_{j+1} x_j^2 \in [1/2, 2]$; that estimate is not multiplied across stages. Its compression bound and (80) give $p_2 \cdot M_{j-1} p_2 < -1$, while $|\mathcal{E}_j| < 1$. Also $(C_M h_{j-1} + 1)/h_j < \zeta$. These prove the two actual activation conditions for the next endpoint problem. The central gradient and Hessian bounds on the entire history and $t_j^{-1} \leq h_j$ prove the next endpoint-selection hypotheses. The global one-sided mean bound and polynomial label coefficients prove the full spatial history inverse hypotheses. These estimates establish every component of the induction hypotheses for the subsequent solution.

11 The limiting datum and endpoint

The initial estimates of Section 9, applied to the same joined finite families and the actual source envelope, give

$$\sum_{j \geq J} \|u_j(0) - u_{j-1}(0)\|_{H^m} < \infty \quad \text{for every fixed } m.$$

The seed and the first normal forward increment are two fixed smooth increments; including the latter as the first term of this series or in the initial finite datum gives the same limit. Lemma 7.2 places all these initial velocities in one compact set. Sobolev embedding at arbitrarily high fixed orders therefore proves that their single limit u_0 is smooth, compactly supported and divergence free.

The target times increase to a finite positive $T_\infty \leq S_{\text{base}}$. At the central target the exact expansion gives

$$|\nabla u_j(t_j, 0)|_F \geq h_j - C_M h_{j-1} - k_j^{-1/4} \longrightarrow \infty.$$

Let u be the unique maximal smooth Sobolev solution from u_0 . If its lifetime T^* exceeded T_∞ , it would be smooth on a closed interval containing all t_j , with bounded H^{s+1} norm. Apply Lemma 2.3 on the actual varying intervals $[0, t_j]$ to u_j and u . Their initial H^s difference tends to zero, so their gradients converge uniformly on those intervals. The displayed divergence of the central gradients would contradict boundedness of ∇u there. Thus $0 < T^* \leq T_\infty < \infty$. This argument does not assert equality of T^* and T_∞ . Lemma 2.4 now supplies the exact Frobenius gradient limsup and the once-counted antisymmetric-vorticity integral in H_d . Lemma 2.2 fixes the Newton pressure normalization, and Lemma 2.1 supplies every required Sobolev order on each shorter closed time interval.

The quantitative joined-source estimate and the prescribed scale sequence verify the hypotheses at every finite stage. The initial limit, finite lifetime and continuation conclusions therefore prove Theorem 1.1, with no additional assumption on the dimension or on a prospective solution.

Use of artificial intelligence

All mathematical derivations in this manuscript were produced using artificial intelligence.

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